

Chapter 3

Pump Hydraulics and Physical Concepts



3.1 One-Dimensional Calculation with Velocity Triangles

The objective of the flow calculation for a centrifugal pump is to determine the main dimensions and blade angles of the impeller and the collector for a specified pumping task. To do so, delivery flow, head and speed for the calculation point (usually the best efficiency point) must be available. For this calculation secondary flows and uneven velocity distributions in the impeller and diffusing elements are ignored and the real flow is replaced by an idealized one-dimensional flow (streamline theory). The one-dimensional flow theory is essential for the understanding of the basic flow processes and for the initial design of the impeller, the volute casing or diffuser and the return vanes - even if these components are subsequently optimized by means of numerical methods.

The hydraulic calculation always follows the course of the fluid through the machine. In this process, we consider the inlet and outlet of each component in the flow path. This results in the calculation stations 1 to 6 below. The impeller, diffuser and volute casing with the respective main dimensions, angles and calculation stations are shown in Tables 0.1 and 0.2.

Multistage diffuser pumps:	Volute casing pumps:
(1) Impeller inlet (LE)	(1) Impeller inlet (LE)
(2) Impeller outlet (TE)	(2) Impeller outlet (TE)
(3) Diffuser inlet (LE)	(3) Volute inlet
(4) Diffuser outlet (TE)	(4) Volute outlet
(5) Return vane inlet (LE)	(5) Outer volute
(6) Return vane outlet (TE)	(6) Discharge nozzle

LE means the leading edge and TE the trailing edge of the blades or vanes.

In the case of volutes, the calculation stations 5 and 6 are only required for the outer channel of double volutes, (Table 0.2).

Pump calculation follows the direction of pressure increase (ignoring the minor pressure drop in the inlet nozzle or inlet casing and in the return vanes if applicable). Likewise, turbine operation of a pump (Chap. 12) is calculated in the direction of the flow, but follows the course of diminishing pressure. The numbering of the calculation stations is retained: the turbine calculation sequence therefore is from station 6 to 1. The following shall be determined:

- Circumferential speeds: $u = \omega \times r = \pi \times d \times n/60 = \pi \times d \times n^{(s)}$ (n in rpm; $n^{(s)}$ in s)
- Absolute velocities: c
- Relative velocities: w
- Angle in the absolute reference frame: α
- Angle in the relative reference frame (rotating reference frame): β
- Blade or vane angles on a component: subscript B. Unless otherwise specified this means the *camber* angle of the impeller blades, the diffuser vanes, the return vanes or the volute cutwater.
- Passage width between the blades or vanes: a (shortest distance, Table 0.2)
- Width in the meridional section: b
- Meridional velocity components: subscript m , where: $c_m = Q_{La}/(f_q \times \pi \times d \times b)$; note that $w_m = c_m$ applies.
- Circumferential velocity components: subscript u , where $u = c_u + W_u$
- c_u is counted as positive when acting in the direction of u , negative when opposite to the u -direction
- w_u is counted as positive if acting opposite to the direction of u
- Outer, mean and inner streamlines are described by the subscripts a, m, i ; for example $c_{1m,m}$, $\beta_{1B,a}$. The subscript m always refers to the streamline which starts or ends on the inlet or outlet diameter calculated as the *geometric* mean value (d_{1m} and d_{2m} on the impeller and d_{3m} and d_{4m} on the diffuser). The one-dimensional calculation assumes representative conditions on the mean streamlines defined in this way. At low specific speeds frequently only the outer and inner streamlines are considered while three, five or more streamlines are advisable at high specific speeds.
- When the fluid flow enters or leaves a cascade there is a blockage effect due to the finite blade thickness. The flow velocity in the meridional section undergoes (arithmetically) a sudden change at these points. It is, therefore, possible to define velocities with and without blockage at any calculation station; quantities with blockage are described by a raised stroke, e.g. c_{1m}' .
- Two velocities must be distinguished at the inlet of a blading: (1) the velocity vector, which is calculated from the velocity triangle (vector diagram); (2) the mean velocity resulting from the continuity equation. This is given the subscript q to avoid confusion; hence $c_q = Q/A$ and $w_q = Q/A$, where A is the local cross section under consideration.

- The flow rate of the impeller Q_{La} consists of the useful flow rate Q plus leakage through the annular seal at the impeller inlet Q_{sp} and the balance flow Q_E , resulting in $Q_{La} = Q + Q_{sp} + Q_E$. In addition, there may be an intermediate take-off in the case of multistage pumps.

The flow around a blade is considered from the view of an “observer sitting in the blade”. Consequently, the *relative* velocities are relevant for the impeller, while the *absolute* velocities are used for the volute or diffuser calculation. The relationship between circumferential impeller speed u , relative velocity w and absolute velocity c is obtained from the rules of vector addition which can be illustrated as velocity triangles (refer also to Chap. 1.1). The equations describing the geometry of triangles therefore apply to the calculation of all velocities, their components in circumferential or meridional direction and the angles α and β .

Inlet triangle: Consider the velocity relationships on the impeller inlet shown in Table 3.1. The meridional velocity immediately in front of the impeller blade leading edges is $c_{1m} = Q_{La}/A_1$, where A_1 is calculated in accordance with the position of the leading edge, Eq. (T3.1.2). Immediately after the leading edges the meridional velocity is increased to $c_{1m}' = \tau \times c_{1m}$ due to the blade blockage. The latter is obtained from $\tau_1 = t_1 / \{t_1 - e_1 / (\sin \beta_{1B} \times \sin \lambda_{La})\}$, Eq. (T3.1.7), using the sketch in Table 3.1 (with the pitch of the blades $t_1 = \pi d_1 / z_{La}$).¹ If the blades are not perpendicular to the front shroud ($\lambda_{La} \neq 90^\circ$), additional fluid is displaced, since the blade blockage obviously becomes larger than with $\lambda_{La} = 90^\circ$. This influence is taken into account by λ_{La} as shown by the sketch in Table 0.1. The *circumferential* components of the absolute or relative velocity are not affected by the blockage, as follows from the conservation of angular momentum.

The fluid flow to the impeller is mostly axial ($\alpha_1 = 90^\circ$). The circumferential component of the absolute inflow velocity is therefore $c_{1u} = 0$. However, if a device for pre-rotation control is installed or when the inlet casing or the return vanes generate an inflow with $\alpha_1 \neq 90^\circ$, the circumferential component is obtained from Eq. (T3.1.3). Figure 3.1 shows the inlet triangles for non-swirling inflow $\alpha_1 = 90^\circ$, pre-rotation $\alpha_1 < 90^\circ$ (“pre-swirl”) and counter-rotation $\alpha_1 > 90^\circ$. It can be seen that the approach flow angle β_1 of the blades is increased by pre-swirl and reduced by counter-swirl. As a result of the blade blockage c_{1m} grows to c_{1m}' so that the approach flow angle increases from β_1 to β_1' .

The difference between blade angle β_{1B} and flow angle β_1' is known as incidence: $i_1' = \beta_{1B} - \beta_1'$. If the incidence is zero, the blade has only a displacement effect on the flow; local excess velocities are correspondingly low. They grow with increasing incidence until the flow separates, since incident flow generates a circulation around the leading edge at $i_1 \neq 0$, Fig. 3.1d. The flow angles without and with blockage are calculated from Eqs. (T3. 1.6 and 3.1.8).

¹With profiled blades, the effect of the blockage may not be easily defined in certain conditions. Generally, e_1 must be selected where the cross section is narrowest relative to the next blade. The effect of the angle λ (if any) must be considered iteratively.

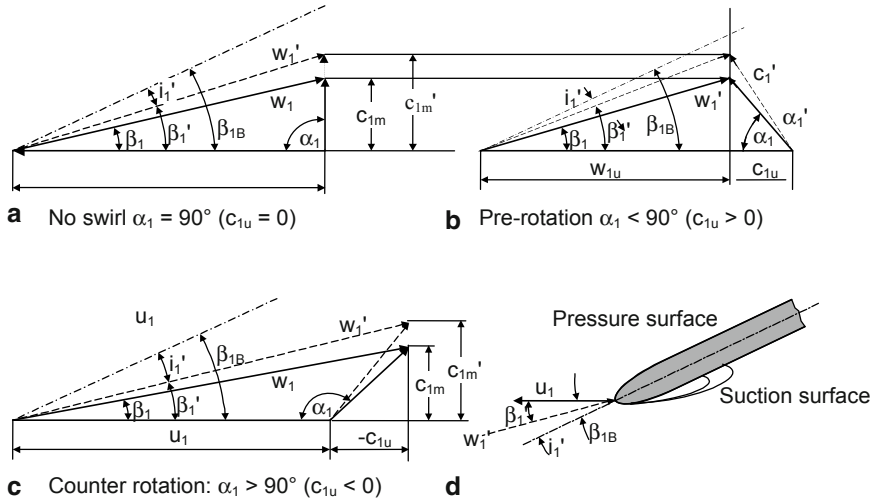


Fig. 3.1 Velocity triangles at the impeller inlet

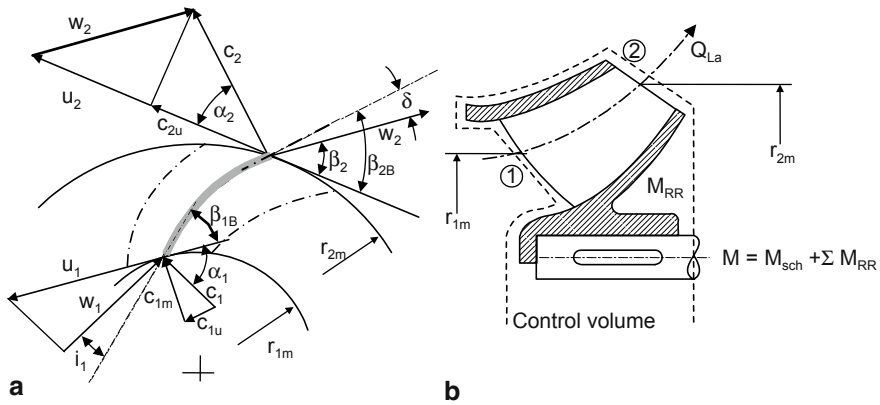


Fig. 3.2 Balance of angular momentums acting on the impeller. **a** Velocity vectors; **b** control volume

With a certain flow rate (i.e. a specific c_{1m}') blade and flow angles are identical ($\beta_{1B} = \beta_1'$) and the incidence becomes zero. This flow situation, called “shockless entry” or “shock-free”, is calculated from the condition $\tan \beta_1' = \tan \beta_{1B}$ from Eq. (T3.1.10), where $\phi_{1,SF} = c_{1m}'_{SF}/u_1 = \tan \beta_{1B}/\tau_1$ is the flow coefficient introduced in Chap. 3.4. If the approach flow angle drops below the blade angle ($i_1 > 0$), the stagnation point is situated on the pressure surface of the blade. If the pump flow rate exceeds the value of the shockless entry, the incidence is negative and the stagnation point is located on the blade suction surface.

Outlet triangle: The velocity relationships at the impeller outlet are shown in Fig. 3.2a and Table 3.2. The meridional velocity downstream of the impeller is

obtained from Eq. (T3.2.2). Blade blockage is still present immediately upstream of the impeller outlet and the velocity is correspondingly greater than downstream of the trailing edge: $c_{2m}' = c_{2m} \times \tau_2$, Eq. (T3.2.3). Again, the blockage does not affect the *circumferential* component. The absolute velocity c_2 and outflow angle α_2 are relevant for the design of the diffusing elements, Eqs. (T3.2.10 and 3.2.13).

3.2 Energy Transfer in the Impeller, Specific Work and Head

The head H measurable between discharge and suction nozzles has been defined in Table 2.2 as the useful total head difference generated by the pump. However, Eq. (T2.2.5) does not specify how the impeller converts the power applied to the rotor into pumping work.

To calculate the energy transfer from the impeller to the fluid the conservation of momentum according to Eq. (1.10) can be used, since it establishes a relationship between flow quantities and external forces. If suitable control surfaces are selected, the momentum balance allows a global statement without knowledge of the flow details within the control volume.

The control volume is selected according to Fig. 3.2b around the impeller as presented in the meridional section. The following quantities are applied to its boundary:

- The flow Q_{La} enters the impeller through area 1 with the angular momentum $\rho \times Q_{La} \times r_{1m} \times c_{1u}$.
- The flow leaves the impeller through area 2 with the angular momentum $\rho \times Q_{La} \times r_{2m} \times c_{2u}$.
- The control surface cuts through the shaft. Therefore, external forces in the form of $M = M_{sch} + \sum M_{RR}$ must be applied to this interface.
- Viscous shear stresses cause friction moments M_{RR} on front and rear shrouds that can be estimated according to Chap. 3.6.1. But these need not be considered when calculating the moment M_{sch} acting on the blades.
- Turbulent shear stresses occur in the areas 1 and 2 in circumferential direction, since the control surfaces are not positioned perpendicularly to the velocity vectors c_1 and c_2 . These shear stresses generate a moment M_τ , which is neglected when calculating according to the streamline theory. From numerical calculations it can be inferred that M_τ amounts to typically 1% of the moment transmitted by the impeller when no recirculation is present².

The static pressures in the areas 1 and 2 do not generate any forces in the circumferential direction and therefore do not enter the momentum balance. Likewise, the radial velocity components do not contribute to the angular momentum acting

²Recirculation normally occurs at partload below $q^* = 0.5$ to 0.7 (Chap. 5). It may, however, occur even near the best efficiency point when extremely wide impellers are used (for example in dredge pumps).

on the blades; consequently, only the circumferential components c_{1u} and c_{2u} need to be considered. When applying one-dimensional streamline theory, the non-uniform velocity distributions at the control surfaces are replaced by representative average velocities. They generate exactly the same angular momentum on the mean radii r_{1m} , r_{2m} as the integral over the real flow distributions. Subject to all these prerequisites, the conservation of momentum provides the angular momentum acting on the blades in the form of “Euler’s turbine equation”:

$$M_{sch} = \rho Q_{La} (r_{2m} c_{2u} - r_{1m} c_{1u}) \quad (3.1)$$

The mean radii are defined in such a way that they divide inlet and outlet cross sections into two areas which both receive identical flow rates (where c_m is assumed constant over the cross section):

$$r_{1m} = \sqrt{\frac{1}{2}(r_{1a}^2 + r_{1i}^2)} \quad \text{and} \quad r_{2m} = \sqrt{\frac{1}{2}(r_{2a}^2 + r_{2i}^2)} \quad (3.2)$$

M_{sch} is the torque which must be applied to the shaft in order to create the flow conditions shown in Fig. 3.2a. According to Newton’s law (“action equals reaction”), M_{sch} is also equal to the moment transmitted to the fluid. At an angular velocity ω of the shaft the corresponding driver power is (with $u = \omega \times r$):

$$P_{sch} = M_{sch}\omega = \rho Q_{La} (u_{2m} c_{2u} - u_{1m} c_{1u}) \quad (3.3)$$

The specific work produced by the blades is obtained from dividing P_{sch} by the mass flow rate $\dot{m}$ passing through the impeller ($\dot{m} = Q_{La} \rho$), Chap. 1.2.2:

$$Y_{sch} = Y_{th} = \frac{P_{sch}}{\rho Q_{La}} = u_{2m} c_{2u} - u_{1m} c_{1u} \quad (3.4)$$

As demonstrated by Eqs. (3.1), (3.3) and (3.4) a pre-swirl ($\alpha_1 < 90^\circ$) reduces blade moment, power consumption and head while a counter-swirl increases these values. Since the specific work of the blades according to Eq. (3.4) does not include any losses (although it applies to flows with losses!), it is also called “theoretical specific work”. Accordingly, $H_{th} = Y_{sch}/g$ is the “theoretical head”.

The energy Y_{sch} transmitted to the fluid comprises both the useful energy in the discharge nozzle Y according to Eq. (2.1) and the hydraulic losses (see Chap. 1.2.2 and Eq. (1.8)). The following therefore applies: $Y_{sch} = Y + g \times \Sigma Z_h$. These losses are described by the “hydraulic efficiency” η_h defined by Eq. (3.5):

$$\eta_h = \frac{Y}{Y_{sch}} = \frac{H}{H_{th}} = \frac{H}{H + \Sigma Z_h} \quad (3.5)$$

The hydraulic efficiency includes all hydraulic losses between the suction and discharge nozzles, i.e. in the inlet, impeller, diffuser and discharge casing. Details concerning η_h can be found in Chap. 3.9, Eq. (3.28a) and Figs. 3.30, 3.31 and 3.32.

From geometrical relationships of the velocity triangles $u \times c_u = \frac{1}{2} (u^2 + c^2 + w^2)$ is obtained. Inserting this expression into Eq. (3.4) produces Eq. (T3.3.2). According to this equation, the specific work consists of three components: centrifugal component $u_2^2 - u_1^2$, deceleration of the relative velocity $w_2^2 - w_1^2$ and acceleration of the absolute velocity $c_2^2 - c_1^2$. The impeller outlet velocity c_2 is decelerated in the collector. There, its kinetic energy is for the most part converted into static pressure. With reference to Eq. (1.15) the first two terms stand for the increase of the static pressure in the impeller plus the impeller losses, Eq. (T3.3.3).

The energy Y_{sch} transmitted to the fluid by the impeller causes a total pressure increase $Y_{tot,La}$ at the impeller outlet. A fraction of Y_{sch} is dissipated in the impeller in the form of losses. Consequently $Y_{sch} = Y_{tot,La} + g \times Z_{La}$ (Chap. 1.2.2) applies. Y_{tot} is the *useful* hydraulic energy present at the impeller outlet (less the inlet energy at the suction nozzle). The specific work $Y = g \times H$ measured between discharge and suction nozzles is obtained as $Y = Y_{tot,La} - g \times Z_{Le}$ when the losses in the diffuser are subtracted. The relationships between specific work and the increase of the total pressure and the static pressure are summarized in Table 3.3. The increase of the static pressure in the impeller H_p (expressed as static head) is obtained from Bernoulli's equation in the relative reference frame, Eq. (1.15). The ratio of the static head to the head of the pump, $R_G = H_p/H$, is called the "degree of reaction". Table 3.3 also includes equations which describe the energy transfer in the relative reference frame.

According to Eq. (3.4) the blade forces create a flow condition which is described only by the circumferential components of the absolute velocity at the impeller inlet and outlet. The static pressures need not be considered. According to Chap. 1.4.1 the static pressures result from the streamline curvature imposed by the blades on the fluid in such a way that they maintain equilibrium with the centrifugal forces.

As will be discussed in Chap. 5, recirculations occur at partload. Their angular momentums acting on the impeller inlet and outlet are *not* included in the above equations. *It is most important to note that one-dimensional pump theory as discussed above applies to pump operation without recirculation only. Ignoring this fundamental fact often leads to erroneous conclusions.* This problem will be discussed in greater detail in Chap. 8.4.

3.3 Flow Deflection Caused by the Blades. Slip Factor

As mentioned above, only the average velocities prevailing at the chosen control surfaces are considered when applying the theorem of conservation of momentum. The complicated flow conditions in the interior of the control volume are ignored. The conservation of momentum consequently fails to show the manner in which the flow observed at the control surfaces was generated; and how the impeller, its outlet

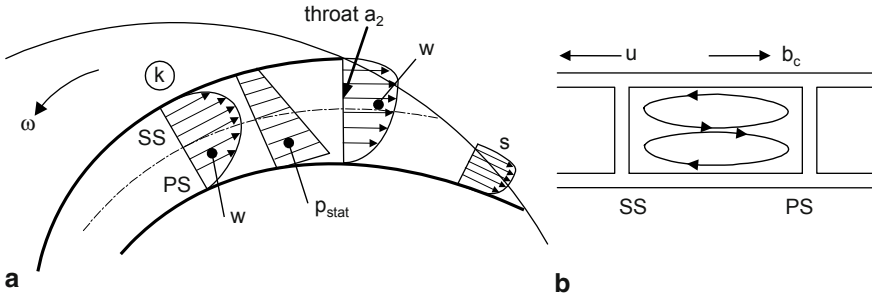


Fig. 3.3 Slip phenomenon. **a** Flow between the blades; **b** Secondary flow

width and blade angles must be designed so that these average velocities actually occur at given values of Q_{La} and ω .

The blade moment results from blade forces which can be imagined as an integral of the pressure and shear stress distributions over the blade surface - much like the lift of a wing is integrated from its pressure distribution. If a blade (or a wing) is to generate a net force, this integral obviously cannot be zero, i.e. greater pressures must be present on the blade pressure surface than on the suction surface. Since the pressure distributions in a turbo machine solely result from the velocity distributions around the blades, it must be concluded that different flow conditions are present on blade pressure and suction surfaces. Hence the flow is not able to follow the blades exactly (the flow is not “blade-congruent”). The work transfer is only made possible through a deviation between the blade and flow angles. In Fig. 3.2a, the flow angle β_2 is therefore represented smaller than the blade angle β_{2B} . The described phenomenon is quantified by the “slip factor” or by the “deviation angle” $\delta = \beta_{2B} - \beta_2$. Both terms implicitly assume the idea of a blade-congruent flow and consider the deviation of the real flow from the blade outlet angle.

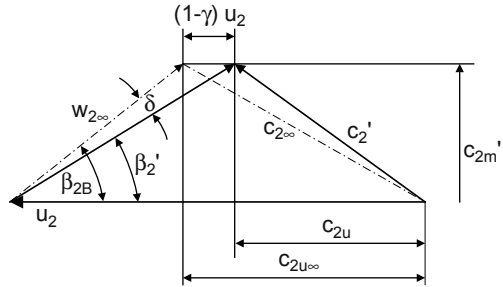
As will be discussed in detail in Chap. 5, the flow distribution at the impeller outlet - and consequently the mean flow angle and the slip factor - results from a complex equilibrium of forces. The deviation of the real flow from the blade-congruent flow is essentially influenced by the following mechanisms:

- The velocity differences between the pressure and suction surfaces of the blades are induced by the transfer of work, Fig. 3.3a, Profile k .
- The Coriolis acceleration b_c is opposite to the direction of rotation and causes a secondary flow which transports fluid towards the pressure surface, thus reducing the flow angle β (refer to Fig. 3.2a and Chap. 5).
- Immediately downstream of the trailing edge, the differences in static pressure acting on the blade pressure and suction surfaces vanish, because pressure differences in the free flow can only be maintained through different streamline curvature. The velocity distribution already adapts in the triangular section downstream of the throat at the impeller outlet in a way that this outflow condition is satisfied and that the circulation around the trailing edge is not excessive. Ahead of the throat, in the actual impeller channel, the flow is guided more effectively

Fig. 3.4 Streamline of relative flow in a radial impeller (Navier-Stokes calculation). The broken lines correspond to blade-congruent flow



Fig. 3.5 Slip and deviation angle



and deviates less from the blade angle (refer to profiles k and s in Fig. 3.3a). Figure 3.4 illustrates these effects by the paths of the relative velocity calculated with a 3D-Navier-Stokes program. Up to the throat a_2 the flow paths are almost congruent with the blade contour marked by broken lines; in the inclined section after a_2 the flow paths curve in the direction of the pressure surface and the flow angle β is correspondingly reduced toward the impeller outlet. *With backward curved blades, the slip factor is to a large extent created in the triangular section at the impeller outlet.* Conversely, with radial blades (having inlet and outlet angles of 90°) the slip factor is primarily caused by the Coriolis force.

Since these flow processes cannot be calculated by simple means, empirical data have to be used when calculating the flow outlet angles according to the streamline theory. Figure 3.5 shows the relevant outlet triangle immediately ahead of the trailing edge (i.e. with blockage), where subscript ∞ is used for blade-congruent flow. Known are u_2, c_{2m}' from Q_{La} and the blade outlet angle β_{2B} . The difference between $c_{2u\infty}$ and c_{2u} is defined as:

$$c_{2u\infty} - c_{2u} = (1 - \gamma) u_2 \quad (3.6)$$

Here γ is defined as slip factor, while the quantity $(1 - \gamma)$ is the “slip”. Therefore $\gamma = 1.0$ means blade-congruent flow. The smaller γ , the greater the deviation between flow and blade angles.

With Eq. (3.6) and $\tan \beta_{2B} = c_{2m}' / (u_2 - c_{2u\infty})$ the following definition equation for the slip factor is obtained:

$$\gamma \equiv 1 - \frac{c_{2u\infty} - c_{2u}}{u_2} = \frac{c_{2u}}{u_2} + \frac{c_{2m} \tau_2}{u_2 \tan \beta_{2B}} \quad (3.7)$$

To obtain data for the impeller design, the coefficient γ has to be calculated from test data and correlated with geometrical quantities. This calculation is performed at the best efficiency point or in its vicinity. In this context, the measured values of Q , H , P , η are known, and c_{1u} can be calculated from Eq. (T3.1.3) in case of $\alpha_1 \neq 90^\circ$. The hydraulic efficiency η_h is determined by loss analysis according to Chaps. 3.6 and 3.7 so that $H_{th} = H/\eta_h$ is also available. The circumferential component of the absolute velocity c_{2u} at the impeller outlet is obtained from Eq. (T3.2.8). This value can be used to calculate the coefficient γ from Eq. (3.7).

A modified form of the Wiesner equation [44] is used for correlating the data obtained in this way. Based on the calculations by Busemann [3], Wiesner developed a formula for the prediction of slip factors and compared it with measurements on compressors and pumps. Busemann's calculations assumed frictionless flow, zero blade thickness, two-dimensional blades between parallel shrouds and constant blade angle from inlet to outlet in the shape of a logarithmic spiral. To adapt the slip factor correlation to a wider database of pumps, the full blade thickness was used as blockage for the calculation and the correction factor f_1 was introduced. This evaluation produced Eq. (T3.2.6) for slip factor prediction. The correlation reflects the test results of γ for radial impellers with a standard deviation of about $\pm 4\%$ which implies a 95%-confidence limit of approximately $\pm 8\%$. Since the tolerances of the slip factor are emphasized in the calculation of the head, Eq. (T3.3.7), considerable uncertainties must be expected in the performance prediction according to the streamline theory, unless more accurate test data of similar impellers are available. Even an attempt to improve the accuracy of the correlation through additional parameters such as b_2^* or d_1^* has failed so far. Although these parameters have some influence, it is not sufficiently systematic and, consequently, obscured through additional 3D-effects of the flow.

Guided by the idea that a long blade would improve flow guidance and thereby reduce the slip, Pfleiderer developed a slip factor formula incorporating the static moment of the streamline (projected into the meridional section) [B.1]. However, with the analyzed pumps no reduction of the standard deviation below the mentioned level of $\pm 4\%$ was obtained. These findings (at first glance surprising) are explained thus: as mentioned above, the flow guidance in the actual blade channel is quite good so that the length of the blade in this area has only a minor influence. The slip mainly occurs in the "triangle" between a_2 and d_2 . Slip is determined to a far greater degree by Coriolis forces and the development of the blade angles and cross sections than by the blade length projected into the meridional section.

The uncertainties inherent to the streamline theory are due to the fact that the 3-dimensional impeller flow cannot be described by a one-dimensional approach - regardless of the number of geometrical parameters taken into account. The main sources of the uncertainties are discussed below:

- The calculation of the slip factor according to Eqs. (T3.2.4–6) only takes into account the parameters outlet angle β_{2B} , blade number z_{La} , blade blockage and indirectly the inlet diameter d_{1m}^* . However, the entire blade and channel development determines the velocity distribution at the impeller outlet and therefore also the integral which constitutes the mean value c_{2u} in Eq. (3.4).
- The secondary flow in the impeller increases with growing outlet width, which tends to increase the deviation between the blade and flow angles. Impellers with large relative outlet widths b_2^* generate very uneven outflow profiles. As mentioned above, the streamline theory breaks down when there is flow recirculation, and so does the slip factor concept. This may be the case for extremely wide impellers.
- To evaluate the work transfer in the impeller, not only the nominal outlet angle must be used, but the whole blade development near the impeller exit needs to be considered. To be relevant for slip and head prediction, β_{2B} should be approximately constant over an extended range - for example, up to the cross section at a_2 . The blade distance a_2 at the throat according to Table 0.2 is an important dimension in the design and for checking a casting. An angle β_{a2} can be defined by Eq. (3.8):

$$\beta_{a2} = \arcsin \frac{a_2}{t_2} = \arcsin \frac{a_2 z_{La}}{\pi d_2} \quad (3.8)$$

- The smaller the ratio $\tan\beta_{a2}/\tan\beta_{2B}$, the lower is the head and the greater the deviation from the slip factor calculated with β_{2B} according to Eq. (T3.2.6). This is demonstrated by the measurements in [42] where the outlet width a_2 was varied in a wide range, while inlet and outlet angles were kept constant. The resulting Q-H-curves varied from very steep to very flat. Applying any of the published slip factor formulae would completely fail to predict these results.
- Profiling the trailing edge has an effect on the impeller flow and, consequently, the head achieved. Depending on the trailing edge profile, various definitions are possible for the outlet angle, since camber angle β_{2B} , pressure side angle $\beta_{2B,DS}$ and suction side angle $\beta_{2B,SS}$ can be quite different. In the case of un-profiled blades, all three angles are approximately identical. With symmetrical profiling the camber angle constitutes a representative mean value. Conversely, with increased under-filing the camber angle is locally significantly greater than the angle on the pressure side. Finally, with profiling on the pressure side $\beta_{2B} < \beta_{2B,SS}$ applies, (Fig. 3.6).
- The head coefficient tends to decrease with increasing impeller inlet diameter (at otherwise identical design parameters). This fact is not sufficiently represented in the slip factor formula which predicts an influence of the inlet diameter only above some limiting value.
- The pumping action of the shrouds (discussed in Chap. 3.10.3) increases the work done by the impeller - and thus the slip factor. The incidence at the impeller inlet also has an effect on the flow turning achieved by the blades.

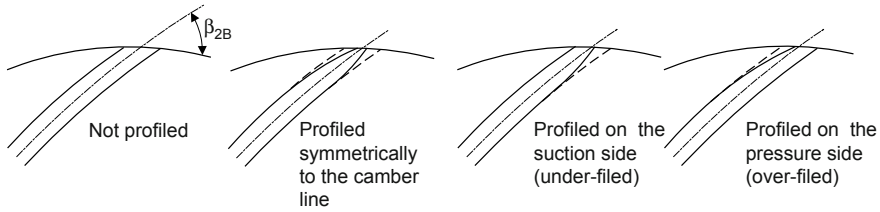


Fig. 3.6 Shapes of impeller blade trailing edges

- In summary, the meridional section, specifically the variation of the width $b(r)$, and the blade angle development $\beta_B(r)$ or $\beta_B(L)$ have an impact on the flow deceleration in the impeller channels and thus on the head. Such effects cannot be taken into account by 1D-theory and the slip factor concept, see also Chaps. 7.2.2 and 7.14.

3.4 Dimensionless Coefficients, Similarity Laws and Specific Speed

Turbulent flows in complex geometries cannot be described accurately by simple analytical means. In practice, such flows are treated with similarity characteristics (“model laws”) and dimensionless coefficients by means of which test results can be generalized and applied for prediction purposes in new applications. This approach even applies to geometries as simple as a straight pipe, where Reynolds number, relative roughness and friction coefficients are used for pressure drop prediction, Chap. 1.5.1.

Model tests and their conversion to other speeds and pump sizes consequently form a most important basis in pump design and application. A principle prerequisite for applying similarity laws are geometrical and dynamic similarity. *Geometric* similarity is fulfilled when *all* dimensions of the parts *conducting the flow* of two machines (subscript “a” for the prototype, subscript “M” for the model) are scaled to the same ratio $\lambda = D_a/D_M$. *Dynamic* similarity requires identical values of Euler number, Reynolds number, and Froude number for prototype and model. Which specific similarity characteristics have to be fulfilled depends on the physical process being examined. In pump design, it is only necessary to maintain identical Euler numbers in model and prototype. The effect of the Reynolds number is generally minor and is therefore only taken into account for detailed loss investigations, Chap. 3.10.³

If gravity exercises a substantial effect, the similarity according to Froude must be maintained. This applies to flows with a free surface, for example when modeling pump sumps, Chap. 11.

³When pumping oils or other media of high viscosity, head and efficiency are severely reduced compared with the pumping of water, Chap. 13.1.

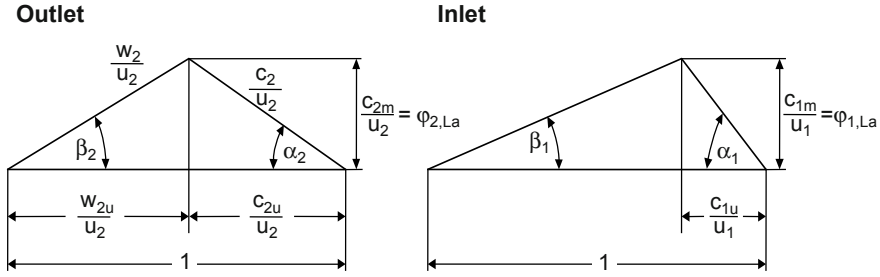


Fig. 3.7 Velocity triangles in dimensionless representation

Similarity or model laws can be developed according to various methods, [1, 12, 45]. Without going into the details of similarity theory, the model laws for pumps (and turbines) can be derived directly from the velocity triangles. To this end, consider the dimensionless inlet and outlet triangles where all velocities are referred to the circumferential velocities u_1 and u_2 respectively, Fig. 3.7. When doing so, all flow angles remain the same. Thus the representation according to Fig. 3.7 is independent of the circumferential velocity - and consequently independent of impeller diameter and speed. Since geometrical similarity is an indispensable requirement, also u_1 and u_2 are in a fixed relationship. All velocities are proportional to $n \times d$. *For geometrically similar impellers the velocity triangles at the inlet and outlet are determined completely by the quantities c_m/u and c_u/u alone* (Fig. 3.7).

Since the meridional velocity is proportional to the volumetric flow rate, the ratio c_m/u describes the influence of the flow rate. Consequently it is possible to define two flow coefficients: $\varphi_1 = c_{1m}/u_1$ for the impeller inlet and $\varphi_{2,La} = c_{2m}/u_2$ for the impeller outlet, Eqs. (T3.4.6) and (3.4.7). With geometrically similar impellers the same kinematic flow conditions prevail at a given flow coefficient regardless of impeller diameter and speed. The flow coefficients strictly refer to the volumetric flow rate $Q_{La} = Q/\eta_v$ passing through the impeller. Often, however, the flow coefficients are calculated with the useful flow rate Q . This is a good approximation, provided that the volumetric efficiency is not very different between model and prototype.

Since the quantities c_{2u}/u_2 and c_{2m}/u_2 according to Fig. 3.7 are identical in model and prototype because of identical kinematic conditions, it follows from Eq. (3.7) that the slip factor is the same in both machines. From Eq. (T3.3.1) it can be deduced further that the specific work in model and prototype is proportional to the square of the circumferential speed. The following head coefficient is therefore defined: $\psi = 2 \times Y/u_2^2$, Eq. (T3.4.8). $\psi = f(\varphi)$ is also a dimensionless characteristic which is independent of speed and impeller diameter.

Solving Eq. (T3.4.6) for the flow Q and Eq. (T3.4.8) for the head H and inserting the equations obtained in $P_{st} = \rho \times g \times H_{st} \times Q/\eta$ according to Eq. (2.4), a power coefficient λ according to Eq. (T3.4.9) can be defined.

The following proportionalities for geometrically similar pumps result directly from Eqs. (T3.4.6 to T3.4.9): $Q \sim n \times d^3$; $H \sim n^2 \times d^2$ and $P \sim n^3 \times d^5$. The similarity or model laws between model and prototype according to Eqs. (T3.4.1 to T3.4.4) are obtained from these relationships.

Since all forces are proportional to the product of pressure $\times$ area, the following also applies: $F \sim n^2 \times d^4$, Eq. (T3.4.5).

The above relationships presuppose the same efficiencies in model and prototype. If this prerequisite is not satisfied with sufficient accuracy, corrections have to be made by means of an appropriate loss analysis according to Chaps. 3.6–3.10. The similarity conditions for cavitating flows are covered in Chap. 6.2.3.

Specific speed: According to Eqs. (T3.4.8 and T3.4.2) we can write for the head:

$$H = \psi \frac{u_2^2}{2g} = k_1 n^2 d_2^2 \psi \quad (3.9)$$

Likewise, Eqs. (T3.4.6 and T3.4.1) yield for the flow rate:

$$Q = \varphi_2 \pi d_2^2 b_2^* u_2 = k_2 \varphi_2 b_2^* n d_2^3 \quad (3.10)$$

where k_1 and k_2 are constants. Eliminating the impeller diameter from these relationships, Eq. (3.11) is obtained after several transformations (written for the best efficiency point):

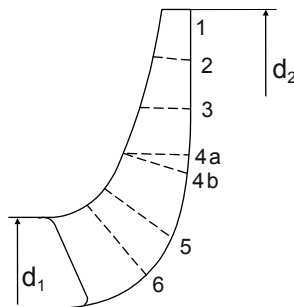
$$\frac{\sqrt{k_2} \sqrt{\varphi_2 b_2^*}}{k_1^{0.75} \psi_{\text{opt}}^{0.75}} = n \frac{\sqrt{Q_{\text{opt}}}}{H_{\text{opt}}^{0.75}} = \text{ISC} \quad (3.11)$$

The left side of this equation is a constant “ISC” for geometrically similar pumps; it depends on neither the impeller diameter nor the speed. The middle term describes the relationship between speed, volumetric flow and head of the pump.

Considering the wide range of applications of centrifugal pumps with their various impeller shapes (radial, semi-axial, axial, Table 2.1) described in Chap. 2, such a relationship between the flow characteristics Q , H and n is of fundamental significance for the selection of a suitable type of pump for a given application.

The quantity ISC (impeller shape coefficient) is therefore termed “specific speed” or “type number”. Its relevance is explained by way of Fig. 3.8. Consider an impeller at a given speed n_x , providing the flow rate Q_x . If the impeller is designed with various outer diameters or trailing edge shapes (1 to 6 in Fig. 3.8), the head decreases roughly with the square of the diameter according to Eq. (3.9). Since n_x and Q_x are constant, the relationship between Q , H and n changes; this means that the specific speed increases. The shape of the impeller in Fig. 3.8 changes from radial impellers (1 to 4a) to semi-axial impellers. Finally an almost axial impeller is

Fig. 3.8 Development of the impeller shape with the specific speed



achieved with trailing edge 6. As can also be seen from Fig. 3.8, the relationships b_2/d_2 and d_1/d_2 increase with growing specific speed.

The specific speed must invariably be calculated with the performance data at the best efficiency point and the head per stage. With double-entry impellers the specific speed is defined in Europe with the flow rate *per impeller side*, Eq. (T3.4.15). However, in North America the total flow rate is used (omitting the factor f_q , refer also to Chap. 2.2). Unfortunately n_q is not dimensionless. To obtain a dimensionless characteristic it is possible to divide Q_{opt} by the reference quantity $Q_1 = 1 \text{ m}^3/\text{s}$ and likewise introduce $H_1 = 1 \text{ m}$ and $n_1 = 1 \text{ rpm}$. Formulated thus, n_q is a dimensionless figure. In practice, n_q is considered as dimensionless without explicitly writing the reference quantities. A really dimensionless characteristic is ω_s according to Eq. (T3.4.16).

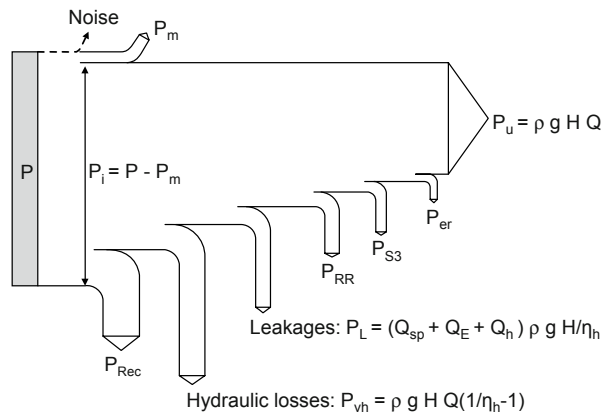
The specific speed permits a comparison of geometrically *dissimilar* impellers. Hence it can be used for comparing pumps from different vendors for a given application. But the specific speed provides only a rough indication concerning the hydraulic features, configuration and impeller shape of a pump. Depending on the application, a pump with given data n , Q_{opt} and H_{opt} - consequently with a given specific speed - can be equipped with impellers of quite different shapes. Examples are: (1) Sewage or dredge pumps typically have impeller outlet widths twice as large as pumps for non-clogging fluids with identical specific speed. (2) Depending on the required suction capability, the ratio of impeller inlet to outlet diameter d_1/d_2 can vary appreciably for a given n_q , (Chap. 6). (3) The head coefficient can be varied in relatively wide limits in order to obtain flat or steep characteristics without great effect on the specific speed. In this way the head rise H_o/H_{opt} can be adapted to specific requirements.

3.5 Power Balance and Efficiencies

Losses arise whenever a fluid flows through (or components move in) a machine. The useful power P_u introduced in Chap. 2.2 is therefore always smaller than the power P supplied at the pump shaft. The sum of all losses $\Sigma P_v = P - P_u$ is

dissipated into heat. Apart from the hydraulic losses according to Eq. (3.5) additional losses occur in a pump (“secondary losses”) so that altogether the following sources of loss must be considered:

1. Mechanical losses P_m in bearings and shaft seals. These do not generally result in a heating of the fluid and are therefore termed *external* losses.
2. Volumetric losses are due to all leakages which are pumped by the *impeller*. These include: (a) The leakage Q_{sp} through the annular seal at the impeller inlet. (b) The leakage Q_E through the device for axial thrust balancing. These leakages (Q_{S2} in Fig. 3.16) either occur through balance holes in single-stage or multi-stage machines with individual impeller balancing or through central balance devices such as pistons or disks (Q_E in Fig. 3.16), refer also to Chap. 9.2. In special cases additional fluid (Q_h) may be circulated within the pump which is branched off for auxiliary purposes such as feeding a hydrostatic bearing, flushing, sealing or cooling. Such flows must be taken into account when calculating volumetric efficiency and pump performance. To pump all of these leakages the power $P_L = \rho \times g \times H_{th} \times (Q_{sp} + Q_E z_{st1}/z_{st} + Q_{CB} z_{st2}/z_{st} + Q_h) = \rho \times g \times H_{th} \times Q/\eta_v$ must be supplied by the impeller, where η_v is the volumetric efficiency, Eq. (T3.5.9). Back-to-back pumps (as shown in Figs. 2.9 and 2.10) comprise two groups of stages: the first group has z_{st1} stages which are opposed by the second group with z_{st2} stages. From the 1st group there is a balance flow rate Q_E over the piston to the external balance line. Between both groups a center bush breaks down the pressure difference between the last stage and the first group of stages; the flow rate is Q_{CB} . The equations in Table 3.5 apply to both in-line and back-to-back pumps. For inline pumps set $z_{st1} = z_{st}$ and $z_{st2} = 0$.
3. Disk friction losses P_{RR} are generated on rear and front shrouds of the impellers which rotate in the fluid as hydraulically smooth or rough disks.
4. Similar friction losses P_{er} are created by the components of axial thrust balance devices (balance piston or disk, Chap. 9.2).
5. In multistage pumps leakages also occur in the “interstage seal”, i.e. the annular seal separating two stages, Fig. 3.16. These leakages do not flow through the impeller; consequently, they are not imparted the full blade work. They constitute a throttling loss P_{s3} which should be evaluated with only about 40% of the head, Chap. 3.6.3.
6. Hydraulic losses due to friction and turbulent dissipation in all components between suction and discharge nozzle are covered by the hydraulic efficiency η_h (Chap. 3.7). The dissipated power is $P_{vh} = \rho \times g \times H \times Q \times (1/\eta_h - 1)$.
7. Fluid recirculation at partload creates high losses P_{Rec} due to an exchange of momentum between stalled and non-separated fluid zones. If correctly designed, these are zero near the best efficiency point (and above). When operating against a closed valve or at a low flow rate, they make up the greatest part of the power consumption (Chaps. 4 and 5).

Fig. 3.9 Power balance of a pump

The loss components 2 to 7 are generated within the pump. In the case of the balancing flow this applies only if Q_E is returned into the suction nozzle. All these losses are called “internal losses” because they heat up the fluid in the machine.

The overall power balance of a pump according to Eq. (T3.5.1) is depicted in Fig. 3.9.⁴ Table 3.5 contains the relevant equations and rules of thumb to estimate the secondary losses.

In addition, the following efficiencies are used: the mechanical efficiency according to Eq. (T3.5.6); the internal efficiency according to Eq. (T3.5.5) and the volumetric efficiency according to Eq. (T3.5.9).

The secondary losses are discussed in Chap. 3.6 and the hydraulic losses in Chap. 3.8. The calculation of all types of losses is subject to uncertainties of ± 20 to 30%. If the individual loss under consideration amounts to only a few percent of the coupling power - as with pumps having more than 80% overall efficiency - such an uncertainty is not too serious. However, in the case of small pumps and low specific speeds it is significant. Loss analysis is necessary not only for new developments; it can also be very helpful in determining the cause of a performance deficit observed in a plant or on the test stand. When performing this task, often the focus is not so much on the absolute value of the calculated loss; the goal is rather to determine trends caused by a deviation from design or to evaluate the effect of an intended or observed modification. In such cases, consistent consideration of the losses is more important than the choice between different calculation methods.

⁴The sound power radiated as air-, fluid-, and solid-borne noise is negligibly small compared to all other losses.

3.6 Calculation of Secondary Losses

In the following, disk friction losses and leakage losses are considered as separate phenomena that can be measured in special test stands outside, or even inside, a pump. However, a close interaction is in effect between the main flow at the impeller outlet, the flow in the impeller sidewall gaps, leakage through annular seals and the disk friction losses. The interaction is difficult to capture either experimentally or numerically (this topic is discussed extensively in Chap. 9.1).

3.6.1 Disk Friction Losses

When a circular disk or a cylinder rotates in a fluid, shear stresses corresponding to the local friction coefficient c_f occur on its surface, Chap. 1.5.1. On a disk rotating in an extended *stationary* fluid (without the influence of a casing) the shear stress is $\tau = \frac{1}{2}\rho \times c_f \times u^2$ with $u = \omega \times R$, Eq. (1.31). The friction force on a surface element $dA = 2\pi \times r \times dr$ is then $dF = 2\pi \times \tau \times r \times dr$ and the torque exercised by friction becomes: $dM = r \times dF = \pi \times \rho \times c_f \times \omega^2 \times r^4 \times dr$. The friction power *per side* of the disk $P_{RR} = \omega \times M$ is obtained from the integral $P_{RR} = \omega \times \int dM$ (between inner radius r_1 and outer radius r_2) as $P_{RR} = (\pi/5) \times \rho \times c_f \times \omega^3 \times r_2^5 \times (1 - r_1^5/r_2^5)$. The friction coefficient c_f depends on the Reynolds number and the surface roughness. It has roughly the same magnitude as found on a plate in parallel flow, Eq. (1.33).

In the case of a rotating cylinder (with the length L and radius r_2) no integration is necessary since the radius is constant. In the same way as above the power $P_{RZ} = \pi \times \rho \times c_f \times \omega^3 \times r_2^4 \times L$ is obtained.

In agreement with the similarity laws in Table 3.4, the disk friction power increases with the third power of the speed and the fifth power of the diameter (assuming geometrical similarity $L/r_2 = \text{constant}$ and neglecting the influence of the Reynolds number).

If the body rotates in a casing (as is the case in a pump) the velocity distribution between casing and rotating body depends on the distance between the impeller shroud and the casing wall as well as on the boundary layers which form on the stationary and rotating surfaces. A core flow with approximately $c_u = \frac{1}{2}\omega \times r$ is obtained (in other words, $u = \omega \times r$ can no longer be assumed). In the case of turbulent flow, the power absorbed by a disk in a casing therefore amounts to roughly half of the power of a free disk rotating in a stationary fluid.⁵

The disk friction losses of impeller shrouds or of cylindrical bodies are calculated according to Eqs. (T3.6.2 and 3.6.4). The friction coefficients k_{RR} and k_{RZ}

⁵This experimental finding follows from data in [35]; it cannot be deduced by inserting half the angular velocity in the equation given earlier.

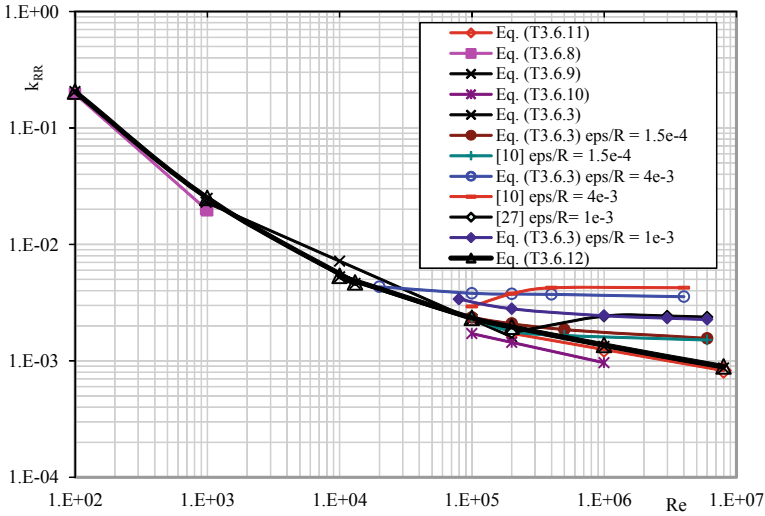


Fig. 3.10 Disk friction coefficients for smooth and rough surfaces calculated for $s_{ax}/R = 0.08$ (without leakage through the impeller sidewall gap); ε = roughness

required for this purpose are determined from correlations derived from experiments. All relevant equations are compiled in Table 3.6. They were derived from measurements on hydraulically smooth disks.⁶

The equations for turbulent flows were expanded by correction factors f_R and f_L by means of which the effect of the roughness and the leakage flow through the impeller sidewall gaps can be estimated. The increase of the friction on rough disks compared with hydraulically smooth disks is determined as follows: the friction coefficient c_f is calculated for the rough surface and for the hydraulically smooth surface $c_{f,0}(\varepsilon = 0)$ from Eq. (1.33) with $Re = u \times r_2/\nu$ and ε/r_2 . This produces the correction factor $f_R = c_f/c_{f,0}$ according to Eq. (T3.6.6).

Equation (T3.6.3) produces similar friction coefficients as Eqs. (T3.6.8 to T3.6.11), Fig. 3.10. However, Eq. (T3.6.3) has the advantage that the entire range from laminar to turbulent can be covered with a single equation without any unsteadiness at the transitions.

Numerous tests confirm that the axial casing clearance s_{ax} has only a very small effect on the disk friction; Eq. (T3.6.11) overestimates this effect. With s_{ax} tending to zero it yields too small and with s_{ax} tending to infinity too large friction coefficients. In contrast, Eq. (T3.6.3) produces sensible values even with extreme axial clearances s_{ax} . With s_{ax} tending towards infinity Eq. (T3.6.3) supplies factors

⁶Equations (T3.6.3) and (T3.6.5) are based on [33] and [25], Eqs. (T3.6.8) to (T3.6.11) on [6], Eq. (T3.6.12) on [14].

roughly corresponding to a free disk whose friction factors are given by $k_{RR} = 0.0365/\text{Re}^{0.2}$, [35]. Refer to Table 3.6 for further definition.

In [14] a method was developed to calculate disk friction losses by taking into account the fluid rotation in the impeller sidewall gap, Eqs. (T3.6.12 and T3.6.13). This equation covers the effects of the impeller sidewall gap geometry as well as the roughness of the casing and the impeller shrouds. It is useful particularly when the casing and the impeller disk have different roughness values. For a detailed discussion refer to Chap. 9.1.

The various correlations from Table 3.6 are compared in Fig. 3.10. It can be seen that in particular Eqs. (T3.6.3) and (T3.6.12) are equivalent and suitably reflect Eqs. (T3.6.8 to T3.6.11). The effect of the roughness is appropriately covered by Eq. (T3.6.6).

Literature overviews concerning disk friction losses can be found in [14, 17, 34]; the basic theory is treated in [35]. When comparing with other sources, the different definitions of the torque coefficients must be considered (frequently the coefficients are defined in a way that they comprise the torque or the friction power for both sides of the disk).

The disk friction losses in a pump depend on the following parameters:

1. **Reynolds number:** As with the flow over a plate or through a pipe, the friction coefficient drops with increasing Reynolds number. When pumping water, the flow in the impeller sidewall gap is generally turbulent. In the case of oils or other fluids with high viscosity the flow becomes laminar and the disk friction increases greatly (Chap. 13.1).
2. **Roughness of the rotating disk:** The roughness of the stationary or the rotating surface increases the friction power, provided the roughness peaks protrude from the boundary layer thickness. In this context, grooves in circumferential direction such as generated by machining the shrouds on a lathe are less harmful than an unstructured roughness as typically found on a cast surface, [11]. According to the measurements in [11] the disk friction of a machined disk with $\varepsilon_{\max} = 120 \mu\text{m}$ was practically identical in magnitude to that of a polished disk with $\varepsilon \approx 0$. Machined disks can therefore be calculated in general with $\varepsilon \approx 0$. For cast or sandblasted disks $\varepsilon = \varepsilon_{\max}/c_{\text{eq}}$ can be assumed. The equivalence factor is $c_{\text{eq}} = 2.6$ and $\varepsilon_{\max}$ is the maximum roughness height, Chap. 3.10. Disk friction losses are minimized if hydraulically smooth surface conditions are achieved. To this end, the permissible roughness that must not be exceeded can be evaluated according to Table D1.1 where $w = c_u \approx \frac{1}{2} \times \omega \times r$ must be employed. The roughness problem is also discussed in Chap. 3.10.
3. **Roughness of the casing wall:** When rotor and casing have the same roughness the rotation of the fluid in the impeller sidewall gap is independent of the roughness, Eq. (T3.6.13). However, if $\varepsilon_{\text{impeller}} \neq \varepsilon_{\text{casing}}$, the roughness has an effect on the rotation and, consequently, the disk friction. With $\varepsilon_{\text{impeller}} > \varepsilon_{\text{casing}}$ the rotation increases, while it is reduced if $\varepsilon_{\text{impeller}} < \varepsilon_{\text{casing}}$. At the same time the disk friction changes as per Eqs. (T3.6.12) and (T3.6.13). For a more detailed calculation refer to Chap. 9.1.

4. **Axial sidewall gap s_{ax} :** With a very small distance between impeller shroud and casing $s_{ax} < \delta$ (δ = boundary layer thickness) the velocity distribution in the axial sidewall gap is linear (Couette flow) and the mean velocity is approximately $\frac{1}{2}\omega \times r$. When increasing the axial sidewall gap, the friction loss drops to a minimum (which has no practical significance since it cannot be executed in reality) before it rises again. As already mentioned, the effect of the axial sidewall clearance is minimal in turbulent flow, while it is significant in laminar flow. The larger the wetted casing surfaces of the sidewall gap, the slower is the fluid rotation.
5. **Shape of the casing and size of the impeller sidewall gap:** have an effect on the rotation of the fluid and consequently the disk friction. *Complicated shapes of the casing, large casing surfaces, ribs and other structures reducing fluid rotation increase the disk friction losses in a way similar to rough surfaces*, Chap. 9.1.
6. **Influencing the boundary layer:** The turbulence structure of the boundary layer can be influenced by fine grooves in the flow direction. The friction coefficient could thus be lowered by up to 10% in the tests reported in [2]. However, producing grooves of the necessary fineness and with sharp tips is expensive and scarcely feasible in practice.
7. **Leakage flow:** through the impeller sidewall gap (Chap. 9.1): A radially inward-directed leakage introduces an angular momentum of $\rho \times Q_{sp} \times c_{2u} \times r_2$ in the impeller sidewall gap which reduces the disk friction, if $c_{2u}/u_2 > 0.5$. Conversely, a radially outward-directed leakage flow decelerates the fluid rotation in the impeller sidewall gap (if it has no or little pre-rotation) so that the disk friction increases. The effect of leakage can be estimated by the correction factor according to Eq. (T3.6.7)⁷ which was derived from the calculations in Chap. 9.1. It roughly agrees with the measurements in [38]. In practice, the effects of radially inward leakage (reducing the disk friction) and roughness (increasing disk friction) compensate to some extent so that $f_L = f_R = 1.0$ can often be used for simplification.
8. **Partload recirculation:** slows down the fluid rotation if it enters the impeller sidewall gap with low circumferential velocity, Fig. 5.30. The disk friction increases as demonstrated by Fig. 9.10.
9. **Exchange of momentum:** An exchange of momentum and, consequently, an interaction occur between main flow and impeller sidewall gap flow, Chap. 9.1.

The last three parameters can severely influence the disk friction loss and render its calculation quite uncertain, especially with large leakage flows and at partload. Even at the best efficiency point, the tolerance of the calculated disk friction loss is estimated to be about $\pm 25\%$. The physical mechanisms of sidewall gap flow are

⁷The impeller sidewall gap integration according to Table 9.1 is recommended for more thorough investigations.

treated comprehensively in Chap. 9.1, since they are very important for calculating the axial forces acting on the impeller

When determining the disk friction losses in a pump, the contributions of all surfaces rotating in the fluid must be calculated using the equations from Table 3.6. These include impeller front and rear shrouds, their cylindrical end faces, annular seals on the impeller and the balance device if any. In the case of radial impellers, friction on the shrouds by far exceeds the contribution of all other components. The size of the impeller inlet diameter has little impact since $(r_1/r_2)^5 \ll 1$, Eq. (T3.6.2).

For typical radial impellers the disk friction loss can be calculated from the useful power P_u and the quantities n_q , Re and ψ_{opt} . This leads to Eq. (T3.5.12) for turbulent flows, or (more generally) to Eq. (T3.5.13) for laminar and turbulent flows. In these equations the contributions of front and rear shrouds, including cylindrical surfaces and annular seals, have been taken into account for typical impeller geometries. The equations show:

- The share of disk friction losses in the power consumption of a pump drops exponentially with increasing specific speed and increasing head coefficient. With low specific speeds disk friction is the main source of loss: at $n_q = 10$, $\psi_{opt} = 1$ and $Re = 10^7$ the disk friction is approximately 30% of the useful power of the impeller.
- With low specific speeds the head coefficient must be selected as high as possible in order to improve the efficiency (a high head coefficient leads to a reduced impeller diameter and consequently less disk friction, since $P_{RR} \sim d_2^5$).

3.6.2 Leakage Losses Through Annular Seals

Close running clearances (“annular seals”) between impeller and casing limit the leakage from the impeller outlet to the inlet, see Figs. 3.15, 3.16 and Table 3.7. Any leakage reduces the pump efficiency. Since the *entire* mechanical energy transferred by the impeller to the leakage flow (i.e. the increase of the static head and the kinetic energy) is throttled in the seal and converted into heat, one percent of leakage flow also means an efficiency loss of one percent. Leakage flows likewise occur on axial thrust balance devices in which the entire differential head of the pump is throttled to suction pressure (here, too, the overall efficiency is reduced by one percent for every percent of leakage).

The annular seal consists of a case ring and a rotating inner cylinder. The radial clearance s between the rings is small compared to the radius of the rotating parts ($s \ll r_{sp}$). Due to the pressure difference across the seal, an axial flow velocity c_{ax} is generated. With the rotor at rest, this axial flow can be treated according to the laws of channel flow if the hydraulic diameter $d_h = 2 \times s$ is used.

Through the rotation of the inner cylinder a circumferential flow is superimposed on the axial flow. To describe these flow conditions two Reynolds numbers are required: Re for the axial and Re_u for the circumferential flow. Both definitions are

given in Table 3.7, Eq. (T3.7.9). The transition from laminar to turbulent occurs at a Reynolds number $Re^* = 2000$, which is formed with the mean vectorial velocity from c_{ax} and $u_{sp}/2$, as given by Eq. (T3.7.10).

The circumferential velocity c_u in the annular seal depends on the pre-rotation at the annular seal inlet, $k_{in} = c_u/(\omega \times r)$; c_u develops in the annular seal with increasing flow path length z according to Eq. (T3.11a), [5]. It reaches an asymptotic value given by $k_{out} = c_u/(\omega \times r) = 0.5$ for long annular seals.

$$k \equiv \frac{c_u}{u_{sp}} = 0.5 + (k_{in} - 0.5) \exp \left\{ -\frac{\lambda z}{4s} \left(1 + \frac{0.75}{1 + 4 \left[\frac{Re}{Re_u} \right]^2} \right) \right\} \quad (3.11a)$$

In laminar flow the velocity distribution in circumferential direction is linear with $c_u = 0$ at the stator and $c_l = \omega \times r_{sp} = u_{sp}$ at the rotor. Since c_u decreases from the rotor to the stator with growing radius, the centrifugal forces, too, decrease from rotor to stator. Consequently, the velocity distribution is inherently unstable. Above a specific circumferential velocity “Taylor vortices” are generated in the circumferential direction. These manifest as pairs of vortices with opposite swirl structures. The stability criterion of the vortex development is determined by the Taylor number Ta : $Ta = u_{sp} \times s/v \times (s/r_{sp})^{0.5} = Re_u/2 \times (s/r_{sp})^{0.5}$, [35]. Taylor vortices occur at $Ta > 41.3$. However, the flow remains laminar up to approximately $Ta = 400$ with an adequately small axial Reynolds number Re . In laminar flow the resistance coefficient increases considerably due to the Taylor vortices (by a factor of 2 to 3) refer also to [35, 34].

In stable laminar flow ($Ta < 41.3$) the rotation has no noticeable effect on the friction coefficient λ which is calculated as $\lambda = 96/Re$ for a concentric gap which is confirmed by the tests in Fig. 3.11 with $Re_u = 0$. Conversely, in turbulent flow λ depends on the ratio of the circumferential to the axial Reynolds numbers Re_u/Re . As follows from Eq. (T3.7.10) the flow is always turbulent for $Re_u > 4000$, even if there is no axial flow ($Re = 0$). Equation (T3.7.14) can be used to estimate the friction coefficient for flows with $Re < 2000$.

In practical applications, the transition between laminar and turbulent cannot be related clearly to either the criterion $Re^* = 2000$ or the Taylor number. The transition between laminar and turbulent flow depends on the roughness and surface structure (e.g. serrations), the axial Reynolds number Re and the circumferential Reynolds number Re_u . As yet, there are no reliable correlations to describe this transition. Test data in pipes and on flat plates demonstrate that the friction factors in laminar flow usually cannot be lower than in turbulent flow at any given axial Reynolds number. In order to avoid a discontinuity in the curve $\lambda = f(Re)$ it is therefore suggested to calculate all friction factors by taking the maximum values of the laminar and turbulent flows using the logic: $\lambda = \text{Max}(\lambda_{lam}; \lambda_{tur})$.

An exception to the above statement are surfaces with a *uniform* roughness where a local minimum can be observed in the curve $\lambda = f(Re)$. Surfaces with technical roughness do not show this behavior. This minimum should therefore be

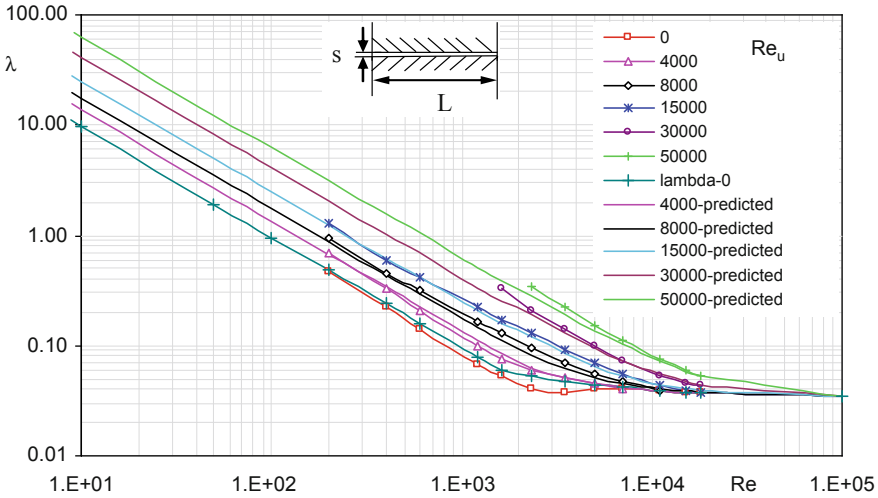


Fig. 3.11 Friction coefficients of smooth annular seals, test data from [36, 41] $\varepsilon/s = 0.01$; $L/s = 413$; $s = 0.315$ mm. The curves labeled “predicted” were calculated from Eqs. (T3.7.11) to (T3.7.14)

ignored. Structured surfaces (grooves, serrations and honeycomb) have dimensions greater than the turbulence length scales. Therefore, they do not show the same behavior as a uniform roughness.

To calculate the annular seal leakages, experimentally determined friction coefficients are used for turbulent flow. Figure 3.11 shows measurements on a straight, plain annular seal, Fig. 3.12 an example for a straight serrated seal with large grooves [41] (see Fig. 3.15 for various seal geometries). Figure 3.11 was supplemented by tests from [36] for $Re_u = 30,000$ and $50,000$. As with all turbulent flows, the resistance of an annular seal depends on the roughness of the wall. In the case of small clearances this dependency is particularly high since the *relative* roughness ε/d_h is high due to the small hydraulic diameter.

The required high-pressure differentials and problems of measuring accuracy aggravate the experimental determination of the friction coefficients of annular seals in a wide range of Reynolds numbers. It is therefore frequently necessary to extrapolate the available test data.

The resistance law for the flow in channels, Eq. (1.36), is suitable for extrapolation since the entire range of turbulent flows from hydraulically smooth to hydraulically rough is described by it. The influence of the roughness in this context is expressed by ε/s . This better describes the fine structure of the flow in the annular seal than $\varepsilon/(2s)$ and reflects more accurately the test results at least in this case.⁸

⁸It is open to debate whether ε/s or $\varepsilon/2s$ is more relevant for small clearances. Undoubtedly the hydraulic diameter is $d_h = 2s$. Considering the structure of the flow one may rather tend intuitively to select ε/s , which could be viewed as a local obstruction to the flow.

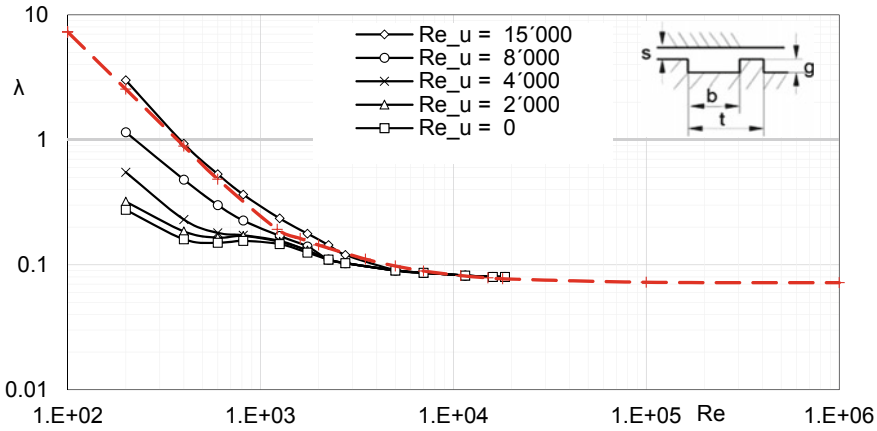


Fig. 3.12 Friction coefficients of serrated annular seal, test data from [41]; $s = 0.31$ mm; $b/t = 0.7$; $t/s = 16.1$; $g/s = 3.2$

The resistance coefficients calculated in this way apply to a stationary rotor ($u = 0$) and are therefore termed λ_o , Eq. (T3.7.12). The effect of the rotation in turbulent flow is covered by an experimentally determined factor λ/λ_o , Eq. (T3.7.13), which harbors considerable uncertainties. As can be derived from Eq. (3.11b), $Re_u/Re < 2$ mostly applies to the annular seals on the impeller. Consequently, the correction for the influence of the rotation according to Eq. (T3.7.13) is below 25% in most applications so that this uncertainty is often not really serious.

$$\frac{Re_u}{Re} = d_{sp}^* \sqrt{\frac{\zeta_{EA} + \lambda \frac{L}{2s}}{R_G \psi - k^2(1 - d_{sp}^{*2})}} \quad (3.11b)$$

The test data from Fig. 3.11 were corrected for the rotation influence according to Eq. (T3.7.13). In this way λ_o was obtained which can be compared with Eq. (T3.7.12). The results of this calculation demonstrate that the Eqs. (T3.7.12 to T3.7.13) are suitable for extrapolating the test data. This even applies in the laminar range, Fig. 3.13.

The friction coefficients of smooth annular seals and of seals with coarse or fine serrations and isotropic patterns can be estimated from Fig. 3.14. In addition to the friction loss, inlet and outlet losses must be considered in an annular seal. These are covered by ζ_E and ζ_A or by the sum of these coefficients $\zeta_{EA} = \zeta_E + \zeta_A$. Without rotation, $\zeta_E = 0.5$ and $\zeta_A = 1$ would apply to genuinely sharp-edged inlets. As a result of the rotation both values drop so that $\zeta_{EA} = 1$ to 1.2 can be used. The existing tests exhibit a major scatter since the inlet and outlet losses depend on Re and Re_u and even a tiny radius or chamfer at the seal inlet severely reduces the coefficient ζ_E . For example, if the clearance is $s = 0.3$ mm, a radius of 0.2 mm produces $r/d_h = 0.33$ and reduces the inlet loss coefficient from $\zeta_E = 0.5$ to $\zeta_E = 0.03$ (see Table 1.4).

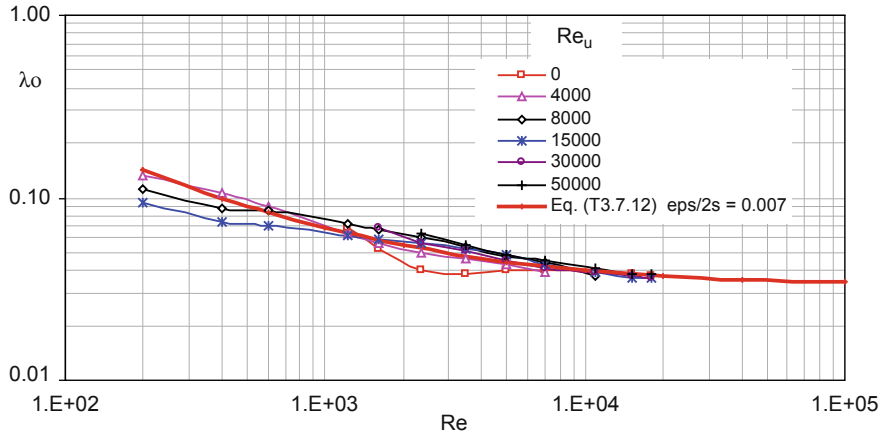


Fig. 3.13 Friction coefficients of smooth annular seals with $u_{sp} = 0$ according to Fig. 3.11 (ϵ s stands for the sand roughness ϵ)

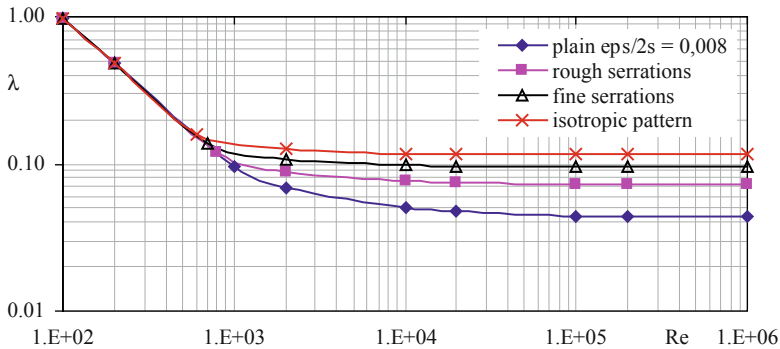


Fig. 3.14 Friction coefficients of annular seals, calculated from Table 3.7 for $Re_u/Re = 2$

In the case of plain seals (without serrations) the greatest possible roughness should be aimed at. However, there are practical limits to this, since the close manufacturing tolerances required for the seal diameters cannot be achieved with very rough surfaces. To reduce the leakage flow, grooves and isotropic patterns have been developed which will increase the resistance coefficient. Large grooves were examined in [41]; fine and coarse grooves in [9]; honeycomb patterns in [4]. The effect of such types of structures is based on increased energy dissipation through turbulent exchange of momentum between the leakage flow and the fluid in the grooves or recesses. A sort of “form resistance” is obtained which causes the transition from hydraulically smooth to fully rough flow being shifted towards lower Reynolds numbers than with a seal having no grooves. For $Re > 10,000$ it is therefore possible with such geometries, as a first approximation, to assume

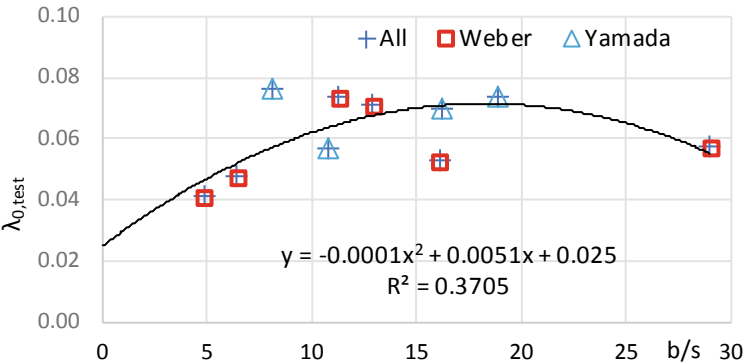


Fig. 3.14.1 Friction factors of serrated seals from test data for $Re_u = 0$

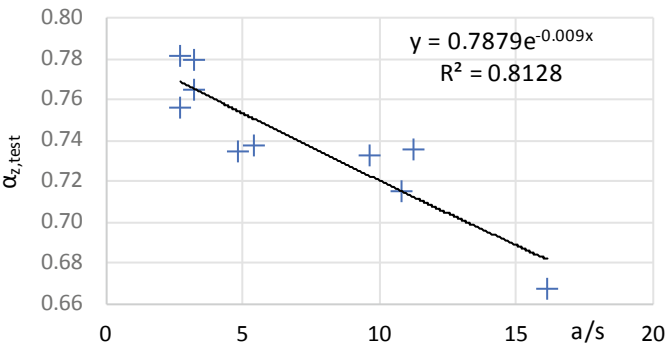


Fig. 3.14.2 Jet contraction factor

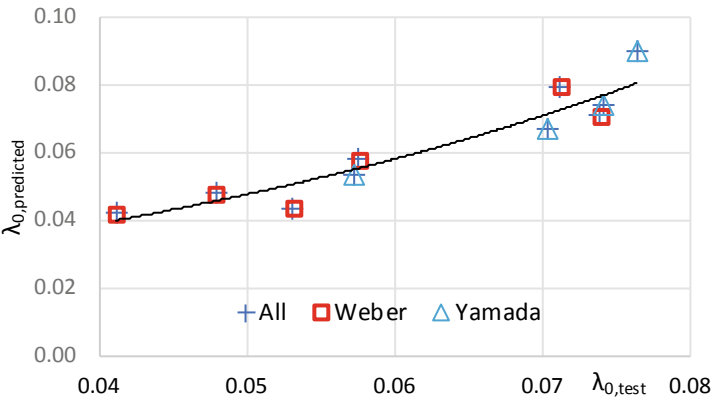


Fig. 3.14.3 Serrated-seal friction factors calculated from Eq. (T3.7.46) compared to test data derived from [41, 47]

Re-independent resistance coefficients for the calculation. As explained above, the influence of rotation on λ/λ_0 is relatively small. In many cases it is possible to describe the attainable level of resistance coefficients for $Re > 10,000$ as follows:

Honeycomb patterns: $\lambda = 0.1-0.18$; Fine grooves: $\lambda = 0.07-0.09$

Hole patterns: $\lambda = 0.07-0.1$; Coarse grooves: $\lambda = 0.04-0.1$

These are only rough indications since the effective resistance coefficients sensitively depend on the surface structure.

Friction factors of seals with deep serrations ($g/s > 2$):

Grooves with width “b” and smooth portions (“lands”) with $a = t - b$ make up the seal; the nominal clearance is “s” (or s_0), see sketch in Fig. 3.12 or Table 3.7(4). A method has been developed for predicting the friction coefficients as function of the groove dimensions; it is based on test data in [41, 47]. The physical model involves the following assumptions and steps: (1) When the fluid enters the gap width s, there is a streamline (or “jet”) contraction (similar to the flow through a sharp orifice). The velocity in the resulting vena contracta is $c_{VC} = c_{ax}/\alpha_z$ where c_{ax} is the nominal velocity in the seal $c_{ax} = Q_{sp}/(\pi d_{sp}s)$. (2) When entering a groove, the fluid expands like a plane *wall jet* (formulae given in Table 1.3). (3) The loss in a groove is calculated from Eq. (1.12); i.e. as a shock loss created by decelerating the flow from c_{VC} to c_b where c_b is assumed to half the center-line velocity of the expanding jet at the length $x = b$. (4) An empirical factor β has been introduced to account for effects not captured by the jet model. (5) The friction loss on the lands is calculated as a plain seal with λ from Eq. (T3.7.12). (6) The friction loss is increased due to the additional surface given by the depth of the grooves. Admittedly, the effect is small and the assumption open to debate, but it is an attempt to enter the groove depth into the model. Overall, the results are very little influenced. (7) The resulting formula for calculating the equivalent friction factor $\lambda_{0,eq}$ of the serrated seal is given by Eq. (T3.7.46). The formula ensures a smooth transition between serrated and plain seals (where $a = t$ and $b = g = 0$). (8) Eq. (T3.7.46) represents the friction factor for the seal *without rotation* at high Reynolds numbers where the curve levels off as seen in Fig. 3.12. The effect of rotation is subsequently introduced by Eq. (T3.7.13); the dashed curve in Fig. 3.12 shows the curve calculated in this way from Table 3.7(4). (9) The published friction factors have been analyzed by these steps: (A) Test result λ was divided by the factor λ/λ_0 from Eq. (T3.7.13); these test data are plotted in Fig. 3.14.1 as function of the groove width b/s . (B) The friction factor λ_{smooth} for the un-serrated seal was taken from the applicable tests (its influence is moderate). (C) The contraction factor was calculated by solving Eq. (T3.7.46) for α_z . The result is plotted in Fig. 3.14.2 as function of the land a/s . The effect of the contraction increases with increasing a/s (surprisingly?). The resulting correlation, Eq. (T3.7.45) was

introduced in Eq. (T3.46) to calculate $\lambda_{0,\text{eq,test}}$. (D) A slight adaptation of the empirical factor β was useful to improve the data fit.

The resulting friction factors are compared in Fig. 3.14.3 to the test data. In view of the many uncertainties (which include the reading of the data from the published graphs), the standard deviation of 9.5% seems acceptable.

Results, findings and recommendations:

- The groove width ratio b/s is relevant for the expansion of the jet. The optimum seems in the range of $b/s = 8$ to 20 , Fig. 3.7.14-2. Considering that the “core length” of the wall jet is $x_1/s = 6.4$ (Table 1.2), the groove width b should be greater than x_1 in order to enable the jet to sufficiently expand. Above $b/s > 20$, there are too few grooves in a given seal length and the dissipation of kinetic energy per unit length of the seal diminishes.
- The width of the grooves should be as large as possible $b/t = 0.75$ to 0.9 in order to maximize dissipation by vortices. The minimum land “a” is influenced by manufacturing and mechanical reasons as well as wear of the seal. The minimum land may be $a_{\min} = 0.4$ mm.
- Jet expansion in the groove demands a sufficiently large expansion angle ε . With $\tan \varepsilon = g/b$ and assuming a minimum $\varepsilon = 10^\circ$ (see Chap. 1.7) one gets $(g/b)_{\min} = 0.18$. The optimum from test data in the literature seems near $g/b = 0.25$ and $g/s = 3$ to 6 . If the pump is designed for operation at double the new clearances, selecting $g/s = 5$ to 6 will still provide high friction factors at worn conditions. Too deep grooves do not increase the friction factors because the friction on the outer wall of the groove is then reduced.
- The grooves can be machined with an angle other than 90° to the rotor axis in order to use a slightly trapezoidal tool for cutting the grooves.

The correlations cover a fair range of tested geometries as shown in Table 3.7(4). However, there may be geometries with finer grooves which generate higher flow resistances.

Annular seals are designed in a variety of shapes according to Fig. 3.15:

- Straight plain annular seals require the least manufacturing effort.
- Stepped annular seals with 1 to 2 stages or intermediate chambers offer increased resistance since a head corresponding roughly to one stagnation pressure is dissipated in one chamber: $\zeta_k = 1$ to 1.3 . Annular seal length and diameter can possibly be better optimized with stepped seals.
- Z-shaped annular seals provide a slightly higher resistance coefficient ζ_k than stepped seals.
- The multiple seal (labyrinth seal) is the most complex design solution. With this type of seal it must be noted that clearance “a” is designed large compared with clearance s to avoid rotor instabilities (Chap. 10).
- Radial and inclined or diagonal seals are designed according to Chap. 3.6.5.

The required seal clearance depends on considerations of mechanical design. The main objective is to prevent contact between rotor and stator. Design criteria are:

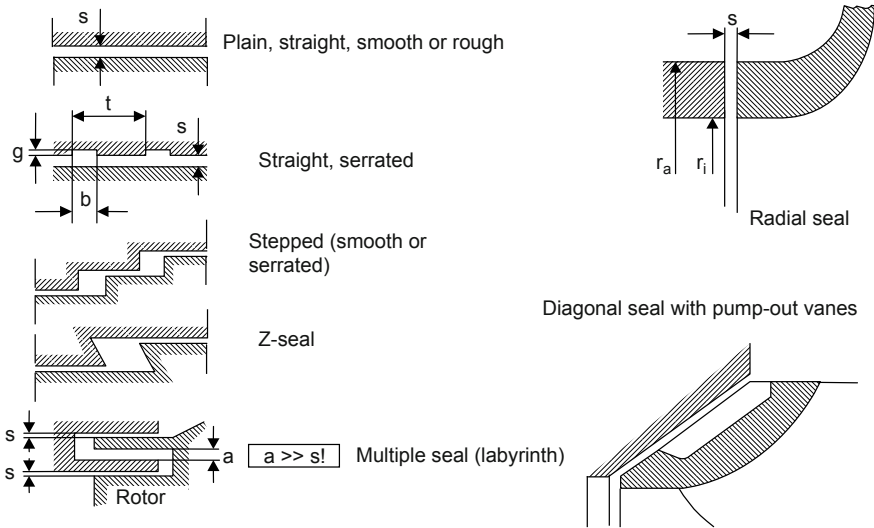


Fig. 3.15 Types of seals

the shaft bending under the rotor weight and radial thrust, possible thermal deformations of rotor and casing and the tendency of the materials used to seize.

The diametrical clearance $\Delta d = 2 \times s$ is in the order of:

$$\frac{\Delta d}{d_{sp}} = 0.004 \left\{ \frac{d_{Rref}}{d_{sp}} \right\}^{0.53} \quad \text{with } d_{Rref} = 100 \text{ mm} \quad (3.12)$$

Equation (3.12) roughly corresponds to the minimum clearance values from [N.6] *plus* tolerances according to half the tolerance fields H7/h7. Executed clearances as a rule are in the range of $\pm 30\%$ of the clearances calculated from Eq. (3.12). Some pump manufacturers use internal standards, for example defined as 75% [46] to 80% of clearances from [N.6]. In [N.6] an allowance of 0.125 mm is specified for materials with a high tendency to seize and in pumps for media above 260 °C. This allowance is added to the clearances according to Eq. (3.12).

When starting pumps for hot media without preheating, the differential heating of rotor and stator must be considered. The impeller wear rings heat up faster than the casing because of the thin material thicknesses involved. The clearances are thus reduced due to thermal expansion.

All types of seals can be designed either with smooth surfaces or with many different types of grooves or isotropic patterns (honeycomb or hole patterns) in order to increase the flow resistance. A “smooth” surface in terms of manufacturing is not the same as “hydraulically smooth”. The roughness must always be considered when calculating the leakage loss, provided the flow is turbulent. For example, machining a seal with a clearance of 0.4 mm to quality N7 (i.e. a sand

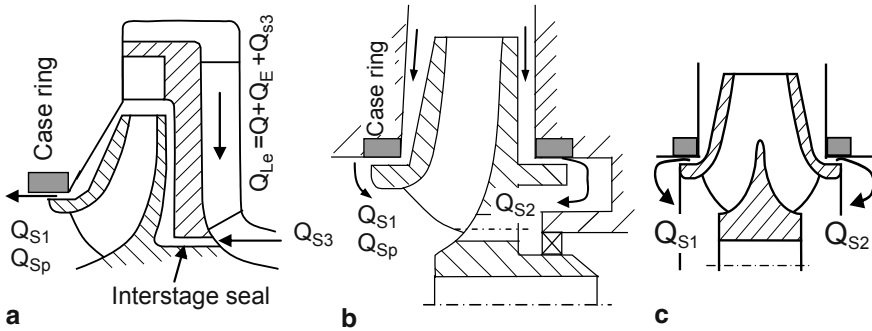


Fig. 3.16 Leakage flows: **a** multistage pump; **b** impeller with balance holes; **c** double-entry impeller

roughness of 4 μm) gives a relative roughness of $\varepsilon/s = 0.01$, which is hydraulically very rough, see the graph in Table 1.6).

The seal diameter is chosen as small as possible, the seal length is typically $L_{sp} = (0.1 \text{ to } 0.14) \times d_{sp}$.

Calculation procedure for leakage losses (Table. 3.7): To start with, the pressure difference acting over the seal must be established. The static pressure H_p prevailing at the impeller outlet can be calculated from Eq. (T.3.7.1). In many cases H_p can be estimated accurately enough with the help of the degree of reaction from $H_p = R_G \times H$. For moderate specific speeds $R_G = 0.75$ is a good assumption. With radially inward directed leakages (Q_{s1} and Q_{s2} in Fig. 3.16) the pressure between the impeller outlet and the seal drops due to the rotation of the fluid in the impeller sidewall gap (Chap. 9.1). This is described by the rotation factor $k = \beta/\omega$ (β is the angular velocity of the fluid).

As per Eq. (1.27) the pressure reduction in the impeller sidewall gap can be calculated from $\Delta p = 1/2 \rho \times \beta^2 \times (r_2^2 - r_{sp}^2) = 1/2 \rho \times k^2 \times u_2^2 \times (1 - d_{sp}^{*2})$; Eq. (T3.7.3) is obtained for determining the pressure difference over the seal at the impeller inlet.

The greater the fluid rotation (i.e. the greater k) the greater is the pressure drop in the sidewall gap and the smaller is the pressure difference over the seal and the resulting leakage flow. Since k increases with growing leakage (Chap. 9.1), a *radially inward* leakage depends on itself and partly limits itself.

However, if the leakage Q_{s3} flows *radially outward* as in the case of the interstage seals of multistage pumps, the pressure differential due to the rotation in the impeller sidewall gap is added according to Eq. (T3.7.4) to the pressure recovery in the diffuser $\Delta H_{Le} = H - H_p = (1 - R_G) \times H$. Again the leakage tends to limit itself, since a large flow through the sidewall gap has the effect of reducing the fluid rotation or the pumping action of the rear shroud and thus the pressure difference across the seal.

The rotation factor k can be estimated in different ways: (1) according to Eq. (T3.7.2) as a function of the Reynolds number and the seal geometry; (2) by reading from Figs. 9.8 or 9.9; (3) according to Eq. (T9.1.4); (4) by the calculation according to Table 9.1.

The leakage through the axial thrust balance device of a multistage pump (Chap. 9.2.3) is generally returned to the suction nozzle. In that case the head generated by $(z_{st} - 1)$ stages, plus the static head calculated from Eq. (T3.7.3), is throttled by the balance device. The balancing flow must therefore be calculated with the head difference defined by Eq. (T3.7.5).

The axial velocity c_{ax} in the annular seal has to be calculated iteratively according to Eq. (T3.7.6) since the friction coefficient depends on the Reynolds number (i.e. on c_{ax}). Stepped seals or multiple seals are calculated as resistances connected in series according to Chap. 1.9. To this end, the resistances of the individual steps or seals are added as per Eq. (T3.7.6).

Pumping grooves: Instead of *annular* grooves it is also possible to machine *threads* on the rotating internal cylinder which work against the pressure to be sealed (“pumping grooves”). The leakage flow decreases with falling c_{ax}/u_{sp} . Pumping grooves always give less leakage than plain seals, however they make sense only when $c_{ax}/u_{sp} < 0.7$. Annular grooves similar to Fig. 3.12 are more effective above this limit. The ratio of seal length to clearance should be above $L_{sp}/s = 50$. Pumping grooves are geometrically similar if the following parameters are identical: pitch angle α , groove width “ a ” to pitch $(a + b)$: $a/(a + b)$, groove depth h to clearance s : h/s and pitch to clearance: $(a + b)/s$. For given values of these parameters the friction coefficient is independent of the clearance.

Since the selection of favorable parameters depends on the ratio of the axial velocity in the seal to the circumferential speed of the rotor, an iterative calculation is necessary. Table 3.7(3) supplies all the required information and equations derived from tests in [18]. Equations (T3.7.39) and (T3.7.40) apply to the geometric parameters specified in Eqs. (T3.7.35) to (T3.7.38) from which a deviation of $\pm 10\%$ is possible without Eqs. (T3.7.39) and (T3.7.40) losing their validity.

Assessment and optimization of the annular seal leakage:

- The seal clearance is the most important parameter since the leakage flow increases approximately with the 1.5th power of the clearance. Seemingly minor deviations from the assumed dimensions have a noticeable effect on the efficiency at low specific speeds. In the case of a pump with $n_q = 15$, for example, a clearance change by 0.03 mm from 0.25 to 0.28 mm means a reduction of efficiency of one percentage point. Usually the clearance calculated from the *nominal* diameters corresponds to the *minimum* clearance. The (statistically) expected *average* clearance is larger than this minimum. Therefore, leakage calculations should be based on an average clearance which is calculated by adding half the tolerance field to the minimum clearance.
- The clearance has practically no effect on the friction coefficient, provided the relative roughness ϵ/s is used in the calculation. For $Re > 10,000$, the roughness

must always be considered for leakage flows, since the actual ratio of ε/s is usually quite large.

- When optimizing the seal geometry in terms of shape (Fig. 3.15), seal length and surface structure, the rotor-dynamic characteristics must be considered in addition to the efficiency (Chap. 10).
- With turbulent flow an eccentricity of the inner cylinder can be neglected. In contrast, with laminar flow the resistance of a seal drops with increasing eccentricity, Eq. (T3.7.11).
- The efficiency impairment caused by the seal leakage decreases with growing specific speed since the leakage flow is referred to an increasingly larger useful flow rate (Q_{sp}/Q_{opt}) and because the pressure difference over the seal drops. Equation (T3.5.10) provides a relationship for estimating the leakage. Above $n_q > 60$ the seal loss is roughly 1% or below (and 2% if balance holes are provided).
- Close interaction exists between leakage flow, pressure distribution in the impeller sidewall gap, disk friction and main flow (Chap. 9.1). With unusually high leakage, i.e. with clearances appreciably above those calculated from Eq. (3.12), the calculations should be done according to Chap. 9.1 (Table 9.1).

Because of the many influencing factors and the limited amount of relevant test data, leakage calculations are subject to uncertainties of approximately $\pm 30\%$ due to the following factors:

- Turbulence and roughness structure (and their interaction)
- Pressure difference over the seal (impeller sidewall gap flow)
- Effective clearance which can vary over the circumference and the length because of manufacturing tolerances.
- The actual clearance can be influenced by: (1) the stator being at a slightly different temperature from the rotor; (2) differences in material implying dissimilar coefficients of thermal expansion between rotor and stator; (3) centrifugal force: the widening of the rotating part reduces the clearance
- Stator deformation under load
- Inlet and outlet loss coefficients ζ_E and ζ_A depend on the pre-rotation at the seal entry. The edges are nominally sharp but even tiny rounding or chamfering will reduce the loss coefficients.

3.6.3 Power Loss Caused by the Inter-stage Seal

The leakage flow through seals placed between the stages of a multistage pump is calculated according to Table 3.7(1). However, the relative loss of efficiency is smaller than with leakages around the impeller, because the head increase in the diffuser only amounts to about a quarter of the work of the impeller. Consequently, the loss obtained by throttling the leakage flow Q_{s3} over the head differential ΔH_{s3}

is calculated as power loss in this seal: $P_{s3} = \rho \times g \times Q_{s3} \times \Delta H_{s3}$. This loss is *not* incorporated into the volumetric efficiency. It scarcely depends on the operating point of the pump. The ratio P_{s3}/P_u decreases with growing specific speed. Equation (T3.5.11a) serves to estimate this loss. Often this approximation is sufficient. For more accuracy ΔH_{s3} is calculated from Eq. (T3.7.4), the leakage according to Eqs. (T3.7.7–13) and the power loss P_{s3} from Eq. (T3.5.11a).

3.6.4 Leakage Loss of Radial or Diagonal Seals

Seals with radial or diagonal flow (Fig. 3.15) are used for centrifugal pumps operating with fluids which are laden with abrasive solids. Such designs avoid flow deflections upstream or downstream of the seal which would result in local metal loss due to abrasion.

The flow in a radial or diagonal seal is affected by centrifugal forces generated in the seal through the circumferential component of the flow velocity. This effect is covered by the rotation factor k as in Chap. 3.6.2. The pressure difference Δp present across the seal is suitably corrected with an additional element $\frac{1}{2}\rho \times (k \times \omega \times r_a)^2$. If the flow is from the inside to the outside, this term is positive and the leakage increases accordingly. Conversely, if the fluid flows radially inwards, centrifugal forces oppose the flow. The pumping action in the seal thus reduces the leakage.

The leakage loss of radial-flow seals can be calculated according to [40]:

1. The pressure difference across the seal is determined from Table 3.7(1) on the analogy of Eqs. (T3.7.1 to T3.7.4).
2. The friction coefficients $\lambda = f(\text{Re}, \varepsilon)$ must be determined iteratively from Eqs. (T3.7.12) and (T3.7.13) in the same way as for axial seals. The Reynolds numbers are calculated with radial and circumferential velocities, which are determined as the geometric averages between inlet and outlet, Eq. (T3.7.21). For this reason the radius ratio $a_r = r_a/r_i$ appears in this equation.
3. The radial velocity in the seal and the leakage are obtained from Table 3.7(2), Eqs. (3.7.22) and (T3.7.23).

When pumping abrasive suspensions, the clearances can increase considerably during service because of wear. The consequent decrease in head and efficiency is not only caused by the volumetric loss. Large leakage flows also generate a pre-swirl at the impeller inlet and thereby reduce the head according to the Euler equation, Table 3.3. Strong pre-swirl has an impact on cavitation so that NPSH_R is apt to increase appreciably. The leakage flow can be reduced through pump-out vanes (as shown in Fig. 3.15). The effect of the leakage on head, efficiency and NPSH_R can be roughly estimated according to Eqs. (T3.7.18 to T3.7.20).⁹

⁹In order to derive these relationships, the measurements in [21] were recalculated into a more general form.

3.6.5 Leakage Losses in Open Impellers

In the case of open or semi-open impellers (see Table 2.1) fluid flows from the pressure to the suction surface of the blades through the clearance between casing and blades. In the process, the kinetic energy of the leakage flow is largely dissipated and the main flow deflection is reduced (the slip factor dropping accordingly). Consequently, efficiency and head suffer with growing clearance. However, the kinetic energy of the leakage flow is fed into the low-energy zone on the suction surface (the “wake”, as per Chap. 5). This can have a favorable effect on the stability of the head-capacity characteristic of a semi-axial pump. The pumping of free gas is also facilitated due to the mixing of the leakage flow with the main flow, because the risk of gas accumulations on the suction surface of the blades is reduced (Chap. 13.2). The leakage flow reinforces the secondary flow in the channel, imparting greater three-dimensionality to the flow in the impeller.

Very large clearances allow some fluid to flow back from the volute into the gap between the blades and the casing and consequently back into the impeller. The conditions are then similar to those in a vortex pump according to Fig. 2.20. The sensitivity of the pump to a clearance change depends on the circumferential component of the fluid flowing back from the volute. This interaction with the collector further contributes to the uncertainty of predicting the drop in performance due to the leakage.

Because of the interaction between the leakage and the main flow, a strict separation of the leakage losses from the hydraulic losses and the friction on the casing wall is not possible. In the case of small clearances ($s/b_2 < 0.01$), such as are used in semi-axial pumps with high specific speeds, it therefore appears sensible not to determine the leakage losses separately but to include them in the hydraulic efficiency.

The influence of the clearance on head, power consumption and efficiency was measured in several experiments; [7, 8, 22, 29] are referenced as recent works. According to these investigations the hydraulic behavior of pumps with open impellers can be characterized as follows:

- The head of a pump drops in a linear fashion with increasing clearance according to $(H_{(s=0)} - H_{(s)})/H_{(s=0)} = g \times s$. The subscript $s = 0$ describes the conditions extrapolated to zero clearance and subscript s the behavior with finite clearance. The gradient g describing the decrease in head greatly depends on the impeller design. There is no perfect correlation between gradient g and the impeller parameters since the three-dimensional flow in the impeller cannot be described with simple geometrical parameters. Different impellers react to an increase in clearance with quite different sensitivities. Performance impairment is mitigated when the number of blades and/or the outlet angle is increased. Impellers with a large number of radial blades and $\beta_{2B} = 90^\circ$ therefore react less sensitively to the clearance in spite of the low specific speed. The insensitivity of blades with large outlet angles can be explained by the velocity in the gap having a high circumferential component whose energy contributes to the flow

deflection. In the case of small blade angles, in contrast, the leakage flow has a radially inward directed component opposing the main flow. From the previously mentioned test results it is possible to derive a correlation, Eq. (T3.7.15), for the head decrease. Equation (T3.7.15) describes the above effects qualitatively quite well, but has a considerable scatter.

- Because of the lower flow deflection, the power consumption also drops with increasing clearance. The same applies to the efficiency which drops more than the power consumption. These effects are demonstrated by Eqs. (T3.7.16 and 3.7.17) according to which the efficiency reduction amounts to approximately 2/3 of the head decrease. Accordingly, the relative power-consumption diminishes by roughly 1/3 of the head reduction.
- The head decreases throughout the load range. The relative impairment increases with growing flow rate. The shut-off head drops approximately half as much as the head at the best efficiency point.
- The best efficiency point is shifted towards smaller flow rates with increasing clearance.

According to [B.1] it is possible to very roughly estimate the effect of the clearance on head and efficiency for radial, semi-axial and axial impellers from:

$$\frac{\Delta\eta}{\eta} = \frac{\Delta H}{H} = \frac{\Delta Q}{Q} = \frac{s}{b_2} \quad (3.13)$$

Semi-axial pumps with $n_q > 70$ to 100 generally have semi-open impellers (without a front shroud) because a large-diameter shroud would generate high disk friction losses and considerably impair the efficiency (typically by 3%).

3.6.6 Mechanical Losses

The mechanical losses P_m are generated by the radial bearings, the axial bearing and the shaft seals. Occasionally P_m includes auxiliary equipment driven by the pump shaft. These losses depend on the design of the pump, i.e. on the selection of anti-friction versus journal bearings or stuffing boxes versus mechanical seals.

The mechanical efficiency of large pumps is around 99.5% or even above. In contrast, the mechanical losses of small pumps (say below 5 kW) can use up a considerable portion of the coupling power. Examples are process pumps which are often equipped with dual mechanical seals.

The mechanical losses can be estimated according to Eq. (T3.5.6a) if more accurate information is not available. But usually it is best to consult the documentation of the manufacturer of the mechanical equipment. Exact analysis is indicated in the case of special designs such as wet motors or magnetic couplings.

When pump performance tests are done at a speed lower than that in the plant, it must be taken into account that the mechanical losses have a lower exponent of the

speed than the power of the pump: $P_m \sim n^x$ with $x = 1.3$ to 1.8 is expected rather than $P \sim n^3$. In other words, the mechanical efficiency of a given pump decreases with the speed.¹⁰

3.7 Basic Hydraulic Calculations of Collectors

At the impeller outlet the fluid has the circumferential velocity $c_{2u} = c_2 \times \cos \alpha_2$ and the specific kinetic energy $E_{kin} = \frac{1}{2}c_{2u}^2$. Up to moderate specific speeds, $\cos \alpha_2$ is close to 1.0. Equation (T3.3.9) yields $c_{2u}/u_2 = \frac{1}{2}\psi_{th}$ for swirl-free inflow. The theoretical specific work is $Y_{th} = \frac{1}{2}\psi_{th} \times u_2^2$. Using these expressions, we can derive the kinetic energy at the collector inlet $E_{kin}/Y_{th} = \psi_{th}/4$ as a fraction of the energy transferred by the impeller. An acceptable efficiency can consequently be achieved only if the kinetic energy available at the impeller outlet is effectively decelerated in the diffusing elements. The collector design attempts to convert the greatest possible part of E_{kin} into static pressure.¹¹

In Table 2.1 the following diffusing elements were introduced: volute casings, vaned or un-vaned diffusers or concentric annular chambers, all of which can be designed by relying on similar principles.

According to Newton's law of inertia, a mass retains its momentum if not prevented by external forces to do so. As a consequence, the fluid has the tendency to maintain its angular momentum (its "swirl") $\rho \times Q \times r_2 \times c_{2u}$ downstream of the impeller unless it is influenced by structures or wall friction. In an essentially vaneless casing the flow thus follows the law $c_u \times r = c_{2u} \times r_2 = \text{constant}$ (apart from friction effects). The circumferential component of the flow velocity in any element of the casing cross section is therefore calculated from $c_u = c_{2u} \times r_2/r$. The diffusing elements have to be designed so that the flow field in the collector is compatible with the conservation of angular momentum. If the angular momentum is conserved, rotational symmetry around the impeller is achieved and interaction between the flows in the collector and impeller is minimized.

The diffusing elements are dimensioned for the flow rate $Q_{Le} = Q_{opt} + Q_{s3} + Q_E$ where Q_E is the volumetric flow through a device for axial thrust balancing and Q_{s3} the leakage of the inter-stage seal.

¹⁰The mentioned exponent does not imply a contradiction to Eq. (T3.5.6a) which refers to different pumps with optimally adjusted mechanical equipment, while the scaling of the mechanical losses P_m of a given pump assumes given mechanical components which are over-dimensioned at reduced speed.

¹¹Exceptions are axial pumps with very high specific speeds, which under certain conditions, have to be designed without diffusing elements, Chap. 7.6.6.

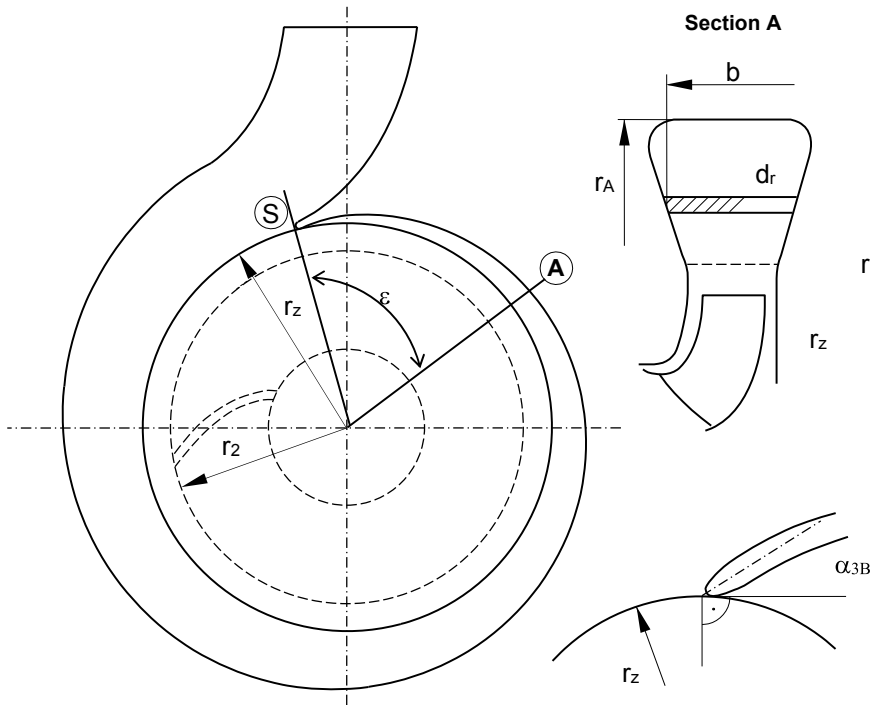


Fig. 3.17 Volute casing

Volute casing: Consider any cross section of a volute located at the circumferential position ε° , Fig. 3.17. The fluid $dQ = c_u \times b \times dr = c_{2u} \times r_2/r \times b \times dr$ flows through an area element $dA = b \times dr$ on the radius r in the section A. The flow rate through the entire cross section is obtained by integration $Q(\varepsilon) = c_{2u} \times r_2 \times \int b/r \times dr$. From the start of the volute (point S in Fig. 3.17) to the cross section A, the impeller discharges the flow $Q(\varepsilon) = Q_{\text{opt}} \varepsilon^\circ/360^\circ$ in the design point. In order to allow the fluid to flow according to $c_u \times r = c_{2u} \times r_2 = \text{constant}$, cross section A must therefore be designed in such a way that Eq. (3.14) is satisfied:

$$\int_{r_z}^{r_A} \frac{b}{r} dr = \frac{Q_{\text{opt}} \varepsilon^\circ}{360^\circ c_{2u} r_2} \quad (3.14)$$

Experience confirms that the losses become approximately minimal if a volute casing is dimensioned so that the fluid at the design point is able to flow in accordance with the angular momentum conservation, see also Chap. 4.2. This means that the cross sections of the volute casing - and most importantly the end cross section or “throat area” - have to be selected according to Eq. (3.14). The throat area then follows from Eq. (3.15):

$$Q = z_{Le} \int c_{2u} \frac{r_2}{r} b \, dr \quad \text{or} \quad Q = c_{2u} r_2 J_{SP} \quad \text{with} \quad J_{SP} = z_{Le} \int_{r_z}^{r_A} \frac{b}{r} \, dr \quad (3.15)$$

z_{Le} is the number of partial volutes forming the casing. Most frequently used are: single volutes with $\varepsilon = 360^\circ$ wrap angle (i.e. $z_{Le} = 1$) and double volutes with 180° each and $z_{Le} = 2$. Occasionally three or four volutes are used to reduce the size of the casing. Double volutes are also designed with wrap angles $\varepsilon_{sp} < 180^\circ$ where $z_{Le} = 360^\circ/\varepsilon_{sp}$ must be used accordingly for each partial volute.

The swirl $r_2 \times c_{2u}$ at the volute inlet is obtained from Eqs. (T3.2.7) or (T3.7.8). If a vaned or vaneless diffuser is installed upstream of the volute, its outlet swirl $c_{4u} \times r_4$ must be inserted into Eqs. (3.14) or (3.15).

The volute cutwater forms a camber angle α_{3B} with the circumferential direction which must be selected in accordance with the approach flow angle. It is obtained from Eqs. (3.16) to (3.19) that apply to both volutes and diffusers. The incidence is selected in the range of $i_3 = \pm 3^\circ$; for volutes set $\tau_3 = 1.0$.

The meridional velocity c_{2m} at the impeller outlet cannot expand suddenly to the inlet width b_3 — in particular if the ratio b_3/b_2 is large. Nevertheless, b_3 is used in Eq. (3.16) with the goal to design the diffuser vanes or cutwaters with low inlet angles. This is advantageous for partload operation in that the tendency to flow separation at the cutwater is reduced.

Meridional component with blockage (Table 0.1); for volutes set $\tau_3 = 1.0$.

$$c_{3m}' = \frac{Q_{Le} \tau_3}{\pi d_3 b_3} \quad (3.16)$$

circumferential component according ($d_z = d_3$): to conservation of angular momentum

$$c_{3u} = c_{2u} \frac{d_2}{d_3} \quad (3.17)$$

approach angle with blockage:

$$\tan \alpha_3' = \frac{c_{3m}}{c_{3u}} \quad (3.18)$$

vane inlet angle:

$$\alpha_{3B} = \alpha_3' + i_3 \quad (3.19)$$

It is advisable to shape the cutwater as an elliptical profile, which is less sensitive to changes in incidence (or pump flow rate). The profile can be applied either symmetrically as in Fig. 3.17 or be made asymmetric to obtain small incidences at partload. Any profiling must be made very short to keep stresses in the cutwater within acceptable levels. Not much profiling is possible with small pumps.

The diffuser which usually follows the actual volute must be designed according to Chap. 1.6. For hydraulic and design reasons, a distance is required between the impeller outlet and the cutwater. In the space thus created the fluid flows according to the laws of the vaneless diffuser, but the strong secondary flow created in the volute has an impact on the wall-near flow.

Diffuser: As shown by Fig. 3.18, a diffuser consists of z_{Le} vanes forming triangular inlet sections and closed channels (“diffusing channels” as per Table 0.2), which can be calculated according to Chap. 1.6. The flow through the triangular inlet section can be imagined to behave in a similar way as a partial-volute. The smaller the number of diffuser vanes, the more this is the case. When the number of diffuser vanes is large the flow does not truly follow the relationship $c_u \times r = c_{2u} \times r_2 = \text{constant}$. The diffuser throat area at a_3 cannot be designed strictly according to Eq. (3.15). Nevertheless Eq. (3.15) forms a physically sound basis to calculate the throat area, if an appropriate correction factor f_{a3} is applied, Chap. 7. For a given inlet width b_3 , the diffuser inlet area can thus be determined from Eq. (3.20).

$$a_3 = f_{a3} \frac{d_3}{2} \left\{ \exp \frac{2 Q_{Le}}{z_{Le} b_3 d_2 c_{2u}} - 1 \right\} \quad (3.20)$$

This relationship corresponds to the integral of Eq. (3.15) for a volute with rectangular cross section. As for the rest, the diffuser can be calculated according to similar principles as a volute: (1) Blade angle α_{3B} according to Eqs. (3.16) to (3.19) where $i_3 = \pm 3^\circ$; (2) Diffusing channel according to Chap. 1.6; (3) Profiling as described above (for details see Chap. 7).

Vaneless diffuser: In a vaneless diffuser, depicted in Fig. 3.19, the flow is able to develop according to the angular momentum conservation and the pressure increases in radial direction following Eq. (1.28). The meridional component is calculated from continuity as per Eq. (3.21).

$$c_m = \frac{Q_{Le}}{2 \pi r b} = \frac{c_{3m} r_2 b_3}{r b} \quad (3.21)$$

Setting $c_u = c_{2u} \times r_2/r$ and neglecting friction, the flow angle is obtained from the following equation:

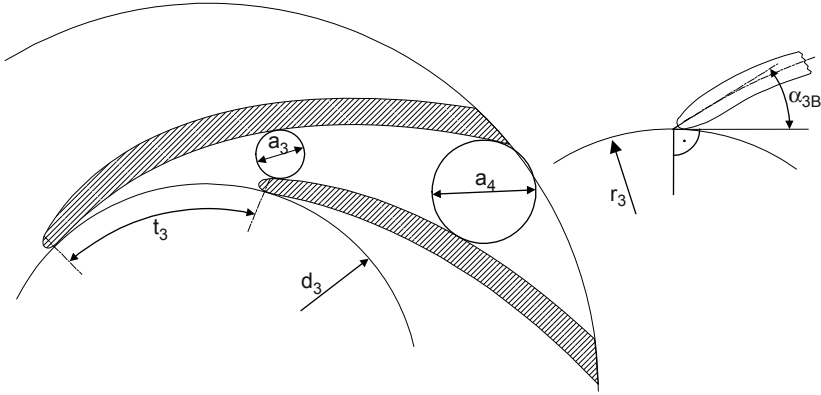
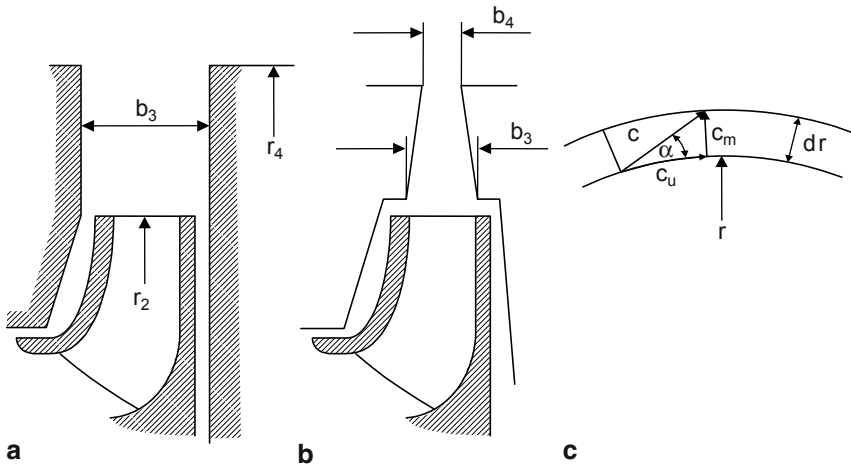


Fig. 3.18 Vaned diffuser

Fig. 3.19 Vaneless diffuser. **a** parallel walls; **b** conical walls; **c** velocity vectors

$$\tan \alpha = \frac{c_m}{c_u} = \frac{c_{3m} b_3}{c_{2u} b} = \tan \alpha_3 \frac{b_3}{b} \quad (3.22)$$

Equation (3.22) shows that the flow angle remains constant in a vaneless diffuser of constant width as long as friction is neglected.

In reality, the circumferential component c_u drops due to wall friction in comparison with the angular momentum conservation employed without shear stresses. As a result, the flow angle grows with increasing radius. According to [B.1] this follows from the relationship:

$$\frac{c_u r}{c_{2u} r_2} = \left\{ 1 + \frac{c_f r_2}{b_3 \tan \alpha_3} \left(\frac{r}{r_2} - 1 \right) \right\}^{-1} \quad (3.23)$$

The flow angle in viscous flow can be deduced from Eq. (3.23) leading to:

$$\tan \alpha_4 = \tan \alpha_3 + c_f \frac{r_2}{b_3} \left(\frac{r_4}{r_2} - 1 \right) \quad (3.24)$$

The meridional velocity results from continuity and is therefore not affected by the friction.

The friction losses in a vaneless diffuser of constant width can be determined approximately as follows: In a ring element (as shown in Fig. 3.19c) the velocity is $c = c_u / \cos \alpha = c_{2u} \times r_2 / (r \times \cos \alpha)$ and the wall shear stress is $\tau = 1/2 \rho \times c_f \times c^2$. Thus, the power $dP_d = c \times dF = 1/2 \rho \times c_f \times c^3 \times dA$ is dissipated, Chap. 1.5. Integrating dP_d from r_2 to r_4 with $\zeta_{LR} = 2 \times P_d / (\rho \times Q \times u_2^2)$ the coefficient of the friction losses ζ_{LR} according to Eq. (T3.8.23) is obtained which comprises *both* sides of the vaneless diffuser.

Equation (T3.8.23) shows that the friction losses *diminish with growing flow rate* since the circumferential component c_u drops. At the same time the flow angle α increases and the friction path length is shortened. For this reason, the best efficiency point shifts towards a higher flow rate if a vaneless diffuser is employed in place of a volute or a diffuser.

There is an optimum flow angle at the impeller outlet that has to be realized in order to place the efficiency optimum at the specified design flow rate. This means that the width b_2 or b_3 cannot be selected arbitrarily.

Since c_u drops with increasing flow, the pressure recovery drops as well. Hence a vaneless diffuser is a stabilizing element - for as long as the flow does not separate. The smaller the outflow angle α_2 from the impeller, the greater the losses become in the vaneless diffuser. Consequently, a vaneless diffuser yields poor efficiencies with low specific speeds. If the vaneless diffuser is designed so that its width decreases from the inside to the outside ($b_4 < b_3$), the flow angle according to Eq. (3.22) grows with increasing radius. Flow path length and losses diminish accordingly.

If the vaneless diffuser width b_3 is designed larger than the impeller outlet width b_2 , shock losses occur due to deceleration of the meridional velocity from c_{2m} to c_{3m} which can be calculated according to Eq. (1.12): $2g \times \Delta H = (c_{2m}' - c_{3m})^2$. The corresponding loss coefficient is shown as Eq. (T3.8.24) in Table 3.8; it includes the effect of the blade blockage at the impeller outlet. This sudden deceleration creates a minor pressure recovery that can be calculated from Eq. (1.11). *According to the principle of angular momentum conservation the circumferential component c_{2u} is not affected by the expansion from b_2 to b_3 .*

Volute casings and diffusers behave like vaneless diffusers with regard to the expansion from b_2 to b_3 .

Annular casing: Annular casings are sometimes used for pumps with low specific speeds or for small pumps. The fluid is discharged from the annular casing via a diffuser. The annular casing differs from the vaneless diffuser in that it is noticeably wider than the impeller outlet and its radial extension accordingly smaller. The deceleration of the meridional velocity from c_{2m} to c_{3m} is high, promoting a vigorous secondary flow and exchange of momentum between the annular casing and the impeller. Since the cross sections in the concentric casing do not correspond to a design according to angular momentum conservation, no circumferential symmetry is obtained at the design point. Consequently, large radial forces are generated as discussed in Chap. 9.3.

3.8 Hydraulic Losses

The hydraulic losses in a pump reduce the useful head according to $H = \eta_h \times H_{th}$ or $H = H_{th} - Z_E - Z_{La} - Z_{Le} - Z_{sp} - Z_A$. They include losses in all components located between the suction and the discharge nozzles, i.e. losses in the inlet casing (Z_E), the impellers (Z_{La}), diffusers (Z_{Le}), volutes (Z_{sp}) and the outlet casing Z_A . Hydraulic losses are generated through friction and vortex dissipation (Chap. 1.5.2):

1. Skin friction losses are the result of shear stresses in the boundary layers on solid structures. Skin friction, which depends on the Reynolds number and the roughness, is important in thin boundary layers and attached flow. These conditions are found in accelerating flows. Friction, Reynolds number and the roughness have less impact on losses in decelerated or separated flow.
2. Flow deceleration leads to thick boundary layers which make the velocity distribution less uniform. In non-uniform flow the exchange of momentum between different streamlines increases due to eddies of various scales. Large eddies break up into small-scale turbulence which finally dissipates into increased molecular movement leading to a slight heating of the fluid.

In the way described under item 2 and in Chap. 1.5.3, non-uniform flows have an altogether profound impact on the pump's hydraulic efficiency. Non-uniform flows are generated by various mechanisms:

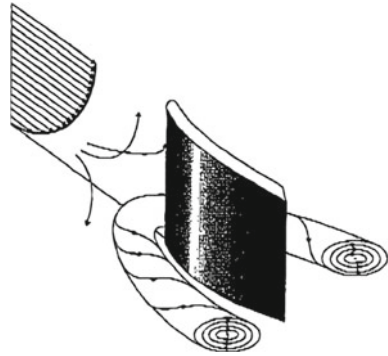
- The work transfer by the blades is inherently linked to non-uniform flow over the pitch of the blades. The non-uniformity of the flow increases with the blade loading (i.e. with increasing pressure coefficient and decreasing blade length).
- Incident flow creates local zones with decelerated flow or even separation on the blades ("shock-losses"). This can occur with or without cavitation.
- A sudden deceleration of the flow, when the impeller or diffuser throat area does not match with the approach flow velocity vector (w_{1q}/w_1 and c_{3q}/c_2), leads necessarily to a non-uniform flow distribution in the channel. The flow mechanism loosely resembles the Carnot-shock of a sudden expansion, Fig. 1.3.

- Any kind of curved flow path, notably the curvature of the meridional section and of the blades, creates a non-uniform flow as illustrated by Fig. 1.12.
- Flow separation implies zones of stalled fluid with local recirculation. The stalled fluid blocks part of the cross section available to the flow. The un-stalled fluid is therefore accelerated and may form a sort of jet. High losses are generated through the strong exchange of momentum between the jet-like through-flow and the stalled zone.
- Secondary flows are generated by Coriolis forces, streamline curvature (Fig. 1.12) and centrifugal forces. These effects are discussed in detail in Chap. 5. Another type of secondary flow is created by different boundary layer flows on the blades and the side walls as illustrated by Fig. 1.6. The strength of the secondary flow (or the cross-flow) may be thought of as establishing itself according to the *principle of least resistance*.
- The mixing of the leakage through the seal at the impeller inlet creates a non-uniform flow near the front shroud in terms of the c_m -distribution, local pre-swirl and incidence angles.
- Wake flows downstream of blades or ribs. The wake of the blades is increased by low energy fluid near the suction side of the impeller blades.
- When the boundary layer flow on the hub or shroud enters the impeller blading (or a vaned diffuser) it is decelerated upon approaching the leading edge. Due to the blockage caused by the blade and by stagnating fluid, liquid is displaced away from the wall and side-ways into the channel. Thereby a “horseshoe” vortex is formed, Fig. 3.20. It has some blockage effect, increases non-uniformity of the flow and can induce cavitation and abrasive erosion. Some photographs of erosion caused by horseshoe vortices in sand-laden water on volute cutwaters and impeller blades are shown in Chap. 14.5.6.
- In semi-open impellers the leakage between the blades and the casing interacts with the main flow in the impeller channels.

These phenomena can hardly be assessed by applying simplified models. For this reason, the hydraulic efficiency is mostly determined from the power balance of a measured pump according to Eq. (T3.5.8). This requires calculating the secondary losses by means of the correlations given in Chap. 3.6. Thus, the hydraulic efficiency is only known once the pump has been built and its performance measured. For designing a pump, it is necessary to rely on earlier tests or correlations which allow a reasonable estimate of η_h .

The hydraulic efficiency calculated from the power balance does not allow any statements as to the contribution of individual pump components to the losses. To answer this question, it can be useful to estimate the loss in individual components. Such calculations have an empirical character since the three-dimensional velocity distributions in the impeller and diffusing elements, which determine both friction and turbulence losses, cannot be described by simple models. Estimations of this type are only meaningful near the best efficiency point. *With pronounced partload recirculation they fail entirely* if the exchange of momentum of the recirculating fluids is not taken into account.

Fig. 3.20 Formation of a “horseshoe” vortex



The **loss model for impellers** according to Table 3.8(1) comprises the following steps and assumptions:

- Equation (T3.8.2) defines an average velocity in the impeller channel which is formed by the throat areas that are defined by a_1 and a_2 at inlet and outlet.
- Equation (T3.8.3) supplies, as a function of the Reynolds number and the roughness, a friction coefficient corresponding to a flat plate in parallel flow. The plate model is preferred to the channel model since undeveloped flows must be assumed because of the short impeller blade lengths. Equation (T3.8.3) includes the roughness. Assessing its effect is frequently required in practice in order to evaluate possible efficiency improvements through additional finishing of the impeller channels.
- Equation (T3.8.5) utilises a dissipation coefficient. According to [B.3] the amount of 0.0015 is added to the friction coefficient since thicker boundary layers, hence greater losses, occur in decelerated flow than on a flat plate in longitudinal flow. The value obtained in this way is further multiplied by an empirical factor containing the relative impeller outlet width. According to this equation, the losses increase considerably with increasing values of b_2^* or specific speed. The empirical factor can thus be interpreted as the effect of uneven velocity distributions and secondary flow.
- Equation (T3.8.6) produces the loss coefficient for the impeller which comprises the friction, deceleration and turbulence effects.
- By means of Eq. (T3.8.9) it is possible to estimate the shock losses at the impeller inlet. This equation describes the deceleration of the vector of the mean inflow velocity w_{1m} (from the velocity triangle) to the velocity w_{1q} in the throat area A_{1q} , calculated from Eq. (T3.8.8). The shock loss in this case is assumed at 30% of a Carnot shock, which supplies plausible results for moderate decelerations. This relationship should not be used for $w_{1q}/w_{1m} < 0.6$.

An alternative model for the friction losses is formulated in Table 3.8(1), Eqs. (T3.8.11–3.8.14). All of the wetted surfaces of the impeller channels, consisting of front and rear shrouds as well as suction and pressure surfaces of the blades, are

taken into account by Eq. (T3.8.11). The mean value of the vectors of the relative velocity at inlet and outlet is used as the mean velocity, Eq. (T3.8.12). The dissipated friction power is obtained according to Eq. (1.34) from Eq. (T3.8.13) and the loss coefficient from Eq. (T3.8.14). This calculation corresponds to the procedure for the loss estimation of volute casings (see below). According to Eq. (T3.8.14) the friction losses in the impeller rise with falling specific speed, which is *not* confirmed by tests and numerical calculations.

The **loss model for diffusers** according to Table 3.8 (2) comprises the following steps and assumptions:

- Mixing losses due to an inflow with a non-uniform velocity profile are *not* considered.
- Equation (T3.8.15): calculation of the velocity vector at the impeller outlet
- Equation (T3.8.16): velocity in the diffuser throat area
- Equation (T3.8.17): determination of the pressure recovery coefficient c_p in the diffusing channel. For plane diffusers use Fig. 1.18. For other diffuser types calculate an equivalent conical diffuser from Eq. (1.45) and determine c_p from Fig. 1.19.
- The friction losses between the impeller outlet and the diffuser throat area can be estimated according to Chap. 1.5.1 from Eqs. (1.32) to (1.35) by calculating the energy dissipated through wall shear stresses on the blades and the sidewalls. After several transformations this results in Eq. (T3.8.18).
- Equation (T3.8.19) supplies the pressure loss coefficient for the diffuser including the return channel. If a volute is provided downstream of the diffuser (i.e. in the absence of a return channel), its loss coefficient ζ_{ov} must be set to zero. The first term in Eq. (T3.8.19) takes into account the loss due to the deceleration from c_2 to c_{3q} . The explanations for Eq. (T3.8.9) apply accordingly. The actual flow deceleration is described by the term $1 - c_p - 1/A_R^2$ (Chap. 1.6).
- The losses of the return channels greatly depend on their configuration. With optimal flow design $\zeta_{ov} = 0.2$ could be attainable for a continuous channel, while a whole stagnation pressure $c_4^2/2g$ could be lost with an unfavorable design (i.e. $\zeta_{ov} = 1.0$).

The **loss model for volute casings** including diffuser and pressure nozzle according to Table 3.8 (2) comprises the following steps and assumptions:

- Mixing losses resulting from an inflow with a non-uniform velocity profile have *not* been considered.
- Since the flow rate in the volute casing varies around the circumference, the dissipation due to friction should be calculated according to Eq. (1.34). This leads to Eq. (T3.8.21), which is written as the sum over all elements ΔA of the wetted surface of the volute casing, since the calculation is as a rule performed section by section. The velocity in the volute changes over the cross section according to $c_u \times r = \text{constant}$ and over the circumference. The division of the

surface elements must be performed accordingly, depending on the desired accuracy. As with the impeller, the friction coefficient is calculated according to the flat plate model and increased by the value of 0.0015 for decelerated flow. In the case of double volutes, the integration (or summation) is performed for both partial volutes. This sum covers the friction effective on all wetted surfaces; it applies to the *entire* flow rate of the pump (i.e. both partial volutes).

- The actual volute is followed by a diffuser/pressure nozzle. The diffuser is treated similarly to Eq. (T3.8.17): an equivalent conical diffuser is calculated as shown in Chap. 1.6 and the pressure recovery coefficient is determined from Fig. 1.19. The pressure loss in the diffuser is then obtained from Eq. (T3.8.22).
- In the case of double volutes, pressure losses in the outer channel are added to the outer volute. The outer channel can be calculated as a bend according to Table 1.4. If cross sectional expansions are involved, it is calculated with additional diffuser losses according to Fig. 1.19. Since outer and inner volutes have different resistances, the flow distribution over both partial volutes does not correspond to the theoretical value obtained from their wrap angles. The individual flow rates through both volute channels can be determined by applying the concept of parallel flow resistances described in Table 1.5. By means of such loss considerations it is also possible to design the outer channel so that approximately the same flow rates are established in both the outer and inner volutes.
- In multistage pumps the fluid moves from the volute into an overflow channel which directs the liquid to the following stage. Because of the severe curvature of that channel, separations and additional losses occur [13]. The deflection losses can be estimated according to Table 1.4.
- The friction losses in a vaneless diffuser are calculated from Eq. (T3.8.23). This is done with the impeller outlet flow angle α_2 – rather than the angle α_3 from Eq. (3.18) – because the flow cannot expand suddenly from b_2 to b_3 .
- The shock losses at the impeller outlet are obtained from Eq. (T3.8.24). These also occur in volute casings. In the case of diffusers, the expansion from b_2 to b_3 is implicitly included in Eq. (T3.8.19). The effect of blade blockage can be considered in a similar way as in Eq. (T3.8.24) by inserting c_2' for c_2 .
- Since volute casing, outer channels and diffuser/pressure nozzles can be designed in different ways, the loss calculation sketched above should be adapted to suit the respective requirements.
- The deceleration and shock-losses for off-design conditions can be estimated from the test data discussed in Chap. 4.1.3, which yield the correlation given by Eq. (T3.8.25). In this equation the diffuser (or volute) loss is correlated successfully against the deceleration ratio c_{3q}/c_2 . The details of the correlations are discussed in Chap. 4.1.3. When applying Eq. (T3.8.25), the diffuser (or volute) losses are first calculated at the BEP according to Table 3.8 (2); this yields $\zeta_{Le, opt}$ which can be used subsequently as $\zeta_{Le, min}$.

The losses in the pump inlet are usually insignificant. They can generally be neglected in end-suction pumps. If an extended pipe section, a suction bell or a

similar component is located between the measuring point in the suction pipe and the impeller inlet, the pressure losses can be calculated according to the customary rules, Tables 1.4 and 1.6. Typical inlet casings of bearing pumps have pressure loss coefficients of $\zeta_E = 0.15$ to 0.4 based on c_{1m} (hence $Z_E = \zeta_E c_{1m}^2 / 2g$), depending on the actual design. According to the measurements in [B.21], the inlet losses decrease with the square of the deceleration of the fluid in the inlet casing following the relationship:

$$\zeta_E = 0.75 \left(\frac{d_1^2 - d_n^2}{d_s^2} \right) \quad (3.25)$$

Outlet losses need not be considered in many pumps since the losses in the diffuser/discharge nozzle are included in the calculation of the volute casing. The losses in the outlet chambers of multistage pumps greatly depend on their design. Typically they are around 1 to 2% of the head of *one* stage.

The procedure described in Table 3.8 has been applied to single-stage end-suction volute pumps in a range of $n_q = 7$ to 85 . All pumps were designed for $Q_{opt} = 0.1 \text{ m}^3/\text{s}$ in a way that they had the same impeller inlet diameter. The factor for decelerating flow was set to zero in Eqs. (T3.8.5) and (T3.8.21). The resultant loss coefficients are plotted in Fig. 3.21. The losses in the volute and the discharge nozzle increase steeply as the specific speed is reduced, while the impeller losses increase almost linearly with n_q .

The hydraulic efficiencies calculated are shown in Fig. 3.22 compared to the values obtained from the statistical data according to Eq. (T3.9.1). Both curves show the same tendency with a flat optimum in the range of $n_q = 30$ to 60 .

In view of the scatter of the statistical data, the agreement between the predicted and statistical hydraulic efficiencies appears to be quite good. The same applies to higher design flow rates, while the discrepancies increase at lower flow rates.

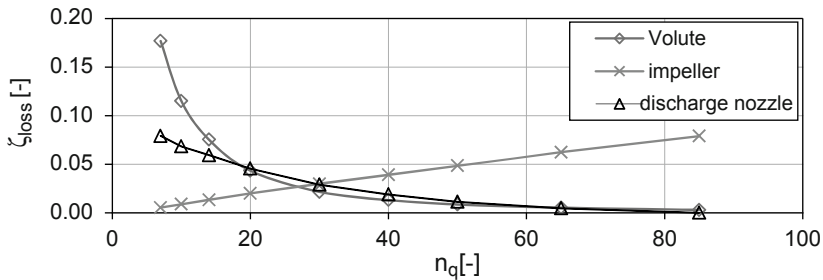


Fig. 3.21 Loss coefficients calculated from Table 3.8

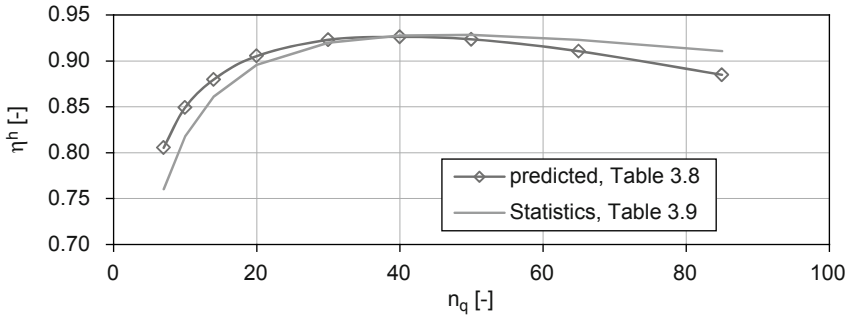


Fig. 3.22 Hydraulic efficiencies calculated from Table 3.8

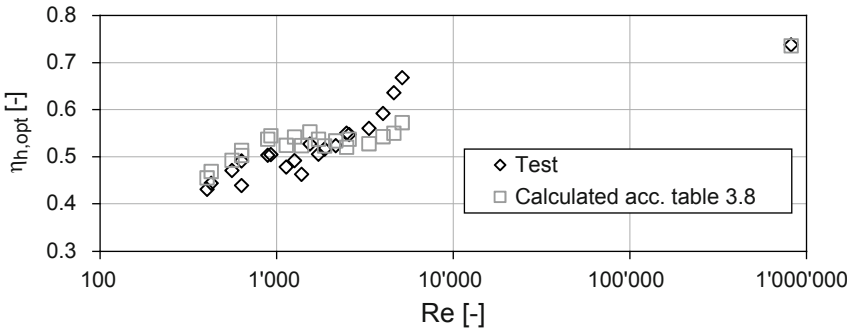


Fig. 3.23 Hydraulic efficiencies tested with various viscosities and speeds compared to calculation according to Table 3.8

Tests with a single-stage end-suction volute pump at various speeds and with viscosities up to 620 cSt are shown in Fig. 3.23 together with the calculation according to Table 3.8. These calculations were done at the actual BEPs measured with viscous flow. The points at about $Re = 800,000$ represent the tests with water.

There are some problems with this comparison: (1) The determination of the hydraulic efficiency of a pump with $n_q = 7$ in viscous flow is quite uncertain; (2) The calculation of the friction factor in the transition region between laminar and turbulent flow is uncertain; (3) no data on diffuser performance were available at low Reynolds numbers. Nevertheless, the results suggest that Table 3.8 may be used to estimate the hydraulic efficiency in viscous flow. The reason is that flow at low Reynolds numbers is dominated by friction effects which - to a large extent - form the basis for the calculation in Table 3.8.

The good agreement between the curves in Fig. 3.22 is considered to be to some extent coincidental because the loss models in Table 3.8 are not specifically adapted to the flow through pumps.

3.9 Statistical Data of Pressure Coefficients, Efficiencies and Losses

Based on numerous measurements in all types of centrifugal pumps, head coefficients and efficiencies were published which allow quite a reliable evaluation of the achievable values. As with all statistics, the figures supply mean values and mean errors but do not permit any statement on possible deviation of the *individual* measurement which, in principle, can be of any magnitude.

Figure 3.24 furnishes the pressure coefficient ψ_{opt} as a function of the specific speed. Here, ψ_{opt} is always based on the outlet diameter d_{2a} at the *outer* streamline. In contrast, ψ_i is based on the outlet diameter at the *inner* streamline d_{2i} according to [19]. The hydrodynamic blade loading at the hub limits the achievable pressure coefficient with semi-axial and axial impellers.

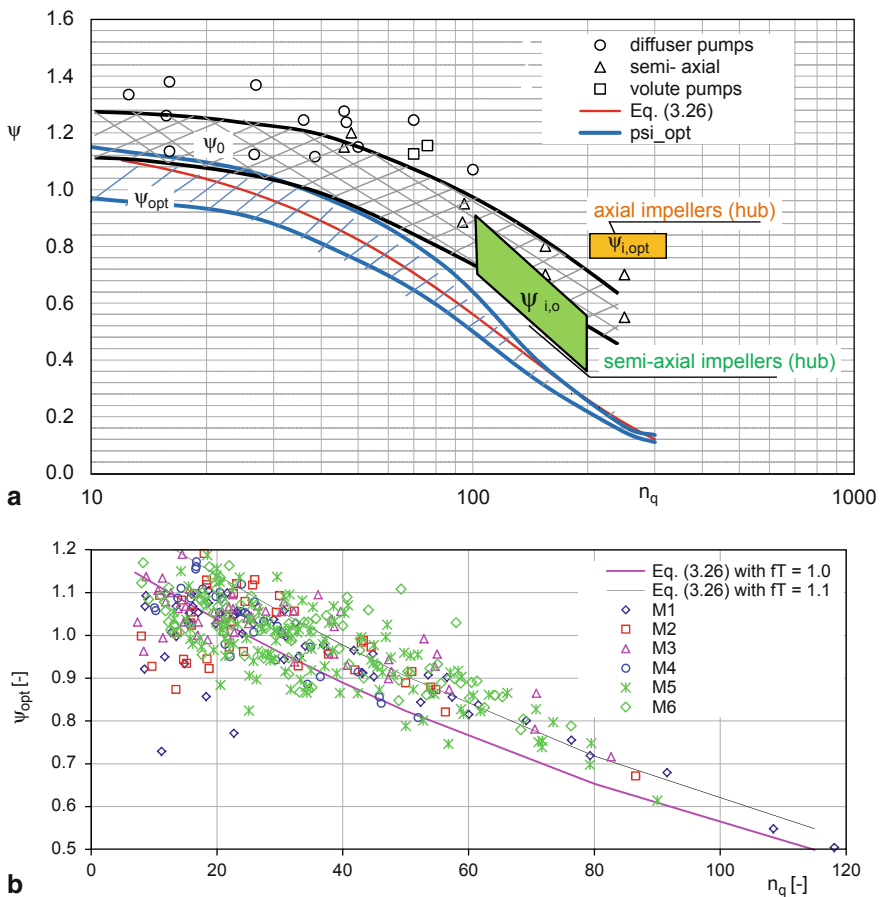


Fig. 3.24 Pressure coefficients: **a** general tendencies (values for hub according to [19]); **b** ψ_{opt} for single-stage volute pumps from various pump manufacturers

The drop of the achievable pressure coefficient with increasing specific speed is largely due to the fact that the centrifugal component of the head according to Eq. (T3.3.2) drops with growing n_q and becomes zero in axial pumps.

Analytical functions for the pressure coefficients according to Fig. 3.24 for the best efficiency point and for $Q = 0$ are provided by Eqs. (3.26–3.28).

The pressure coefficients to be selected to achieve a competitive design depend on the pump type and application. While Fig. 3.24a gives general trends, Fig. 3.24b shows an evaluation of the BEP pressure coefficients of various pump vendors (M1 to M6) for single-stage volute pumps with axial inlet or as inline pumps. It is seen that these low-head pumps (which include standard pumps according to EN733) are designed with head coefficients higher than calculated from Eq. (3.26) with $f_T = 1.0$. Rather $f_T = 1.1$ would be suitable for this application. A similar evaluation of data from various pumps vendors revealed that $f_T = 1.0$ is a suitable selection for multistage pumps.

Selecting high ψ_{opt} yields flatter characteristics and the risk of an instability of the Q-H-curve increases. The more so, the higher the value of ψ_{opt} is chosen. If a steep characteristic is required, ψ_{opt} is selected near the lower limit curve (or even below). In addition, Fig. 3.24 contains pressure coefficients ψ_o for operation against a closed valve, Chap. 4.

Pressure coefficients for designing sewage pumps (single-, two- or three channel impellers) and vortex pump are given Chap. 7.4.

$$\text{BEP: } \psi_{opt} = 1.21 f_T e^{-0.77 n_q / n_{q,Ref}} = 1.21 f_T e^{-0.408 \omega_s n_{q,Ref}} = 100 \quad (3.26)$$

$$Q = 0: \quad \text{diffuser pumps: } \psi_o = 1.31 e^{-0.3 n_q / n_{q,Ref}} \quad n_{q,Ref} = 100 \quad (3.27)$$

$$Q = 0: \quad \text{volute pumps: } \psi_o = 1.25 e^{-0.3 n_q / n_{q,Ref}} \quad n_{q,Ref} = 100 \quad (3.28)$$

The velocity components c_{2u}^* and w_{2u}^* can be calculated on the basis of the curve $\psi = f(n_q)$ given in Fig. 3.24, Eq. (T3.2.8) and Eq. (T3.2.11). If an inflow angle is specified, it is possible to determine the relative velocity w_1^* at the impeller inlet. Such data can be useful for estimations without details of the pump having to be known or calculated.

Figure 3.25 provides this type of data; the assumptions made for calculating the graph are given in the figure caption. In addition, the mean relative velocity $w_{sp}^* = (c_{ax}^2 + 1/4 u_{sp}^2)^{0.5}$ in annular seals has been plotted to ease the estimation of abrasive wear as per Chap. 14.5. With $\alpha_1 = 90^\circ$ we have $d_1^* \approx w_1^*$ since the inlet flow angle is usually small. Hence the impeller inlet diameter can be estimated from $d_1 = d_2 \times d_1^*$ when the outer impeller diameter has been determined.

In Fig. 3.26, overall efficiencies of single-stage radial pumps are shown as functions of the specific speed with the design flow rate as parameter. The overall efficiency reaches a maximum in the range from $n_q = 40$ to 50 for a given pump size. The maximum is caused by two tendencies: (1) To the left of the maximum the

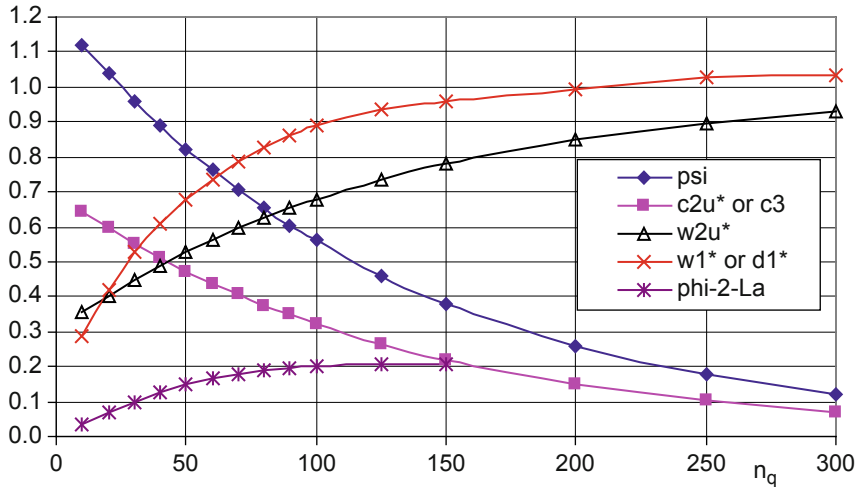


Fig. 3.25 Velocities at impeller inlet and outlet and in annular seals, calculated with $\eta_h = 0.87$, $k_n = 1$, $\beta_{1a} = 15^\circ$, clearance s from Eq. (3.12), leakage flow rate from Eq. (T3.5.10). All velocities are referred to u_2 . For $\alpha_1 = 90^\circ$ it is $d_1^* \approx w_1^*$

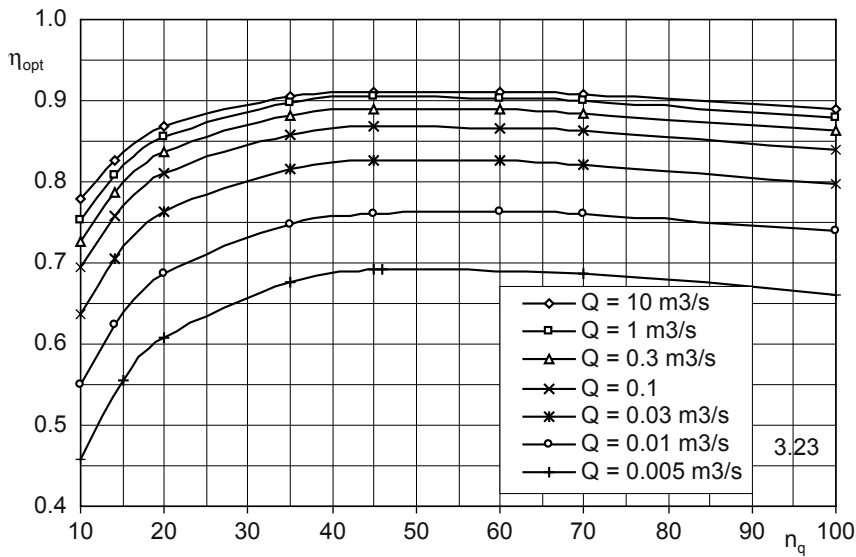


Fig. 3.26 Efficiencies of single-stage, single-entry, radial pumps

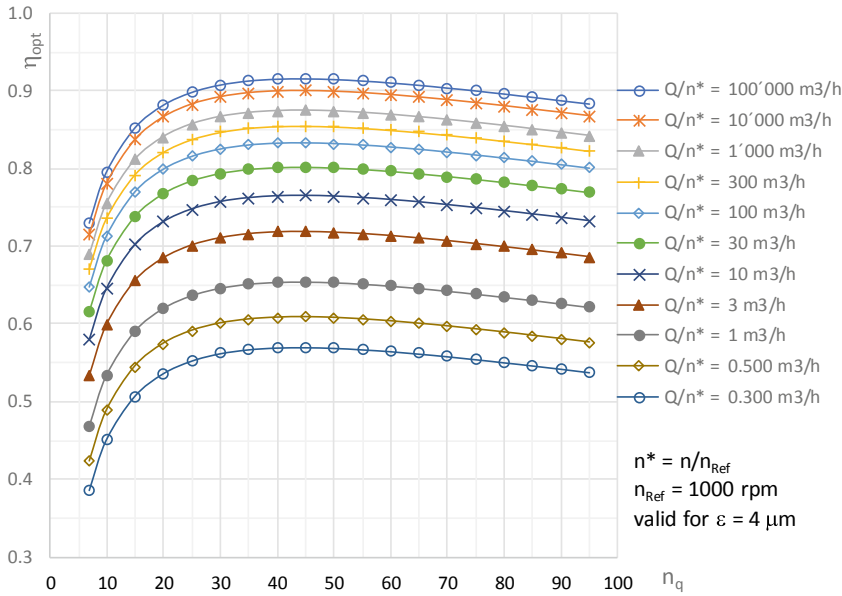


Fig. 3.26.1 Efficiencies of single-stage, single-entry, radial pumps, calculated from Eq. T3.9.1A; parameter Q_{opt}/n^*

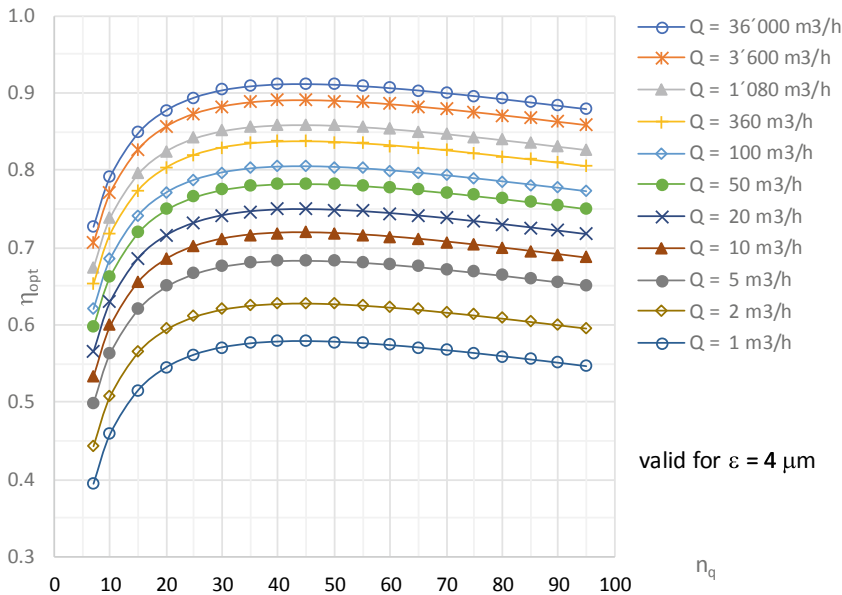


Fig. 3.26.2 Efficiencies of single-stage, single-entry, radial pumps, calculated from Eq. T3.9.1A; parameter Q_{opt}

secondary losses increase exponentially with falling n_q according to Eqs. (T3.5.10 to 3.5.12). (2) At high specific speeds ($n_q > 70$) the hydraulic losses increase with growing n_q due to increasing mixing losses mainly caused by non-uniform flow distributions over the blade height and by secondary flows.

As shown by Fig. 3.26, the overall efficiency increases with the flow rate. This dependency is caused by the influence of pump size and speed since the flow rate follows the proportionality $Q \sim n \times d_2^3$ (Chap. 3.4). In the case of very small pumps or very low flow rates, mechanical losses also have a strong impact, Chap. 3.6.4.

The efficiencies according to Fig. 3.26 apply to impellers without axial thrust balancing. If balance holes are provided, the efficiency must be corrected according to Eq. (T3.9.9).

An alternative to Eq. T3.9.1 and Fig. 3.26 is provided as Eq. T3.9.1A and Fig. 3.26.1. This formula has been adapted from [B.15]. It has the advantage that the influence of the surface roughness is included. The parameter in the graph is Q_{opt}/n^* rather than Q_{opt} . But plotting a graph with Q_{opt} as parameter gives very similar results if reasonable speeds are assumed. This is demonstrated by Fig. 3.26.2 which may be more convenient to use.

Equation T3.9.1A is expected to offer a better estimate than Eq. T3.9.1 for small pumps with $Q_{opt} < 50 \text{ m}^3/\text{h}$. However, any efficiency predictions from statistical data are uncertain for small pumps, because of the strong influence of the surface roughness; and, because the mechanical losses can take a large portion of the power at the coupling.

The efficiencies of semi-axial and axial pumps are shown in Fig. 3.27. Above $n_q > 70$ the efficiencies of semi-axial impellers are slightly superior to those of radial pumps. This is caused by differences in the shape of the impellers: Radial impellers have stronger front shroud curvatures than semi-axial impellers. Hence flow distributions over the blade height become more non-uniform with increasing specific speed, which leads to higher turbulent dissipation.

The efficiencies in Figs. 3.26 and 3.27 were calculated according to Eqs. (T3.9.1 and T3.9.3). These empirical equations show that the losses in a wide range of pump sizes and specific speeds cannot be covered by simple power laws - a fact that must be taken into account when converting the efficiencies from a model to a prototype with larger diameter and/or speed, see Chap. 3.10.

The efficiencies of multistage industrial pumps according to Fig. 3.28 and Eq. (T3.9.2) are generally below the values of single-stage pumps because of additional losses in the return channels. However, in the case of storage pumps which are built less compact than pumps for industrial applications similar efficiencies as in Fig. 3.26 are achieved.

Double-entry pumps according to Fig. 3.29 and Eq. (T3.9.4) with $n_q < 40$ achieve higher efficiencies than single-entry impellers *of the same specific speed*, since the disk friction losses, referred to the power, are only half as high. Furthermore, the losses in the volute casing, which has been designed for twice the flow rate, turn out to be smaller.

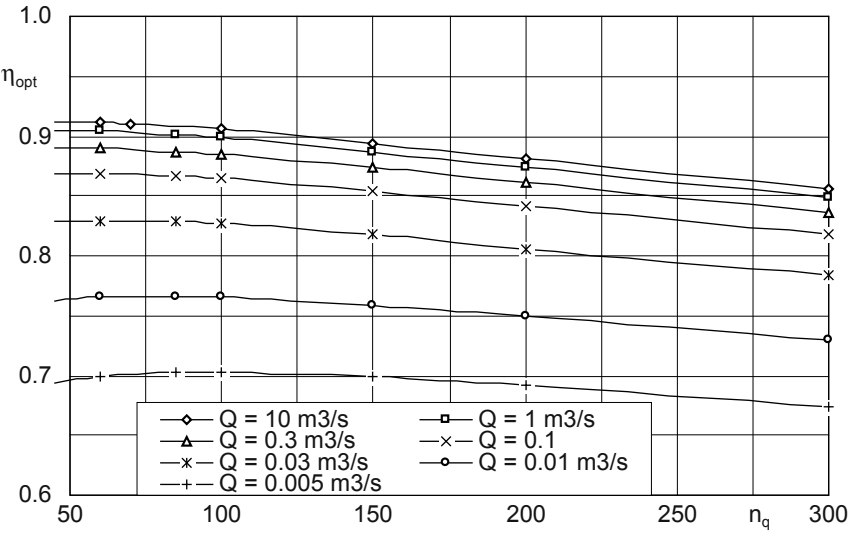


Fig. 3.27 Efficiencies of semi-axial and axial pumps

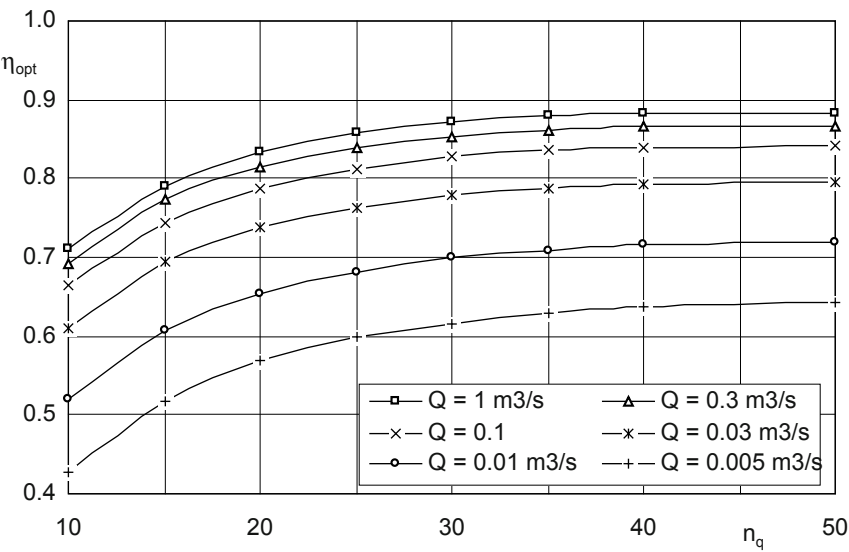


Fig. 3.28 Efficiencies of multistage, single-entry, radial pumps

The hydraulic efficiencies given in Figs. 3.30, 3.31 and 3.32 and Eqs. (T3.9.6) to (T3.9.8) also depend on the specific speeds and the flow rates (or pump sizes and speeds). The maxima are less pronounced than with the overall efficiency. They are caused by two tendencies: (1) The friction and deceleration losses in the diffusing

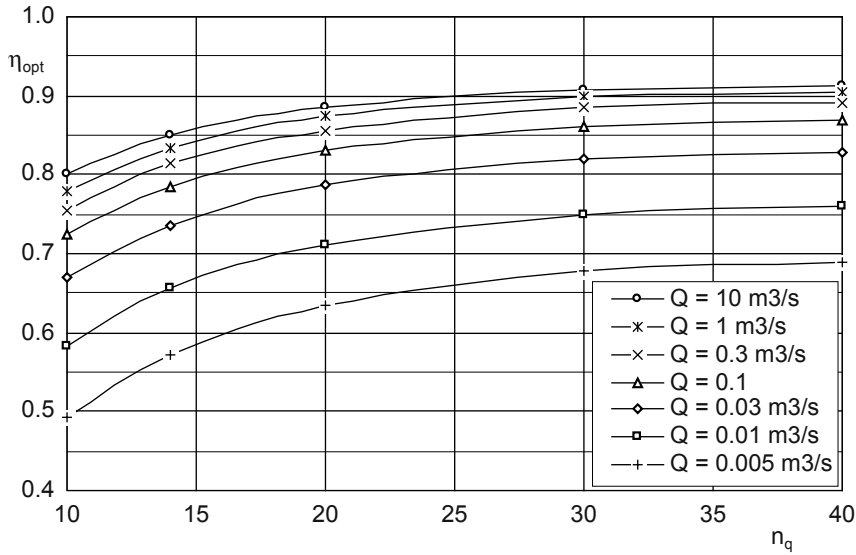


Fig. 3.29 Efficiencies of single-stage pumps with double-entry impeller. Q corresponds to Q_{opt} of the pump, while n_q is calculated with the flow *per impeller side*, Eq. (T3.4.15)

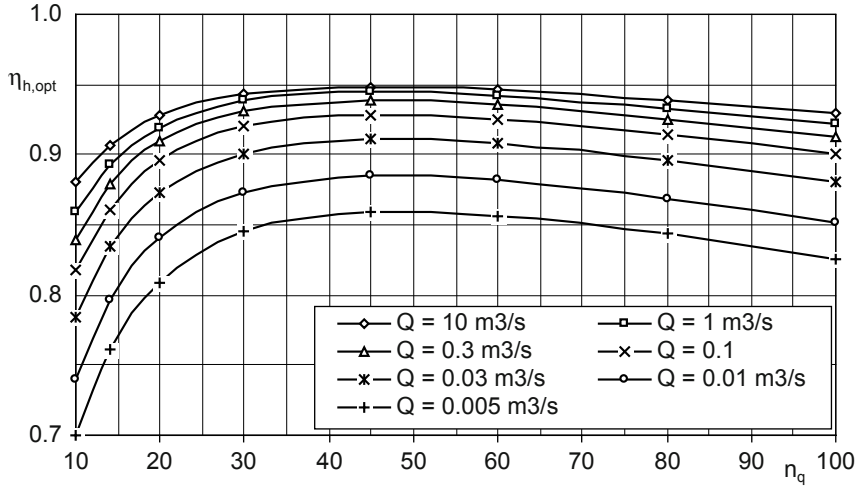


Fig. 3.30 Hydraulic efficiencies of single-stage, single-entry, radial pumps

elements increase strongly at low specific speeds; (2) increasing mixing losses are responsible for the drop of the hydraulic efficiency above $n_q = 70$ (as in the case of the overall efficiency).

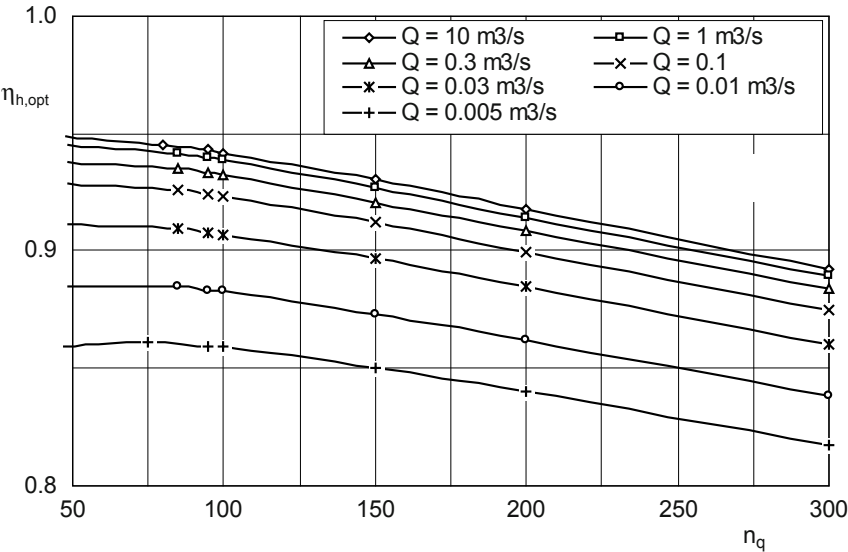


Fig. 3.31 Hydraulic efficiencies of semi-axial and axial pumps

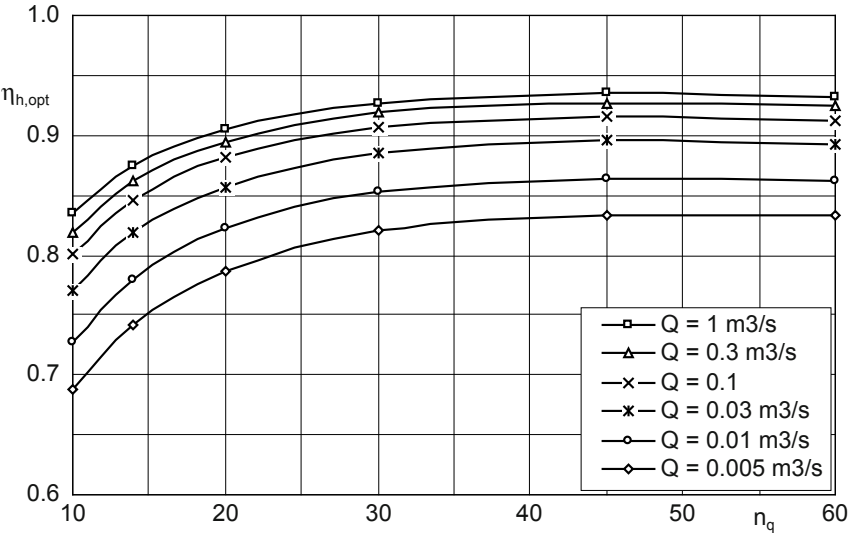


Fig. 3.32 Hydraulic efficiencies of multistage, single-entry, radial pumps

The hydraulic efficiency of very small pumps decreases due to increased *relative* roughness, since the absolute roughness remains more or less constant (it is given by the casting process). In addition, the narrow channels cannot be dressed well. Finally, the economic incentive to improve the efficiency of small pumps is less significant.

When it is difficult to make a reasonable estimate of the hydraulic efficiency, Eq. (3.28a) may be used. It can be particularly helpful with small pumps and at partload.

$$\eta_h = \frac{1}{2}(1 + \eta) \quad (3.28a)$$

Equation (3.28a) supplies realistic estimates even at $Q = 0$ and at partload.¹²

The efficiencies determined according to Table 3.9 or Figs. 3.26, 3.27, 3.28, 3.29, 3.30, 3.31 and 3.32 are subject to a scatter in the magnitude of $\Delta\eta = \pm 0.2 \times (1 - \eta)$. Deviations can be due to the quality of the hydraulic design, seal clearances, axial thrust balancing and mechanical equipment. If sufficient test data of one or several pump ranges are available, the coefficients of the equations in Table 3.9 can be easily adjusted so that the data can be correlated with a smaller standard deviation.

Figure 3.33 shows how the secondary losses and the hydraulic losses in the impeller and the diffusing elements depend on the specific speed. These figures illustrate tendencies and relationships but can also be useful for a rough evaluation. At $n_q < 25$, for example, an optimum design and careful dressing of the diffusing elements is most important. At $n_q < 20$, seal clearance and geometry have to be optimized and the disk friction has to be minimized (as far as these can be influenced). At $n_q > 50$, however, the impeller losses deserve greater attention because of the non-uniform flow distribution. Test data discussed in Chap. 4.1.3 confirm the trends and magnitudes presented by Fig. 3.33.

Since the losses depend on size, type and surface finish of the pump, Fig. 3.33 is obviously unsuitable for an exact loss analysis, for which Chap. 3.6 and Tables 3.5–3.9 should be used.

In order to assess the efficiency potential of a pump, the concept of an “achievable efficiency” may be used. It constitutes the efficiency which would be expected if the pump under consideration were designed and finished in an optimum way. An “achievable efficiency” can be defined with various assumptions and methods depending on the goals of the investigation and the issues at stake. The information on achievable efficiencies can be quite misleading, if mechanical constraints (e.g. safe seal clearances), reliability issues and the impact on costs are disregarded. Therefore, it may often be more useful to perform an analysis as to how sensitively the efficiency reacts to all the parameters which determine the hydraulic and the secondary losses.

Based on the information in Tables 3.5–3.8 the achievable efficiency can be determined (for example) with the following assumptions:

- (1) Calculate the disk friction loss as hydraulically smooth assuming that the lower limit is at 75% of the calculated loss.

¹²The relationship $\eta_h = \eta^{0.5}$ which is sometimes employed does not furnish any useful values either at very low overall efficiencies (small pumps) or at low partload.

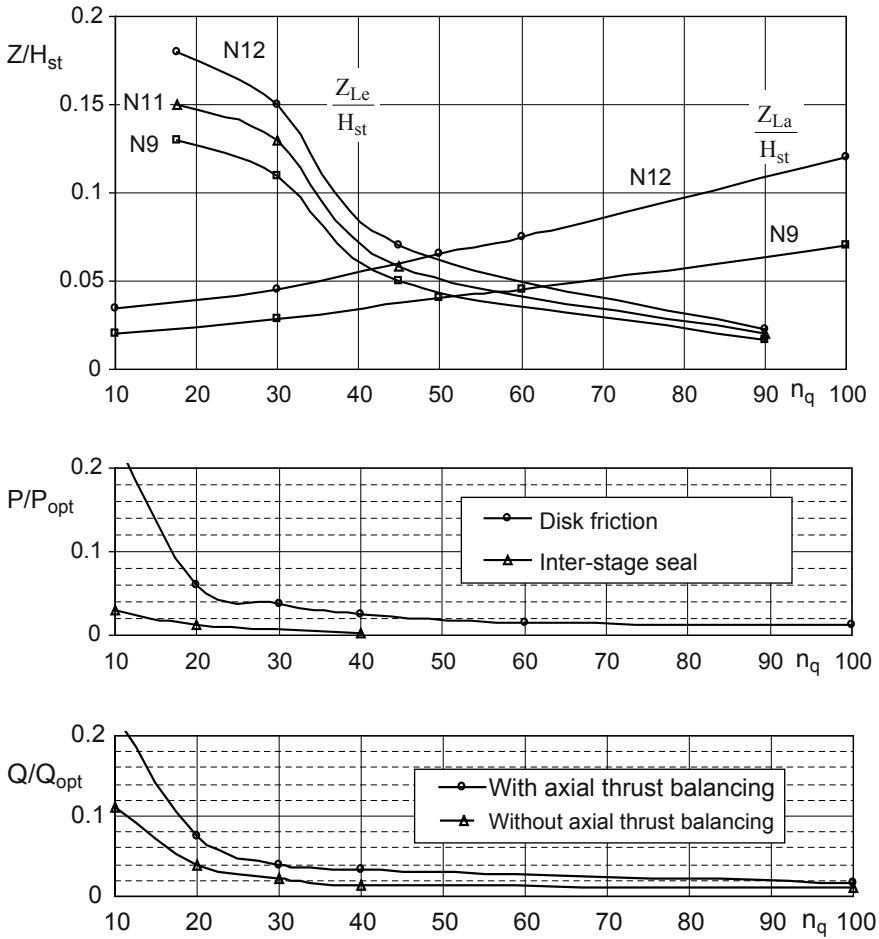


Fig. 3.33 Various losses as functions of the specific speed

- (2) Determine the leakages for the minimum seal clearances and assume only 70% of the calculated value as the lower limit of the correlations used. In this context, it is also possible to examine the effect of a different annular seal design: seal length, stepped seal, grooves or isotropic pattern.
- (3) The hydraulic losses are calculated for hydraulically smooth surfaces and $c_d = c_f$ is inserted in Eq. (T3.8.6). This means the impeller is considered to be without the detrimental effects of deceleration and secondary flow.
- (4) Shock losses in the impeller and diffuser are also assumed as zero.

Theoretical investigations concerning the maximum achievable efficiency of single-stage volute casing pumps are discussed in [23]. This topic is also discussed in detail in [B.21]. The results from [23] can be reflected through Eq. (T3.9.5) if the values for η_{opt} from Fig. 3.26 or from Eq. (T3.9.1) are inserted. The “achievable

efficiency” depends on the design of the pump. In practice it is greatly influenced by considerations of mechanical design, cost and reliability. Careful analysis of all losses and the potential for improvement of an actual pump should therefore follow a comparison with theoretical achievable values.

3.10 Influence of Roughness and Reynolds Number

3.10.1 Overview

The losses in a centrifugal pump, and consequently its efficiency, depend on the roughness of the components and the Reynolds number Re . Re covers the influences of pump size, speed and viscosity. The knowledge of these relationships is of great practical importance for:

1. Determining the efficiency of the prototype (subscript “a”) $\eta_a = f(Re_a, \varepsilon_a/d_a)$ based on the efficiency $\eta_M = f(Re_M, \varepsilon_M/d_M)$ measured in the model test (subscript “M”).
2. Determination of the efficiency expected in the plant based on the efficiency measured in the works test at a lower speed and/or a different viscosity. A typical application would be a boiler feed pump which was measured on the test stand at 1500 rpm with water of 20 °C but is operated at 6000 rpm with water of 180 °C in the power station.
3. Estimation of the characteristics for operation with fluids of high viscosity (e.g. such as oil with $3000 \times 10^{-6} \text{ m}^2/\text{s}$) based on tests with cold water.
4. Evaluation of the expected efficiency gain upon reduction of the roughness.
5. Influence of the roughness in various components (impeller, diffuser, impeller sidewalls and casing) on the efficiency. Or the answer to the question where, in a specific case, additional effort promises the greatest chance of success to improve the efficiency. These effects depend on the specific speed n_q and the pump type.

In the following it is generally assumed that the characteristics and efficiency $\eta_a = f(Re_a, \varepsilon_a/d_a)$ of an *application* (subscript “a”) or prototype are to be determined based on a *model* or *basic* test (subscript “M”) with known (measured) efficiency $\eta_M = f(Re_M, \varepsilon_M/d_M)$ and known characteristics (the “basic test” could, of course, also be a sales curve for example).

Points 1 and 2 are extensively covered in the literature under keywords such as “efficiency scaling”, “efficiency majoration” or “efficiency step-up”. Published formulae for efficiency scaling suffer from a lack of generality because they do not cover the effects of the specific speed and of the roughness. The special aspects of pumping high-viscosity media are covered in Chap. 13.1. On the basis of a simplified loss analysis it is possible to tackle the five tasks mentioned above through a common approach which is discussed in Chap. 3.10.3, while the comparison with test data is documented in [16].

3.10.2 Efficiency Scaling

Efficiency scaling means the prediction of the efficiency of a given pump (“prototype”) from a test done with a *geometrically similar* “model” with different size and/or Reynolds number.

As has been shown in Chaps. 3.6 and 3.8 the disk friction, the leakage losses and a portion of the hydraulic losses (and therefore also the hydraulic and the overall efficiencies) depend on the Reynolds number and the relative roughness. There are different possibilities for evaluating this Reynolds influence or for scaling the efficiencies:

1. Determine the effective mechanical losses and the secondary losses separately for model and prototype according to Tables 3.6 and 3.7. Likewise, calculate the *difference* in hydraulic efficiency between model and prototype according to Table 3.9. Add this difference to the hydraulic efficiency of the model. In this way it is possible to determine the power consumption of the machine according to Eq. (T3.5.1) and its efficiency according to Eq. (T3.5.3). Note that the mechanical losses often differ between model and prototype since different bearings and shaft seals are employed (as mentioned $P_m \sim n^x$ with $x = 1.3$ to 1.8 is applicable and not $P_m \sim n^3$ as for scaling the internal power of the pump).
2. The separate evaluation of all losses according to Chap. 3.10.3 largely corresponds to the procedure suggested under item 1. Scaling the efficiency in this way, is presumably the most accurate method at present, since all relevant design- and plant-specific differences between model and prototype can be taken into account.
3. Proceed as under (1), but calculate the *differences* in the hydraulic losses according to Table 3.8 as caused by the differences in the Reynolds numbers and/or in the relative roughness of model and prototype. Note that the absolute values of the hydraulic losses calculated would be too inaccurate since the mixing losses cannot be evaluated adequately.
4. Use of “majoration formula” (see below).
5. Figures 3.26, 3.27, 3.28 and 3.29 and the equations in Table 3.9 provide a relation between overall efficiency and flow rate Q_{opt} . Since $Q \sim n \times d^3$, such a representation indirectly covers the effect of the pump size and the speed on the efficiency. If the viscosity is constant, this means that the effects of the Reynolds number ($Re \sim n \times d^2$) and the relative roughness are taken into account, provided that the absolute roughnesses of model and prototype are identical. Consequently, the efficiency can be scaled on the basis of Table 3.9 by employing $\eta_a = \eta_M + \Delta\eta$. $\Delta\eta$ is the difference between the efficiencies calculated for model and prototype according to Eqs. (T3.9.1–3.9.4). When proceeding thus, the effect of surface roughness can be taken into account by Eq. (T3.9.1A).
6. The hydraulic efficiencies of model and prototype can be established using numerical methods. The rest of the procedure is the same as in step (1).

Majoration equations: Various equations for efficiency scaling have been published. Overviews can be found in [20, 30–32, 43]. The more relevant equations for *internal* efficiency η_i allow a division into “scalable” (i.e. Reynolds-dependent) and non-scalable losses. Disk friction losses and that portion of hydraulic losses which is caused by friction are “scalable”. Both losses fall with rising Reynolds number and both drop if the relative roughness ε/d_h decreases. These effects make up the efficiency scaling in its proper sense. However, the leakage losses have an opposite tendency: they increase with Re and drop with growing relative roughness ε/d_h .

As is shown in Figs. 3.26, 3.27, 3.28, 3.29, 3.30, 3.31, 3.32 and Table 3.9, changes in efficiency and loss depend on the specific speed and the pump size. Doubling the design flow rate gives rise to quite different efficiency corrections depending on specific speed and design flow rate. The published efficiency scaling formulae don’t do justice to this fact (or only incompletely so). In the light of these considerations it is quite sensible to scale efficiencies according to Table 3.9 as described under point (5).

As mentioned before, the hydraulic losses are composed of friction and mixing losses: Only the friction losses are primarily scalable. Different scaling laws result depending on how the Reynolds numbers and the relative roughnesses of the model and prototype are related with respect to hydraulically smooth or rough behavior. If the surfaces of model and prototype are hydraulically smooth, the scaling follows the relation Re^x , and the exponent drops with increasing Re number. Impeller diameter, speed and viscosity enter into the Reynolds number in the same way. In contrast, if both machines are situated in the fully rough range, the relative roughness has an impact on the efficiency while the Reynolds number has no effect. These relationships become clear from Fig. 1.13, see the points marked M and A. *Majoration formulae with constant exponents of the Reynolds number can offer an approximation only if roughness effects are weak.*

This difficulty can be avoided by the approach implied by Eqs. (T3.9.11 and 3.9.12): the friction coefficients $c_f = f(Re, \varepsilon/L)$ are determined from Eq. (1.33) or Fig. 1.13 for the model and prototype and the hydraulic efficiency is scaled by means of the ratio $c_{f,a}/c_{f,M}$ as given by Eq. (T3.9.11). The coefficient $\zeta_{R,M}$ comprises all friction losses in the pump (especially in the impeller and the diffusing elements). The friction losses can be estimated from the given correlation $\zeta_{R,M}/\psi$, which was derived from calculating various pumps according to Table 3.8. Alternatively, $\zeta_{R,M}/\psi$ can be determined for the case to be examined by means of the equations in Table 3.8.

The disk friction can be treated in a similar manner: The friction coefficients k_{RR} are determined as a function of the Reynolds number and roughness for the model and prototype from Eq. (T3.6.3). Next, the internal efficiency is scaled according to Eq. (T3.9.12) in which the effect of the disk friction and the hydraulic losses are averaged. When scaling the internal and the hydraulic efficiencies, the distribution of the scalable and the Reynolds-independent losses is estimated as a function of the specific speed from Eq. (T3.9.12). Finally, Eq. (T3.9.12) can also be applied to the overall efficiency if the difference in the mechanical losses is neglected.

Equation (T3.9.13) provides the scaling formula according to [N.5]; here, an efficiency η_{hR} which comprises the hydraulic and the disk friction losses is defined and scaled. Annular seal losses and mechanical losses are calculated separately for the model and the prototype and are considered in $\eta_{hR,M}$ and $\eta_{hR,a}$.

The scaling of efficiencies from the model to the prototype harbors a number of uncertainties, which are discussed in Chap. 3.10.3. As a rule, a careful loss analysis according to Chap. 3.10.3 will supply the most accurate prediction, but scaling formulae - such as Eq. (T3.9.11A) - have their justification for rapid estimations. They are important, for example, when contractual formulae for evaluating the guaranteed efficiencies have to be agreed upon between the pump buyer and the manufacturer since the scaling procedure must apply equally to all project bidders.

3.10.3 Calculation of the Efficiency from Loss Analysis

A complete set of equations (Table 3.10) is developed in this chapter by means of which the effect of Reynolds number and roughness on efficiency can be estimated. The process applies to turbulent and laminar flow over hydraulically smooth or fully rough surfaces and covers. It smoothly covers all transitions between the different flow regimes.

The efficiency conversion takes into account the hydraulic and volumetric losses as well as disk friction. These losses depend on the Reynolds number, the roughness and the specific speed in different ways. All the necessary equations are compiled in Table 3.10.

Factors for efficiency conversion: The examination is performed for the best efficiency point. Power consumption and efficiency can be expressed in simplified form (without recirculation) according to Eq. (T3.5.1) as follows:

$$P = \frac{\rho g H z_{st} Q}{\eta_{vol} \eta_h} + P_{RR} + P_m \quad \eta = \frac{P_u}{P} = \frac{\rho g H z_{st} Q}{P} \quad (3.29)$$

In Eq. (3.29) the effect of the interstage seals in multistage pumps is taken into account by a weighted volumetric efficiency defined by Eq. (T3.10.4). Furthermore, the disk friction power P_{RR} in Eq. (3.29) comprises the sum of *all* components of the pump rotor, i.e. it includes the friction at the axial thrust balance device (P_{er}). The efficiency according to Eq. (3.29) can then be expressed by:

$$\eta = \frac{\eta_{vol} \eta_h}{1 + \eta_{vol} \eta_h \left(\frac{P_{RR}}{P_u} + \frac{P_m}{P_u} \right)} \quad (3.30)$$

The quantities η_{vol} , η_h and P_{RR} depend on the Reynolds number and the roughness, while P_m/P_u is assumed to be a constant. To recalculate the pump characteristics

from the base or the model (“M”) to the application (“a”), the factors according to Eq. (3.31) are introduced. These factors *only* take into account the *deviations* from the model laws. Changes in speed and/or impeller diameter, if any, have to be calculated from Table 3.4 before this analysis.

$$f_{\eta} = \frac{\eta_a}{\eta_M} \quad f_{\eta h} = \frac{\eta_{h,a}}{\eta_{h,M}} \quad f_{\eta \text{vol}} = \frac{\eta_{\text{vol},a}}{\eta_{\text{vol},M}} \quad f_{\eta m} = \frac{\eta_{m,a}}{\eta_{m,M}} \quad (3.31)$$

$$f_Q = \frac{\varphi_a}{\varphi_M} \quad f_H = \frac{\psi_a}{\psi_M} \quad \text{or} \quad f_Q = \frac{Q_a}{Q_M} \quad f_H = \frac{H_a}{H_M} \quad (3.32)$$

The disk friction losses in the prototype and in the model amount to:

$$P_{RR,a} = k_{RR,a} \rho_a \omega^3 r_2^5 f_{\text{geo}} \quad \text{respectively} \quad P_{RR,M} = k_{RR,M} \rho_M \omega^3 r_2^5 f_{\text{geo}} \quad (3.32a)$$

The corresponding useful power is $P_{u,a} = \rho_a \times g \times H_a \times Q_a$ and $P_{u,M} = \rho_M \times g \times H_M \times Q_M$.

With f_Q and f_H from Eq. (3.32), it follows that $P_{u,A} = P_{u,M} \times f_H \times f_Q \times \rho_a / \rho_M$. With these relationships the following is obtained:

$$\left(\frac{P_{RR}}{P_u} \right)_a = \left(\frac{P_{RR}}{P_u} \right)_M \frac{k_{RR,a}}{f_Q f_H k_{RR,M}} \quad \text{with} \quad \frac{P_{u,M}}{P_{u,a}} = \frac{1}{f_Q f_H} \quad (3.33)$$

If Eq. (3.30) is written for the application (“a”) and the model (“M”) and the resulting equations are divided by one another, the multiplier for the efficiency is obtained by virtue of Eqs. (3.31–3.33) as follows:

$$f_{\eta} \equiv \frac{\eta_a}{\eta_M} = \frac{f_{\eta h} f_{\eta \text{vol}} \left[1 + \left\{ \left(\frac{P_{RR}}{P_u} \right)_M + \left(\frac{P_m}{P_u} \right)_M \right\} \eta_{\text{vol},M} \eta_{h,M} \right]}{1 + \left\{ \left(\frac{P_{RR}}{P_u} \right)_M \frac{k_{RR,a}}{k_{RR,M}} + \left(\frac{P_m}{P_u} \right)_M \frac{\rho_M f_{\eta m}}{\rho_a} \right\} \frac{f_{\eta h} f_{\eta \text{vol}}}{f_H f_Q} \eta_{\text{vol},M} \eta_{h,M}} \quad (3.34)$$

This equation covers all the losses. If the geometry of the annular seals is known, the Re-dependency of the volumetric efficiency can be calculated according to Table 3.7 and taken into account in the form of $f_{\eta \text{vol}}$. The following assumptions and simplifications were made: (1) $f_{\eta \text{vol}} = f_{\eta m} = 1.0$; $\rho_a = \rho_M$; $f_H = f_{\eta h}$. (2) When pumping viscous fluids with viscosities above $(50 \text{ to } 100) \times 10^{-6} \text{ m}^2/\text{s}$ the best efficiency point moves more or less along the volute characteristic (Chap. 4.2). Hence, we can set $f_Q = f_H = f_{\eta h} < 1$ (this is discussed in detail in Chap. 13). Under these conditions a somewhat simplified equation for the efficiency conversion is obtained, Eq. (T3.10.20).

Factors for head conversion: The assumption $f_H = f_{\eta h}$ is necessary since the factor f_H (in contrast to f_{η}) cannot be determined directly from a test, as shown below.

Exceptions are tests with a viscosity sufficiently high so that the following interference effects are masked:

1. With varying Reynolds number, the seal leakage changes, resulting in a slight shift of the Q-H curve since the flow rate through the impeller $Q_{La} = Q/\eta_{vol}$ changes, Fig. 4.29.
2. The impeller shrouds have a pumping effect similar to that of a friction pump. Boundary layer fluid with $c_{2u} \approx u_2$, flung off the impeller shrouds, contributes to the transfer of energy. The relative contribution of this effect grows with decreasing specific speed, falling Reynolds number and increasing roughness.
3. An *increase* in head was also repeatedly measured when the roughness of the impeller channels was increased. Greater roughness implies a deceleration of the relative velocity near solid walls and a thickening of the boundary layer. A lower *relative* velocity however means a higher *absolute* velocity and consequently an increase of the slip factor and the theoretical head. This increase of H_{th} can (but not always does) exceed the additional losses brought about by the roughness. The same mechanism is active if the viscosity is increased.

Some test results relating to this increase in head are quoted below:

- Double-entry impeller $n_q = 10$, a change of roughness from $\varepsilon = 0.025$ mm to 0.87 mm in the channels and the outside of the impeller shrouds [39]: head increase $f_H = 1.1$ with efficiency loss $f_\eta = 0.84$
- Single-entry impeller, $n_q = 7$, a change of roughness from $\varepsilon = 3.7$ μ m to 46 μ m [15]: increase in head $f_H = 1.01$
- Even with a semi-axial impeller $n_q = 135$ a slight increase in head was measured with increased roughness.
- Tests in [24] showed an increase in head with viscosities up to 45×10^{-6} m²/s; a drop in head was only measured at 100×10^{-6} m²/s. However, this effect cannot be separated from the shift of the Q-H-curve caused by a reduced seal leakage.
- A slight increase in head was observed also in the measurements in [26] and [37] on single-stage pumps with $n_q = 12$ and 20 respectively.

The higher the quality of the hydraulic design (i.e. the impeller operating with thin boundary layers and without separation), the greater is the effect of the roughness on ψ_{th} . In contrast to this, a low-quality hydraulic design with separated flow is less affected by the surface roughness.

Neither the shift of the Q-H-curve due to a change in the annular seal flow, nor the pumping action of the impeller shrouds, nor the change of the slip factor due to changes in the boundary layer thickness or the secondary flows in the impeller can be separated from the friction losses. Attempting to determine the factor f_H directly from a test is therefore too inaccurate.

Hydraulic losses: As explained in Chap. 3.8, the hydraulic losses consist of friction $\zeta_R = f(\text{Re}, \varepsilon)$ and mixing losses ζ_M . All of the friction and mixing losses from inlet, impeller and diffusing elements are included in these two quantities. The theoretical head coefficient in the prototype and in the model can therefore be written as follows (refer also to Table 3.8):

$$\psi_{\text{th}} = \psi_M + \zeta_{R,M} + \zeta_{M,M} = \psi_a + \zeta_{R,a} + \zeta_{M,a} \quad (3.35)$$

In Eq. (3.35) it is assumed that the slip factor, and consequently the theoretical head, remain roughly constant, refer also to Chap. 13.1. If the blade work remains constant, the ratio f_H of the heads must be equal to the ratio of the hydraulic efficiencies, resulting in $f_H = f_{\eta h}$.

If the mixing losses are assumed to be independent of the Reynolds number, $\zeta_{M,M} = \zeta_{M,a}$ in Eq. (3.35) cancel so that a relation between ψ_a and ψ_M in the form of the multiplier $f_{\eta h}$ can be derived:

$$f_{\eta h} = \frac{\eta_{h,a}}{\eta_{h,M}} = 1 - \frac{\zeta_{R,M}}{\psi_M} \left(\frac{\zeta_{R,a}}{\zeta_{R,M}} - 1 \right) = 1 - \frac{\zeta_{R,M}}{\psi_M} \left(\frac{c_{f,a}}{c_{f,M}} - 1 \right) \quad (3.36)$$

Equation (3.36) applies at *constant* flow rate.

If the fraction $\zeta_{R,M}/\psi_M$ of the friction losses in the head is known, Eq. (3.36) can be used to estimate how the head will change with a different roughness and/or Reynolds number. To do so, it is assumed that the friction losses $\zeta_{R,a}/\zeta_{R,M}$ are proportional to the corresponding friction coefficients $c_{f,a}/c_{f,M}$. These friction coefficients depend on the Reynolds number and the roughness. They can be calculated for turbulent and laminar flows according to Eqs. (1.33 and 1.33a).

The portion of the friction losses in the hydraulic losses can be estimated for a given pump according to Table 3.8 and Chap. 3.8. It depends on the specific speed, the pump type and the geometry of the hydraulic components. An estimation of $\zeta_{R,M}/\psi_M$ according to Table 3.8 is complicated and afflicted with uncertainties. For this reason an empirical approach according to Eq. (T3.10.18) was developed on the basis of a large number of tests, [15].

Secondary losses: The disk friction losses are calculated according to Table 3.6 and Chap. 9.1, from Eq. (T3.10.9). The influence of different roughnesses on the casing walls and impeller shrouds can be captured by means of Eq. (T3.10.8). Expeller vanes on the rear shroud, if any, are taken into account by Eq. (T3.10.10), the effect of the leakage through the impeller sidewall gap by Eq. (T3.10.12), the roughness of the impeller disks by Eq. (T3.10.7) and the heating of the fluid in the impeller sidewall gap in case of very high viscosity ($\nu > 400 \times 10^{-6} \text{ m}^2/\text{s}$) by the empirical factor f_{therm} , which is discussed in Chap. 13.1. The volumetric losses are estimated from Eq. (T3.10.4) and the mechanical losses from Eq. (T3.10.14).

Roughness: According to Eq. (T3.10.5) and Chap. 1.5.2 the equivalence factor $c_{eq} = 2.6$ is used to assess the roughness of the various components while the maximum roughness is assumed to be 6-times the average roughness. For simplification the Reynolds number for impeller shrouds and hydraulic channels is calculated with r_2 . The effect of these assumptions is mitigated because only the ratios of friction coefficients occur in Eqs. (T3.10.8, 3.10.19 and 3.10.20). The roughnesses for impeller and collector are averaged according to the empirical expression given by Eq. (T3.10.6). In doing so, a weighted average of the roughness is calculated which depends on the specific speed, [15].

In [15] a collection of 32 tests with different roughnesses and Reynolds numbers were analyzed according to the procedure given in Table 3.10. The empirical coefficients for calculating the relevant mean roughness from Eq. (T3.10.6) and the component of the scalable hydraulic losses from Eq. (T3.10.18) were optimized in order to minimize the standard deviation. The investigation in [15] covers the range: $n_q = 7$ to 135; $d_2 = 180$ to 405 mm; $u_2 = 22$ to 113 m/s; $n = 1200$ to 7000 rpm; $T = 20$ to 160 °C; $Re = 2.5 \times 10^6$ to 9.1×10^7 ; equivalent sand roughness $\varepsilon = 1$ to 130 μm and average roughness $\varepsilon_{CLA} = 0.4$ to 75 μm .

The standard deviation between measurement and calculation for f_η is $\pm 1.0\%$ and for $f_{\eta h}$ it is $\pm 1.5\%$. The scatter increases with decreasing specific speed. A variation of the equivalence factor c_{eq} yielded no improvement; $c_{eq} = 2.6$ is therefore a useful assumption.

The relationship $a_e = 0.98 - 0.0012 n_q f_q^{0.5}$ found for weighting the roughness in the impeller and collector shows that the effect of the roughness in the impeller is quite weak in comparison with the diffusing elements, even at high specific speeds. The portion of scalable losses also drops with increasing specific speed, Eq. (T3.10.18). Both functions are shown in Fig. 3.34. Equation (T3.10.18) supplies quite similar values as the expression (1-V) in Eq. (T3.9.12).

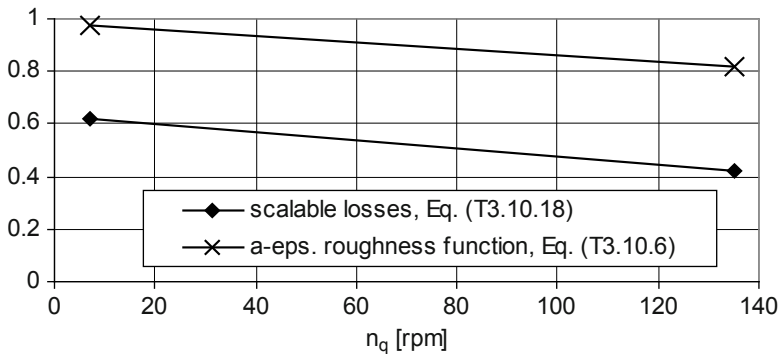


Fig. 3.34 Weighting the roughness a_e acc. to Eq. (T3.10.6) and fraction of scalable losses acc. to Eq. (T3.10.18): $a_1-b_1 \times n_q \times f_q^{0.5}$

Uncertainties of efficiency calculation: Depending on the task set (efficiency scaling, assessment of roughness effects or calculation for highly viscous media), determining the efficiency is subject to a series of uncertainties:

1. The scaling concerns minor differences between two (relative to the scaling amount) large numbers. Even if the efficiency is measured with great accuracy in the test, the uncertainty in the efficiency *differences* $\Delta\eta = \eta_a - \eta_M$ determined on model and prototype is considerable; more so since η_a and η_M generally are measured with different instruments in different test circuits. The experimental verification of scaling formulae is therefore very difficult.
2. The casting and manufacturing tolerances of model and prototype are different.
3. The annular seal clearances are usually selected as functions of the size. They are not always strictly geometrically similar to the model. Even the seal design may not be geometrically similar.
4. For economic reasons, minor design differences are often unavoidable. They can be caused, for example, by the annular seal geometry, the impeller fixation on the shaft, the impeller sidewall gaps, or the inlet or outlet chamber.
5. The calculation of the disk friction losses and annular seal leakages is quite uncertain.

Like turbulence, the roughness calls for a statistical rather than a deterministic description; *the fluid-dynamical relevant quantification of the roughness is one of the main difficulties in determining the efficiency:*

- The effect of roughness on the flow cannot be determined, not even by accurate measurement of the surfaces, which is rarely attempted in practice.
- Frequently the roughness varies locally in the various channels depending on the accessibility for dressing or machining.
- The losses due to roughness are caused by the interactions of the velocity profile near the wall and the turbulence with the roughness elevations. The fine structures of roughness and turbulence are, consequently, responsible for the losses. The interaction between roughness and turbulence is determined on the one hand by the size, shape and number of roughness elevations and, on the other hand, by the size and frequency of the turbulence eddies near the wall. The turbulence structure depends on the local velocity distribution as given by accelerated or decelerated flow, Coriolis and centrifugal forces and flow separation (refer also to Fig. 1.15, Chap. 1.5.2 and [16]). An example of the interaction between roughness and turbulence is the increase in pressure drop caused by the “ripple roughness” quoted in Chap. 1.5.2. Another example is the reduction of the disk friction through fine grooves in circumferential direction (Chap. 3.6.1).
- If the pressure decreases in the flow direction ($dp/dx < 0$), the losses increase with growing wall shear stress (or roughness). In decelerated flow, however, with $dp/dx > 0$ (e.g. in a diffuser) the wall shear stress is reduced with increasing roughness and becomes $\tau_w \approx 0$ during separation. The “friction losses”

therefore decrease with increasing roughness, while the “mixing losses” and the overall losses increase. The concept of distinguishing between “friction” and “form or mixing losses” therefore does not make much sense in decelerated flow.

3.11 Minimization of Losses

In practice, the efficiency achievable is influenced by the boundary conditions imposed by the specific application and the mechanical design. Pump size, the required suction specific speed, demands as to the shape of the Q-H-curve and the manufacturing costs also have a considerable impact. Any attempt at increasing the efficiency of a pump which already reflects the state of the art has to rely on introducing at several locations rather small improvements which sum up to yield the desired effect. To minimize losses, it is necessary to first analyze the secondary losses and the hydraulic losses of the existing pump or the design at hand using Tables 3.5–3.8. At the same time these tables serve to quantitatively evaluate the potential of improvement by possible modifications.

Hydraulic losses (refer also to Chaps. 7 and 8):

- A uniform approach flow reduces the losses in the impeller.
- Optimize blade loading and pressure distribution. Excessively long blades cause friction losses, too short blades mixing losses.
- An excessively large impeller outlet width causes mixing losses.
- An excessively large inlet diameter causes increased mixing, friction and leakage losses through the annular seal and impairs the partload performance.
- In the case of diffusers and volute casings, the diffuser must be optimized according to Chap. 1.6.
- Carefully optimize the flow deceleration in diffusing channels: (1) Design for steady, efficient deceleration downstream of the diffuser or volute throat area where velocities (and friction losses) are high; (2) Avoid a sudden increase of the diffuser opening angles near the diffuser outlet where boundary layers are thick and at the brink of stall.
- Avoid flow separation, sharp curvature and unsteadiness in the development of curvature, angles and flow areas.
- Reduce roughness so that hydraulically smooth characteristics are obtained.
- When replacing an oversized impeller in an existing pump by a retrofit impeller adapted to lower flow, the volute or diffuser remains too large (unless it would be modified too). The increase in efficiency expected from the original performance curves will not materialize.

Secondary losses (refer also to Chap. 3.6):

- Disk friction: Avoid ribs or cavities in the impeller sidewall gaps since these would slow down the rotation of the fluid. Avoid rough walls and large impeller sidewall gap volumes.
- At low specific speeds the disk friction losses can be noticeably reduced by selecting a high head coefficient, but the effects on the Q-H-curve according to Chaps. 4 and 5 have to be considered.
- Annular seal losses: A reduction of the annular seal clearance would be most beneficial, but the reliability of the machine must not be jeopardized under any circumstances. If very tight clearances are selected, it must be ensured that no damage can result from contact between rotor and stator. The risk of seizure can be mitigated by the correct choice of materials or the surface structure of the annular seal, Chap. 14.4.4. Tight seal clearances that rapidly wear in operation are of little benefit to the operator. Fine grooves, hole or honeycomb patterns reduce the annular seal loss. Increasing the length of the annular seal usually brings only a moderate reduction in leakage.
- Flow through the axial thrust balance device: In the case of multistage pumps a balance disk is superior to a balance piston in terms of efficiency. However, a balance piston is less sensitive and therefore in many applications it is more reliable than a disk.
- Mechanical losses: Anti-friction bearings cause fewer losses than plain bearings. Mechanical seals use up less power than stuffing boxes. However design aspects usually determine the choice of these elements.

3.12 Compendium of Equations for Hydraulic Calculations

All calculations discussed in Chap. 3 can be performed using the following tables. Together with the list of symbols and the dimensions and quantities defined in Tables 0.1 and 0.2, these tables are largely self-explanatory.

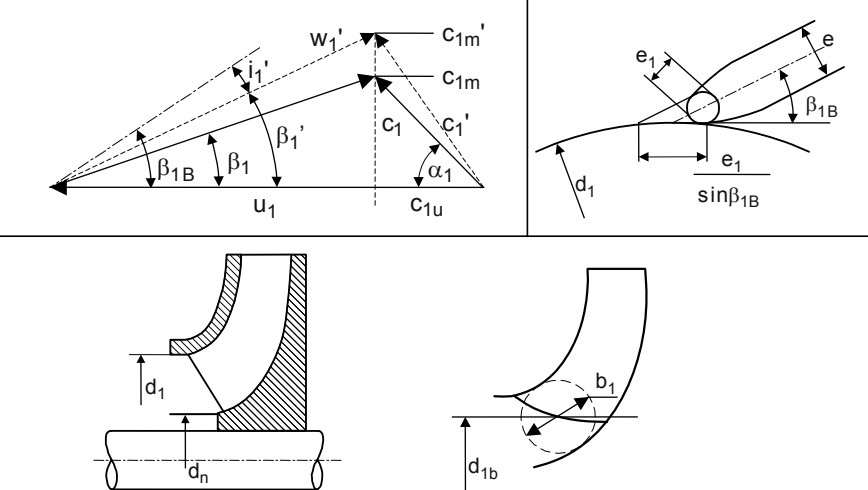
Table 3.1 Velocity triangle at impeller inlet		Eq.
Quantities given or selected	$n, Q_{La}, d_2, d_1, d_n, b_2, z_{La}, \alpha_1, e_1, e, \eta_h$	---
Circumferential speed	$u_1 = \pi d_1 n / 60$	3.1.1
Meridional component of the absolute velocity b_1 see sketch at bottom of table	$c_{1m} = \frac{Q_{La}}{f_q A_1}$	3.1.2
	$A_1 = \frac{\pi}{4} (d_1^2 - d_n^2)$ $A_1 = \pi d_{1b} b_1$	
Circumferential component of absolute velocity	$c_{1u} = \frac{c_{1m}}{\tan \alpha_1}$	3.1.3
Relative velocity	$w_1 = \sqrt{c_{1m}^2 + (u_1 - c_{1u})^2}$	3.1.4
Flow coefficient	$\varphi_1 = c_{1m} / u_1$	3.1.5
Flow angle without blockage	$\beta_1 = \arctan \frac{c_{1m}}{u_1 - c_{1u}}$	3.1.6
Blade blockage	$\tau_1 = \left\{ 1 - \frac{z_{La} e_1}{\pi d_1 \sin \beta_{1B} \sin \lambda_{La}} \right\}^{-1}$	3.1.7
Flow angle with blockage	$\beta_1' = \arctan \frac{c_{1m} \tau_1}{u_1 - c_{1u}}$	3.1.8
Blade angle for selected incidence, i_1 defined by: $i_1' \equiv \beta_{1B} - \beta_1'$	$\beta_{1B} = \beta_1' + i_1'$	3.1.9
Shockless inlet defined by $\beta_1' = \beta_{1B}$ giving c_{1m}	$\varphi_{1,SF} = \left(1 - \frac{c_{1u}}{u_1} \right) \frac{\tan \beta_{1B}}{\tau_1}$	3.1.10
		

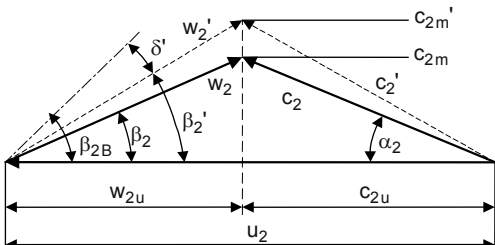
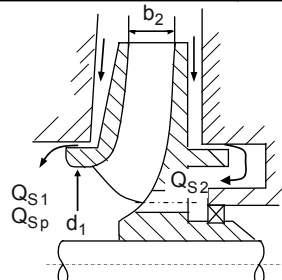
Table 3.2 Velocity triangle at impeller outlet			Eq.
Circumferential speed	$u_2 = \pi \, d_2 n / 60$		3.2.1
Meridional component of absolute velocity $A_2 = \pi \, d_{2b} \, b_2$	$c_{2m} = \frac{Q_{La}}{f_q \, A_2}$	$\varphi_{2,La} = \frac{c_{2m}}{u_2}$	3.2.2
Blade blockage	$\tau_2 = \left\{ 1 - \frac{e \, z_{La}}{\pi \, d_2 \, \sin \beta_{2B} \, \sin \lambda_{La}} \right\}^{-1}$		3.2.3
Influence of impeller inlet diameter on slip factor $k_w = 1$ for $d_{1m}^* \leq \varepsilon_{Lim}$	$\varepsilon_{Lim} = \exp \left\{ - \frac{8.16 \, \sin \beta_{2B}}{z_{La}} \right\}$		3.2.4
	$k_w = 1 - \left(\frac{d_{1m}^* - \varepsilon_{Lim}}{1 - \varepsilon_{Lim}} \right)^3$		3.2.5
Slip factor for $z_{La} \geq 3$ Radial: $f_1 = 0,98$ Semi-axial: $f_1 = 1,02 + 1,2 \cdot 10^{-3} (n_q - 50)$	$\gamma = f_1 \left(1 - \frac{\sqrt{\sin \beta_{2B}}}{z_{La}^{0.7}} \right) k_w$		3.2.6
Circumferential component of absolute velocity	Prediction	$c_{2u} = u_2 \left(\gamma - \frac{c_{2m} \, \tau_2}{u_2 \tan \beta_{2B}} \right)$	3.2.7
	Calculation from measured head H	$c_{2u} = \frac{g \, H}{\eta_h \, u_2} + \frac{u_{1m} \, c_{1u}}{u_2}$	3.2.8
Slip factor from test		$\gamma = \frac{c_{2u}}{u_2} + \frac{\varphi_{2,La} \, \tau_2}{\tan \beta_{2B}}$	3.2.9
Absolute velocity		$c_2 = \sqrt{c_{2m}^2 + c_{2u}^2}$	3.2.10
Relative velocity	$w_{2u} = u_2 - c_{2u}$		3.2.11
	$w_2 = \sqrt{c_{2m}^2 + w_{2u}^2}$		3.2.12
Absolute outlet angle without blockage		$\alpha_2 = \arctan c_{2m} / c_{2u}$	3.2.13
Relative outlet angle with blockage		$\beta_2' = \arctan c_{2m} \, \tau_2 / w_{2u}$	3.2.14
Relative outlet angle without blockage		$\beta_2 = \arctan c_{2m} / w_{2u}$	3.2.15
Deviation angle		$\delta' = \beta_{2B} - \beta_2'$ or: $\delta = \beta_{2B} - \beta_2$	3.2.16
 			

Table 3.3 Work transfer in impeller		Eq.
Specific work of blades Y _{sch} (no losses) Theoretical head: H _{th} = Y _{th} /g	Y _{sch} = g H _{th} = u ₂ c _{2u} - u ₁ c _{1u} = u ₂ ² ⎛⎜ ⎝ u ₂ - d _{1m} [*] c _{1u} u ₂ ⎞⎟⎠	3.3.1
	Y _{sch} = 1/2 ⎛⎜ ⎝ u ₂ ² - u ₁ ² + w ₁ ² - w ₂ ² + c ₂ ² - c ₁ ² ⎞⎟⎠	3.3.2
Specific work related to increase of static pressure in impeller (without losses)	Y _{P,th} = g H _{P, th} = 1/2 ⎛⎜ ⎝ u ₂ ² - u ₁ ² + w ₁ ² - w ₂ ² ⎞⎟⎠ = (p ₂ - p ₁) / ρ + g Z _{La}	3.3.3
	Y _{P,th} ≡ g H _{P,th} = u ₂ c _{2u} - u ₁ c _{1u} - 1/2 ⎛⎜ ⎝ c ₂ ² - c ₁ ² ⎞⎟⎠	3.3.4
Rise of total pressure (enthalpy) in impeller	Y _{tot,La} ≡ Δh _{tot} = (p ₂ - p ₁) / ρ + 1/2 ⎛⎜ ⎝ c ₂ ² - c ₁ ² ⎞⎟⎠ = (p _{2tot} - p _{1tot}) / ρ	3.3.5
Impeller losses	g Z _{La} = Y _{sch} - Y _{tot,La} = Y _{P,th} - (p ₂ - p ₁) / ρ	3.3.6
Head	H = (η _h u ₂ ² / g) ⎛⎜ ⎝ γ - (Q _{La} / (f _q A ₂ u ₂ tan β _{2B})) ⎡ τ ₂ + (A ₂ d _{1m} [*] tan β _{2B} / (A ₁ tan α ₁)) ⎤ ⎞⎟⎠	3.3.7
Static head rise in impeller	H _P = 1/2g ⎛⎜ ⎝ u ₂ ² - u ₁ ² + w ₁ ² - w ₂ ² ⎞⎟⎠ - Z _{La}	3.3.8
Theoretical head coefficient (without losses)	ψ _{th} = 2 ⎛⎜ ⎝ c _{2u} / u ₂ - d _{1m} [*] c _{1u} / u ₂ ⎞⎟⎠	3.3.9
	ψ _{th} = 2 ⎛⎜ ⎝ γ - (φ _{2,La} / (tan β _{2B})) ⎡ τ ₂ + (A ₂ d _{1m} [*] tan β _{2B} / (A ₁ tan α ₁)) ⎤ ⎞⎟⎠	3.3.10
Head coefficient	ψ = 2η _h ⎛⎜ ⎝ γ - (φ _{2,La} / (tan β _{2B})) ⎡ τ ₂ + (A ₂ d _{1m} [*] tan β _{2B} / (A ₁ tan α ₁)) ⎤ ⎞⎟⎠	3.3.11
Static head coefficient	ψ _P = 1 - (u ₁ ² / u ₂ ²) + (w ₁ ² / u ₂ ²) - (w ₂ ² / u ₂ ²) - ζ _{La}	3.3.12
Specific work of impeller in relative system (without losses)	Y _{sch} = u ₂ ² - u ₁ ² - (u ₂ w _{2u} - u ₁ w _{1u})	3.3.13
Total enthalpy rise in relative system	Y _{tot,La} = (p ₂ - p ₁) / ρ + 1/2 ⎛⎜ ⎝ w _{2m} ² - w _{1m} ² + (u ₂ - w _{2u}) ² - (u ₁ - w _{1u}) ² ⎞⎟⎠	3.3.14
	w _{2m} = c _{2m}	

Table 3.4 Scaling laws and dimensionless coefficients

Scaling from model (subscript: M) to prototype (subscript: a)						Eq.
Scaling laws	Flow rate	$Q_a = Q_M \frac{n_a}{n_M} \left(\frac{D_a}{D_M} \right)^3 \frac{\eta_{v,a}}{\eta_{v,M}}$			1. Approximation: $\eta_{v,a} = \eta_{v,M}$	3.4.1
	Head	$H_a = H_M \left(\frac{n_a}{n_M} \right)^2 \left(\frac{D_a}{D_M} \right)^2 \frac{z_{st,a}}{z_{st,M}} \frac{\eta_{h,a}}{\eta_{h,M}}$			1. Approximation: $\eta_{h,a} = \eta_{h,M}$	3.4.2
	NPSH	$NPSH_a = NPSH_M \left(\frac{n_a}{n_M} \right)^2 \left(\frac{D_a}{D_M} \right)^2$				3.4.3
	Power	$P_a = P_M \left(\frac{n_a}{n_M} \right)^3 \left(\frac{D_a}{D_M} \right)^5 \frac{\rho_a}{\rho_M} \frac{z_{st,a}}{z_{st,M}} \frac{\eta_M}{\eta_a}$			1. Approximation: $\eta_a = \eta_M$	3.4.4
	Forces	$F_a = F_M \left(\frac{n_a}{n_M} \right)^2 \left(\frac{D_a}{D_M} \right)^4 \frac{\rho_a}{\rho_M}$				3.4.5
Dimensionless coefficients	Flow coefficient (outlet)	$\varphi_2 = \frac{Q/f_q}{\pi d_{2b} b_2 u_2} = \frac{\psi^{1.5}}{b_2^*} \left(\frac{n_q}{316} \right)^2$			$\varphi_{2.La} = \frac{Q_{La}/f_q}{\pi d_{2b} b_2 u_2} = \frac{c_{2m}}{u_2}$	3.4.6
	Flow coefficient (inlet)	$\varphi_1 = \frac{4Q_{La}}{f_q \pi (d_1^2 - d_n^2) u_1} = \frac{4 b_2^* \varphi_{2.La}}{d_1^{*3} k_n} = \frac{c_{1m}}{u_1}$				3.4.7
	Head coefficient	$\psi = \frac{2 g H}{u_2^2} = \frac{2 Y}{u_2^2}$				3.4.8
	Power coefficient with b_2	$\lambda = \frac{2 P}{f_q \rho \pi z_{st} d_{2b} b_2 u_{2a}^3} = \frac{\varphi_2 \psi}{\eta}$				3.4.9
	Power coefficient with d_2	$\lambda_D = \frac{2 P}{z_{st} \rho d_2^2 u_2^3} = \lambda \pi b_2^* f_q$				3.4.10
	Cavitation coefficient	$\sigma = \frac{2g NPSH}{u_1^2}$				3.4.11
	Reynolds number	$Re = \frac{c D}{v}$	$Re = \frac{w L}{v}$	$Re_u = \frac{u R}{v}$	$Re_u = \frac{2 s u_{sp}}{v}$	3.4.12
	Froude no. Fr Euler no. Eu	$Fr = \frac{c}{\sqrt{g D}}$		$Eu = \frac{2\Delta p}{\rho c^2}$		3.4.13 3.4.14
Specific speeds	Specific speed	$n_q = n \frac{\sqrt{Q_{opt} / f_q}}{H_{opt}^{0.75}} = 315.6 \frac{\sqrt{\varphi_{2opt} b_2^*}}{\psi_{opt}^{0.75}} = 52.9 \omega_s$			$\frac{n \text{ [rpm]}}{Q \text{ [m}^3\text{/s]}}$ $H \text{ [m]}$	3.4.15
	“Type number” DIN 24260 $n \text{ [rpm]}$	$n_q = 5.55 n \frac{\sqrt{Q_{opt} / f_q}}{(g H_{opt})^{0.75}}$		$\omega_s = \frac{\omega \sqrt{Q_{opt} / f_q}}{(g H_{opt})^{0.75}} = \frac{n_q}{52.9}$		3.4.16
	Suction specific speed	$n_{ss} = n \frac{\sqrt{Q_{opt} / f_q}}{NPSH_{opt}^{0.75}} = 157.8 \frac{\sqrt{\varphi_{1opt} k_n}}{\sigma_{opt}^{0.75}}$			$\frac{n \text{ [rpm]}}{Q \text{ [m}^3\text{/s]}}$ $NPSH \text{ [m]}$	3.4.17
	USA-units	$n_q = 0.0194 N_s$	$n_{ss} = 0.0194 N_{ss}$	$N_s, N_{ss} \text{ in: } Q \text{ [gpm]}, H \text{ [ft]}$		3.4.18

Table 3.5 Power balance, efficiencies and losses					Eq.	
Reference values	Q _{Ref} = 1 m³/s n _{Ref} = 1500 min					
Power balance	P = ∑ _{st} $\frac{\rho \text{ g H Q}}{\eta_v \eta_h}$ + ∑ _{st} P _{RR} + ∑ P _{S3} + P _m + P _{er} + P _{Rez}				3.5.1	
Useful power P _u	P _u = ρ g H _{tot} Q				3.5.2	
Overall efficiency (at coupling) η	η = $\frac{P_u}{P} = \frac{\rho \text{ g H}_{tot} Q}{P}$				3.5.3	
	η = $\frac{\eta_v \eta_h \rho \text{ g H}_{tot} Q}{\rho \text{ g H}_{tot} Q + \eta_v \eta_h (\sum_{st} P_{RR} + \sum P_{S3} + P_m + P_{er} + P_{Rez})}$					
Inner power P _i	P _i = P – P _m				3.5.4	
Inner efficiency η _i	η _i = $\frac{\rho \text{ g H}_{tot} Q}{P_i} = \frac{\eta}{\eta_m}$				3.5.5	
Mechanical efficiency η _m	η _m = 1 – $\frac{P_m}{P}$				3.5.6	
Mechanical losses P _m	$\frac{P_m}{P_{opt}} = 0.0045 \left(\frac{Q_{Ref}}{Q} \right)^{0.4} \left(\frac{n_{Ref}}{n} \right)^{0.3}$				3.5.6a	
Stage efficiency	η _{st} = $\frac{\rho \text{ g (H}_{tot} + Z_{EA}) (Q + \frac{Z_{st1}}{Z_{st}} Q_E + \frac{Z_{st2}}{Z_{st}} Q_{CB})}{P - P_m - P_{er}}$				3.5.7	
Calculation of hydraulic efficiency from test (P _{Rec} = 0)	η _h = $\frac{\rho \text{ g H}_{tot} (Q + Q_{sp} + \frac{Z_{st1}}{Z_{st}} Q_E + \frac{Z_{st2}}{Z_{st}} Q_{CB} + Q_h)}{P - \sum P_{RR} - \sum P_{S3} - P_m - P_{er}} = \frac{1 + \frac{Q_{sp}}{Q} + \frac{Z_{st1} Q_E}{Z_{st} Q} + \frac{Z_{st2} Q_{CB}}{Z_{st} Q} + \frac{Q_h}{Q}}{\frac{1}{\eta} - \frac{\sum P_{RR} + \sum P_{S3} + P_m + P_{er}}{P_u}}$				3.5.8	
Volumetric efficiency η _v	η _v = $\frac{Q}{Q + Q_{sp} + \frac{Z_{st1}}{Z_{st}} Q_E + \frac{Z_{st2}}{Z_{st}} Q_{CB} + Q_h} = \frac{1}{1 + \frac{Q_{sp}}{Q} + \frac{Z_{st1} Q_E}{Z_{st} Q} + \frac{Z_{st2} Q_{CB}}{Z_{st} Q} + \frac{Q_h}{Q}}$				3.5.9	
Leakage loss at impeller (API-clearance)	$\frac{Q_{sp}}{Q_{opt}} = \frac{a \text{ } z_H}{n_q^m}$	n _q ≤ 27 ≥ 27	a 4.1 0.15	m 1.6 0.6	With balance: Holes: z _H = 2 Else: z _H = 1	3.5.10
Leakage loss of inter-stage seal	$\frac{Q_{S3}}{Q_{opt}} = \frac{5.5}{n_q^{1.8}}$				$\frac{\Delta H_{S3}}{H_{opt}} \approx 0.4$	3.5.11
Power loss of inter-stage seal	P _{S3} = ρ g ΔH _{S3} Q _{S3}				$\frac{P_{S3}}{P_{u, st, opt}} = \frac{2.2}{n_q^{1.8}}$	3.5.11a
Disk friction power loss of radial impellers. f _{R,La} from Eq. (T3.6.6); f _L from Eq.(T3.6.7)	$\frac{P_{RR}}{P_{u, opt}} = \frac{770 \text{ } f_{R, La} \text{ } f_L}{n_q^2 \text{ } \psi_{opt}^{2.5} \text{ } Re^{0.2} \text{ } f_q}$				Turbulent: Re > 10 ⁵	3.5.12
	$\frac{P_{RR}}{P_{u, opt}} = \frac{38500 \text{ } k_{RR}}{n_q^2 \text{ } \psi_{opt}^{2.5} \text{ } f_q}$				General	3.5.13

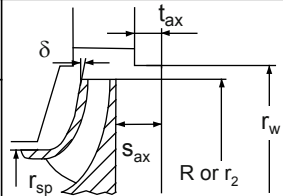
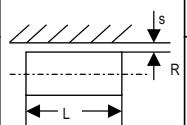
Table 3.6 Friction losses of rotating disks or cylinders				Eq.
Reynolds number	$Re = \frac{u R}{\nu} = \frac{\omega R^2}{\nu}$			3.6.1
Friction power <i>per side</i> of a rotating disk for $\delta < 45^\circ$	$P_{RR} = \frac{k_{RR}}{\cos \delta} \rho \omega^3 R^5 \left\{ 1 - \left(\frac{R_n}{R} \right)^5 \right\}$			3.6.2
Valid for $Re > 10$	$k_{RR} = \frac{\pi R}{2Re s_{ax}} + \frac{0.02}{Re^{0.2}} \cdot \frac{1 + \frac{s_{ax}}{R}}{1 + \frac{s_{ax}}{2R}} f_L f_{R,La}$			3.6.3
Friction power of a rotating cylinder ; valid for $Re > 10$	$P_{RZ} = k_{RZ} \rho \omega^3 R^4 L$			3.6.4
	$k_{RZ} = \frac{2 \pi R}{Re s} + \frac{0,075}{Re^{0,2}} \cdot \frac{1 + \frac{s}{R}}{1 + \frac{s}{2R}} \cdot f_R$			3.6.5
Influence of roughness ε to be calculated with roughness of rotating disk	$f_{R,La} \equiv \frac{k_{RR}(\varepsilon)}{k_{RR}(\varepsilon = 0)} = \left\{ \frac{\log \frac{12.5}{Re}}{\log \left(0.2 \frac{\varepsilon}{r_2} + \frac{12.5}{Re} \right)} \right\}^{2.15}$		The roughness of the casing is not taken into account here, but in Eq. (3.6.13). $\varepsilon = \varepsilon_{max}/c_{eq}$	3.6.6
Influence of leakage valid for: $r_{sp}/r_2 > 0,3$ und $k_e \approx 0,5$	$f_L \equiv \frac{k_{RR}(Q_{sp})}{k_{RR}(Q_{sp} = 0)} = \exp \left\{ -350 \varphi_{sp} \left[\left(\frac{r_2}{r_{sp}} \right)^a - 1 \right] \right\}$		Leakage direction: *radially inwards: φ_{sp} positive; $a = 1.0$ *radially outwards: φ_{sp} negative; $a = 0.75$	3.6.7
Alternatives to Eq. (T3.6.3) hydraulically smooth; without leakage; analogous to [6]	$k_{RR} = \frac{\pi R}{2Re s_{ax}}$	$Re_{lam} \leq 8.7 \left(\frac{s_{ax}}{R} \right)^{-1.87}$	Laminar, merged boundary layers	3.6.8
	$k_{RR} = \frac{0.925}{Re^{0.5}} \left(\frac{s_{ax}}{R} \right)^{0,1}$	$Re_{lam} < Re < 2 \times 10^5$	Laminar, separated boundary layers	3.6.9
	$k_{RR} = \frac{0.02}{Re^{0.25}} \left(\frac{R}{s_{ax}} \right)^{1/6}$	$10^5 < Re < 10^6$	Turbulent, merged boundary layers	3.6.10
	$k_{RR} = \frac{0.0255}{Re^{0.2}} \left(\frac{s_{ax}}{R} \right)^{0.1}$	$Re > 2 \times 10^5$	Turbulent, separated boundary layers	3.6.11
Taking into account all factors which influence rotation in the sidewall gap [14]	$k_{RR} = \frac{\pi R}{2Re s_{ax}} + \frac{0.0625}{Re^{0.2}} (1 - k_o)^{1.75} f_{R,La} f_L$		Valid for $Re > 10$	3.6.12
	$k_o = \frac{1}{1 + \left(\frac{r_w}{r_2} \right)^2 \sqrt{\left(\frac{r_w}{r_2} + 5 \frac{t_{ax}}{r_2} \right) \frac{c_{f,casing}}{c_{f,impeller}}}}$		c_f from Eq. (1.33)	3.6.13

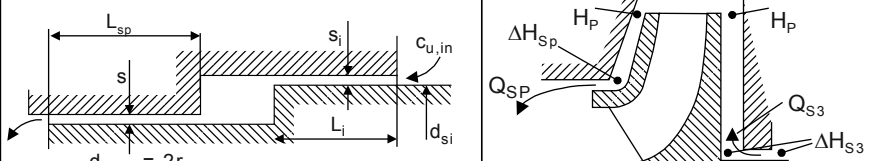
Table 3.7 (1) Leakage losses I: Annular seals				Eq.
Static pressure rise created by impeller	$H_p = \frac{u_2^2 - u_1^2 + w_1^2 - w_2^2}{2g} - Z_{La}$		$H_p = R_G H$ $R_G \approx 0,75$ for $n_q < 40$	3.7.1
Rotation factor k	Q_{sp} radially inward	$k = 0.9 y_{sp}^{0.087}$	$y_{sp} = Re_{u_2}^{0.3} \frac{s d_{sp}}{d_2^2} \sqrt{\frac{s}{L_{sp}}}$	3.7.2
	Q_{sp} radially outward	$k = 0.24 y_{sp}^{-0.096}$		
Pressure difference across seal if leakage flows radially <i>inwards</i>	$\Delta H_{Sp} = H_p - k^2 \frac{u_2^2}{2g} \left\{ 1 - \frac{d_{sp}^2}{d_2^2} \right\}$		$Re_{u_2} = \frac{u_2 r_2}{\nu}$	3.7.3
Pressure difference across seal if leakage flows radially <i>outwards</i>	$\Delta H_{S3} = H - H_p + k^2 \frac{u_2^2}{2g} \left\{ 1 - \frac{d_{S3}^2}{d_2^2} \right\}$			3.7.4
Pressure difference across balance device	$\Delta H_{EK} = (z_{ST} - 1)H + H_p - k^2 \frac{u_2^2}{2g} \left\{ 1 - \frac{d_{EK}^2}{d_2^2} \right\}$			3.7.5
Axial velocity in gap ΔH_{sp} , ΔH_{S3} , or ΔH_{EK} i = number of chambers	$c_{ax} = \sqrt{\frac{2g \Delta H}{\zeta_{EA} + \lambda \frac{L_{sp}}{2s} + \sum_i \left(\frac{d_{sp}}{d_{si}} \right)^2 \left(\frac{s}{s_i} \right)^2 \left\{ \zeta_K + \lambda_i \frac{L_i}{2s_i} \right\}}}$			3.7.6
Leakage flow rate	$Q_{sp} = \pi d_{sp} s c_{ax}$			3.7.7
Loss coefficients	$\zeta_{EA} = 1$ to 1.2 inlet + outlet loss; $\zeta_K = 1$ to 1.3 loss per chamber			3.7.8
Reynolds numbers	$Re_u = \frac{2 s u_{sp}}{\nu} \quad u_{sp} = \frac{\pi d_{sp} n}{60} \quad Re = \frac{2 s c_{ax}}{\nu}$			3.7.9
Transition laminar/turbulent at $Re^* = 2000$	$Re^* = \sqrt{Re^2 + \frac{1}{4} Re_u^2} = Re \sqrt{1 + \frac{1}{4} \left(\frac{u_{sp}}{c_{ax}} \right)^2}$			3.7.10
Friction coefficient for laminar flow, use for $Re < 2000$ ($u_{sp} = 0$)	$\lambda_0 = \frac{96}{Re} \left(1 - 0.6 \frac{e_x}{s} \right) \quad e_x = \text{eccentricity}$			3.7.11
Friction coefficient for turbulent flow, $Re > 2000$ ($u_{sp} = 0$)	$\lambda_0 = \frac{0.31}{\left\{ \log \left(A + \frac{6.5}{Re} \right) \right\}^2}$	Seal surface	A	3.7.12
		Rough	$0.135 \frac{\varepsilon}{s}$	
		Serrated	0.005 to 0.01	
		Isotropic pattern	0.01 to 0.03	
Transition laminar-turb.	use $\lambda = \text{Max}(\lambda_{lam}; \lambda_{tur})$ in order to avoid discontinuity			
Influence of rotation for turbulent flow $Re > 2000$	$\frac{\lambda}{\lambda_0} = \left\{ 1 + 0.19 \left(\frac{Re_u}{Re} \right)^2 \right\}^{0.375}$			3.7.13
Influence of rotation for flow with $Re < 2000$	$\frac{\lambda}{\lambda_0} = \left\{ 1 + 0.2 \left(\frac{Re_u}{2000} \right)^2 \right\}^{1.03}$			3.7.14
				

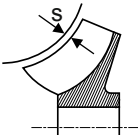
Table 3.7 (2) Leakage losses II: Radial or diagonal gaps, open impellers					Eq.
Semi-open impellers	Head loss	$\frac{\Psi_{(s=0)} - \Psi_{(s)}}{\Psi_{(s=0)}} = \frac{2.5 \frac{s}{d_2}}{\sqrt{b_2^* (1 - d_1^*) Z_{La}} \left(\frac{c}{t_2} \right)^{0.2} n_q^{0.1} (\sin \beta_2)^{1.2} (\sin \beta_1)^{0.4}}$			3.7.15
	Efficiency loss	$\frac{\eta_{(s=0)} - \eta_{(s)}}{\eta_{(s=0)}} = \frac{2}{3} \frac{\Psi_{(s=0)} - \Psi_{(s)}}{\Psi_{(s=0)}}$			3.7.16
	$\Delta\eta$ according to [B.16]	$\Delta\eta = 0.3 \frac{s}{b_2 - s}$	for $\frac{s}{b_2 - s} < 0.13$		3.7.16a
	Reduction of power consumption	$\frac{\lambda_{(s=0)} - \lambda_{(s)}}{\lambda_{(s=0)}} = \frac{1}{3} \frac{\Psi_{(s=0)} - \Psi_{(s)}}{\Psi_{(s=0)}}$			3.7.17
Diagonal gap	Head loss	$\frac{\Psi_{(s=0)} - \Psi_{(s)}}{\Psi_{(s=0)}} = (2.3 \text{ to } 3.3) \frac{s}{b_2}$		Upper value for smooth gap; lower value for gap with pump-out vanes	3.7.18
	Efficiency loss	$\frac{\eta_{(s=0)} - \eta_{(s)}}{\eta_{(s=0)}} = (1.6 \text{ to } 1.9) \frac{s}{b_2}$			3.7.19
	Increase of NPSH ₃	$\frac{\sigma_{(s)} - \sigma_{(s=0)}}{\sigma_{(s=0)}} = (15 \text{ to } 40) \frac{s}{b_2}$		see. Fig. 3.15	3.7.20
Radial gap	Reynolds numbers	$Re = \frac{2 s c_i}{\nu} \sqrt{a_r}$	$Re_u = \frac{2 s \omega r_a}{\nu} \sqrt{a_r}$	$a_r \equiv \frac{r_i}{r_a}$	3.7.21
	Friction coefficient	$\lambda = f(Re, Re_u, \epsilon)$ according to Table 3.7 (1)			
	Pressure difference across gap	according to Table 3.7 (1)			
	Constants	Leakage flow rate	Rotation faktor k		3.7.22
			Re < 10 ⁴	Re > 10 ⁴	
			0.4	0.5	
	Radial velocity in gap at inner radius r _i	Leakage radially inwards	$c_i = \sqrt{\frac{\frac{2\Delta p}{\rho} - k^2 \omega^2 r_a^2 (1 - a_r^2)}{\zeta_E a_r^2 + \frac{\lambda r_i}{2 s} (1 - a_r) + 1}}$		3.7.23
		Leakage radially outwards	$c_i = \sqrt{\frac{\frac{2\Delta p}{\rho} + k^2 \omega^2 r_a^2 (1 - a_r^2)}{\zeta_E + \frac{\lambda r_i}{2 s} (1 - a_r) + a_r^2}}$		
Leakage flow	$Q_{sp} = 2\pi r_i s c_i$				3.7.24

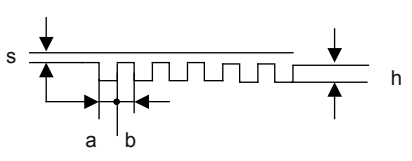
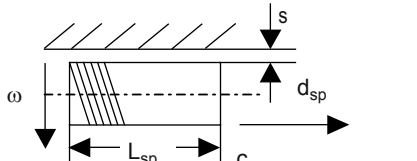
Table 3.7 (3) Screw-type pumping grooves in turbulent flow, [18]				
Pumping grooves should be selected only if: $\frac{c_{ax}}{u_{sp}} < 0.7$ and $\frac{L_{sp}}{s} > 50$		The selection of the geometric parameters depends on $\frac{c_{ax}}{u_{sp}}$		
Given quantities	$d_{sp}, L_{sp}, n, \Delta H, s, \rho, \nu$			Eq.
Axial velocity in gap Estimation for first iteration: $\lambda = 0.06$	$c_{ax} = \sqrt{\frac{2g \Delta H}{\zeta_{EA} + \lambda \frac{L_{sp}}{2s}}}$			3.7.30
Loss coefficient	$\zeta_{EA} = 1.5$ for inlet and outlet loss			3.7.31
Reynolds numbers	$Re_u = \frac{2 s u_{sp}}{\nu}$ $u_{sp} = \frac{\pi d_{sp} n}{60}$ $Re = \frac{2 s c_{ax}}{\nu}$			3.7.32
Flow is turbulent if	$\frac{u_{sp} s}{\nu} > 700$			3.7.33
Number of threads (integer)	$z = \frac{\pi d_{sp} \sin \alpha}{a + b} = \frac{\pi d_{sp} \sin \alpha}{\left(\frac{a + b}{s}\right) s}$			3.7.34
Inclination angle $\alpha = 7$ to 13° (up to 15°)	$\alpha [^\circ] = 13 - 8 \frac{c_{ax}}{u_{sp}}$	Eqs. (3.7.39. and (3.7.40) are valid for the parameters given in the column on the right	$\alpha = 10^\circ$	3.7.35
Depth of grooves $h/s = 2,5$ to 4	$\frac{h}{s} = 1.2 \left(\frac{c_{ax}}{u_{sp}} \right)^{-0,68}$		$\frac{h}{s} = 2.5$	3.7.36
Width of grooves	$\frac{a}{a + b} = 0.5$ to 0.7		$\frac{a}{a + b} = 0.5$	3.7.36
Check that	$\frac{a + b}{s} = 15$ to 20		$\frac{a + b}{s} = 17$	3.7.38
Friction coefficient for $u_{sp} = 0$ in turbulent flow	$\lambda_0 = \frac{0.32}{\left\{ \log \left(0.0022 + \frac{20}{Re} \right) \right\}^2}$		Range tested: $Re = 400$ to $21'000$ $Re_u = 0$ to $37'000$	3.7.39
Friction coefficient with rotation	$\lambda = \lambda_0 \left\{ 1 + k \frac{Re_u}{Re} \right\}^2$	$k = 0.0243 Re^{0.2255}$		3.7.40
With the friction coefficient from Eq. (3.7.40) the axial velocity c_{ax} is calculated iteratively from Eq. (3.7.30)				
Leakage flow rate	$Q_{sp} = \pi d_{sp} s c_{ax}$			3.7.41
				

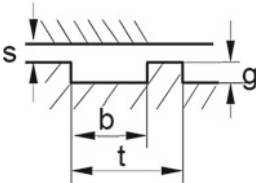
Table 3.7(4) Serrated annular seals				
Land: $a = t - b$				Eq.
Pressure loss of seal without ζ_{EA} Eq. (1.12)	$\Delta p_v = \frac{\rho}{2} (c_{vC} - c_b)^2 = \frac{\rho}{2} c_{ax}^2 \frac{\lambda_{0,eq} L_{sp}}{2s}$			3.7.42
Maximum velocity with jet contraction α_z	$c_{vC} = \frac{c_{ax}}{\alpha_z}$			3.7.43
Velocity at end of groove at length $x = b$	$c_b = 1.75 \beta \left(\frac{s}{b} \right)^{0.5} c_{ax}$			3.7.44
Jet contraction	$\alpha_z = 0.79 \times e^{-0.001(a/s)}$			3.7.45
Equivalent friction factor for seal: $\lambda_{0,eq} = \frac{a}{t} \left(1 + 2 \frac{g}{t} \right) \lambda_{smooth} + 2 \left(\frac{s}{t} \right) \left(\frac{b}{t} \right) \left\{ \frac{1}{\alpha_z} - 1.75 \beta \left(\frac{s}{b} \right)^{0.5} \right\}^2$ Empirical factor $\beta = 0.87$				3.7.46
Recommendations	$\begin{aligned} g/s &= 3 \text{ to } 6 \\ g/b &= 0.2 \text{} \mathbf{0.25} \text{} 0.3 \\ b/t &= 0.7 \text{} \mathbf{0.8} \text{} 0.9 \end{aligned}$			
Test parameters covered by the correlations			min.	max.
	s	mm	0.3	0.9
	g	mm	1.0	5.0
	b	mm	1.5	17.5
	t	mm	5.0	20.0
	$a = t - b$	mm	1.0	15.0
	b/s	-	4.8	29.0
	b/t	-	0.3	0.9
	t/s	-	10.8	32.2
	g/s	-	3.2	5.4
	g/b		0.1	1.0
	a/s	-	2.7	16.2

Table 3.8 (1) Hydraulic losses in impeller		Eq.
Hydraulic efficiency	$\eta_h = \frac{Y}{Y_{sch}} = \frac{H}{H_{th}} = \frac{H}{H + Z_{La} + Z_{Le} + Z_{EA}} = \frac{\Psi}{\Psi + \zeta_{La} + \zeta_{Le} + \zeta_{EA}}$	3.8.1
Average relative velocity in impeller channel	$w_{av} = \frac{2Q_{La}}{f_q z_{La} (a_2 b_2 + A_{1q})}$	3.8.2
Friction coefficient (ϵ = roughness)	$Re = \frac{w_{av} L_{sch}}{\nu}$ $c_f = \frac{0.136}{\left\{ -\log \left(0.2 \frac{\epsilon}{L_{sch}} + \frac{12.5}{Re} \right) \right\}^{2.15}}$	3.8.3
Hydraulic diameter	$D_h = \frac{2(a_2 b_2 + A_{1q})}{a_1 + b_1 + a_2 + b_2}$	3.8.4
Dissipation coefficient	$c_d = (c_f + 0.0015)(1,1 + 4b_2^*)$	3.8.5
Friction and mixing losses	$\zeta_{La,R} \equiv 2g \frac{Z_{La,R}}{u_2^2} = 4c_d \frac{L_{sch}}{D_h} \left(\frac{w_{av}}{u_2} \right)^2$	3.8.6
Relative velocity vector	$w_{1m} = \sqrt{c_{1m}^2 + (u_1 - c_{1u})^2}$	3.8.7
Velocity in impeller throat	$w_{1q} = \frac{Q_{La}}{f_q z_{La} A_{1q}}$	3.8.8
Shock loss at impeller inlet	$\zeta_{La,C} \equiv 2g \frac{Z_{La,C}}{u_2^2} = 0.3 \left(\frac{w_{1m} - w_{1q}}{u_2} \right)^2$ only for: $\frac{w_{1q}}{w_{1m}} > 0.65$	3.8.9
Impeller loss	$\zeta_{La} = \zeta_{La,R} + \zeta_{La,C}$	3.8.10
Alternative calculation with wetted surface and c_f instead of c_d		
Wetted surface in <i>radial</i> impellers per impeller side	$A_{ben} = \frac{\pi}{4} (d_{2a}^2 + d_{2i}^2 - d_1^2 - d_n^2) + z_{La} L_{sch} (b_1 + b_2)$	3.8.11
Average relative velocity	$\bar{w} = 0.5 (w_1 + w_2)$	3.8.12
Dissipated power per impeller side	$P_d = \frac{\rho}{2} c_f \bar{w}^3 A_{ben} = \rho g Q Z_{La,R}$	3.8.13
Friction losses	$\zeta_{La,R} \equiv 2g \frac{Z_{La,R}}{u_2^2} = \frac{c_f A_{ben}}{\varphi_2 A_2} \left(\frac{\bar{w}}{u_2} \right)^3$	3.8.14

Table 3.8 (2) Hydraulic losses in diffuser or volute						Eq.
Hydraulic efficiency		$\eta_h = \frac{Y}{Y_{sch}} = \frac{H}{H_{th}} = \frac{H}{H + Z_{La} + Z_{Le} + Z_{EA}} = \frac{\Psi}{\Psi + \zeta_{La} + \zeta_{Le} + \zeta_{EA}}$			3.8.1	
Diffuser	Velocity vector at impeller outlet	$c_{2u} = \frac{gH}{\eta_h u_2} + d_{1m}^* c_{1u}$	$c_{2m} = \frac{Q_{La}}{\pi d_2 b_2}$	$c_2 = \sqrt{c_{2u}^2 + c_{2m}^2}$	3.8.15	
	Flow velocity in throat	$c_{3q} = \frac{Q + Q_E}{Z_{Le} a_3 b_3}$				3.8.16
	Diffuser c _p from Fig. 1.18 or 1.19	$A_R = \frac{a_4 b_4}{a_3 b_3}$	$\frac{L_{3-4}}{R_1} = L_{3-4} \sqrt{\frac{\pi}{a_3 b_3}}$	$c_p = f\left(A_R, \frac{L_{3-4}}{R_1}\right)$	3.8.17	
	Friction losses in inlet region	$\zeta_{2-3} \equiv \frac{2g Z_{2-3}}{u_2^2} = (c_f + 0.0015) \left(a_3^* + b_3^*\right) \frac{\pi^3 \left(\varphi_2 b_2^*\right)^2}{8 \left(Z_{Le} a_3^* b_3^*\right)^3} \left(1 + \frac{c_2}{c_{3q}}\right)^3$				3.8.18
	Diffuser loss including overflow channel	$\zeta_{Le} \equiv \frac{2g Z_{Le}}{u_2^2} = \zeta_{2-3} + \left(\frac{c_{3q}}{u_2}\right)^2 \left\{ 0.3 \left(\frac{c_2}{c_{3q}} - 1\right)^2 + 1 - c_p - \frac{1 - \zeta_{ov}}{A_R^2} \right\}$				3.8.19
	Overflow channel (alone)	$\Delta \zeta_R = \zeta_{ov} \frac{c_4^2}{u_2^2}$		$\zeta_{ov} = 0.2 \text{ to } 1.5$		3.8.20
Volute	Friction loss ΔA= wetted surface	$\zeta_{Sp,R} = \frac{2gZ_{Sp}}{u_2^2} = \frac{2 P_d}{\rho Q u_2^2} = \frac{1}{Q u_2^2} \Sigma (c_f + 0.0015) c^3 \Delta A$				3.8.21
	Diffuser/discharge nozzle (c _x = diffuser inlet velocity)	$\zeta_{Sp,D} = \frac{c_x^2}{u_2^2} \left(1 - c_p - \frac{1}{A_R^2} \right)$				3.8.22
	Losses of volute including discharge nozzle	$\zeta_{Sp} = \zeta_{Sp,R} + \zeta_{Sp,D} + \zeta_{LS} + (\zeta_{Le} - \zeta_{Le,min})$ $\zeta_{LS} \text{ from Eq. (T3.8.24)}$				3.8.22a
Vaneless diffuser	Friction in vaneless diffuser with constant width	$\zeta_{LR} = \frac{2 c_f r_2}{b_3 \sin \alpha_3 \cos^2 \alpha_3} \left(\frac{c_{2u}}{u_2} \right)^2 \left(1 - \frac{r_2}{r_4} \right)$				3.8.23
	Shock losses	$\zeta_{LS} = \varphi_{2,La}^2 \left(\tau_2 - \frac{b_2}{b_3} \right)^2$				3.8.24
Deceleration and shock losses in vaned diffusers and volutes		$\zeta_{Le} - \zeta_{Le,min} = a + b \frac{c_{3q}}{c_2} + c \left(\frac{c_{3q}}{c_2} \right)^2$		volutes	diffusers	3.8.25
				c = 0.63	c = 0.65	
				x ₀ = 0.87	x ₀ = 0.93	
				a = c x ₀ ²	a = c x ₀ ²	
				b = -2 x ₀ c	b = -2 x ₀ c	
Add losses of vaneless space between impeller and diffuser or volute.						

Table 3.9 Statistical efficiency data and efficiency scaling

Pump type		n_q	$Q_{Ref} = 1 \text{ m}^3/\text{s}$ Valid for $Q \geq 0.005 \text{ m}^3/\text{s}$			Eq.
Overall efficiency	Exponent		$m = 0.1 a \left(\frac{Q_{Ref}}{Q} \right)^{0.15} \left(\frac{45}{n_q} \right)^{0.06}$	$Q \leq 1 \text{ m}^3/\text{s}$	$a = 1$	
				$Q > 1 \text{ m}^3/\text{s}$	$a = 0.5$	
	Radial pumps single-stage	≤ 100	$\eta_{opt} = 1 - 0.095 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.3 \left\{ 0.35 - \log \frac{n_q}{23} \right\}^2 \left(\frac{Q_{Ref}}{Q} \right)^{0.05}$			3.9.1
			$\eta_{opt} = 0.94 - 0.05 \left(\frac{Q_{Ref}}{Q} \frac{n}{n_{Ref}} X \right)^{0.213} - 0.29 \left\{ \log \frac{n_{q,Ref}}{n_q} \right\}^2$			3.9.1A
			$Q_{Ref} = 1 \text{ m}^3/\text{s} = 3600 \text{ m}^3/\text{h};$ $n_{Ref} = 1000 \text{ rpm}; n_{q,Ref} = 44$			
			Roughness influence: $X = \left(\frac{\epsilon}{\epsilon_{Ref}} \right)^2$ Use equivalent sand roughness with $\epsilon_{Ref} = 4 \text{ }\mu\text{m}$ corresponding to N7 in Table D1.3			
Hydraulic efficiency	Radial pumps multistage	≤ 60	$\eta_{opt} = 1 - 0.116 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.4 \left\{ 0.26 - \log \frac{n_q}{25} \right\}^2$			3.9.2
	Semi-axial and axial pumps	≥ 45	$\eta_{opt} = 1 - 0.095 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.09 \left\{ \log \frac{n_q}{45} \right\}^{2.5}$			3.9.3
	Pumps with double-entry impellers	≤ 50	$\eta_{opt} = 1 - 0.095 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.35 \left\{ 0.35 - \log \frac{n_q}{17.7} \right\}^2 \left(\frac{Q_{Ref}}{Q} \right)^{0.05}$			3.9.4
	Theoretically achievable efficiency		$\eta_{th,er} = \eta_{opt} + 0.35 \left(\frac{Q_{Ref}}{Q} \right)^{0.08} (1 - \eta_{opt})$			3.9.5
	Exponent		$m = 0.08 a \left(\frac{Q_{Ref}}{Q} \right)^{0.15} \left(\frac{45}{n_q} \right)^{0.06}$	$Q \leq 1 \text{ m}^3/\text{s}$	$a = 1$	
Hydraulic efficiency				$Q > 1 \text{ m}^3/\text{s}$	$a = 0.5$	
	Radial pumps single-stage	≤ 100	$\eta_{h,opt} = 1 - 0.055 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.2 \left\{ 0.26 - \log \frac{n_q}{25} \right\}^2 \left(\frac{Q_{Ref}}{Q} \right)^{0.1}$			3.9.6
	Semi-axial and axial pumps	≥ 45	$\eta_{h,opt} = 1 - 0.055 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.09 \left\{ \log \frac{n_q}{45} \right\}^{2.5}$			3.9.7
	Radial pumps multistage	≤ 60	$\eta_{h,opt} = 1 - 0.065 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.23 \left\{ 0.3 - \log \frac{n_q}{23} \right\}^2 \left(\frac{Q_{Ref}}{Q} \right)^{0.05}$			3.9.8
Loss due to balance holes		$\Delta \eta_{EL} = 0.018 \left(\frac{25}{n_q} \right)^{1.6} \quad n_q < 40$			$n_q > 40: \Delta \eta_{EL} = 0.01$	3.9.9
Uncertainty of η		$\Delta \eta_{Tot} = \pm 0.2 (1 - \eta_{opt})$				3.9.10
Efficiency scaling	Scaling of hydraulic efficiency	$\frac{\eta_{h,a}}{\eta_{h,M}} = \left\{ 1 - \eta_{h,M} \frac{\zeta_{R,M}}{\psi} \left(1 - \frac{c_{f,a}}{c_{f,M}} \right) \right\}^{-1}$			$\frac{\zeta_{R,M}}{\psi} = b \left(\frac{12}{n_q} \right)^{0.83}$ $b = 0.06 \text{ to } 0.1$	3.9.11
	Quick estimate of overall efficiency	$\eta_a = 1 - (1 - \eta_M) \left(\frac{n_M v_a}{n_a v_M} \right)^{0.07} \left(\frac{d_{2,M}}{d_{2,a}} \right)^{0.2}$				3.9.11A
	Scaling of stage efficiency	$\frac{1 - \eta_{St,a}}{1 - \eta_{St,M}} = V + \frac{1 - V}{2} \left(\frac{c_{f,a}}{c_{f,M}} + \frac{k_{RR,a}}{k_{RR,M}} \right)$			$V = 0.3 + 0.4 \frac{n_q}{200}$	3.9.12
	Efficiency scaling according to [N.5]:	$\eta_{hR} = \frac{\rho g H_{tot} (Q + Q_{sp} + Q_E)}{P - P_m} = \frac{\eta}{\eta_v \eta_m}$			To be calculated for model (M) and prototype (a). $\eta_a = \eta_{hR,a} + \Delta \eta$ η_{hR} includes hydraulic and disk friction losses	3.9.13
		$\Delta \eta = (0.4 \text{ to } 0.6) (1 - \eta_{hR,M}) \left[1 - \left(\frac{Re_M}{Re_a} \right)^{0.2} \right]$				

Table 3.10 (1) Influence of roughness and Reynolds number on efficiency					
Quantities given for “model”, baseline or water	$n_M, Q_{opt,M}, H_{opt,M}, d_{2,M}, \eta_M, v_M$, roughness ε_M of the dif- ferent components			Eq.	
Quantities given for “prototype” or viscous fluid	$n_a, Q_{opt,a}, d_{2,a}, v_a$, roughness ε_a of the different compo- nents				
Specific speed	$\omega_s = \omega \frac{\sqrt{Q_{opt}/f_q}}{(g H_{opt})^{0.75}} = \frac{n_q}{52.9}$		ω_s is dimensionless; n_q is with rpm, m ³ /s, m	3.10.1	
Reynolds number	$Re = \frac{u_2 r_2}{\nu} = \frac{\omega r_2^2}{\nu}$		Re is used for hydraulic chan- nels and disk friction	3.10.2	
Estimated leakage flow, if no details of labyrinths are available	$\frac{Q_{sp}}{Q_{opt}} = \frac{4.1}{n_q^{1.6}} = \frac{7.16 \times 10^{-3}}{\omega_s^{1.6}}$		Can be used for Q_{sp} and Q_E as first approximation	3.10.3	
Volumetric effi- ciency use Eq. (T3.5.9) for back-to-back	$\eta_{vol} = \frac{Q}{Q + Q_{sp} + Q_E + Q_{s3} \frac{\Delta H_{s3} (z_{st} - 1)}{H_{st} z_{st}}}$			$\Delta H_{s3}/H_{st} = 0.4$ is a good estima- tion	3.10.4
Equivalent (sand) roughness $c_{eq} = 2.6$	$\varepsilon = \frac{\varepsilon_{max}}{c_{eq}} = \frac{6 \varepsilon_{CLA}}{c_{eq}}$ with $\varepsilon_{max} = 6 \varepsilon_{CLA}$			Used in Eqs. (3.10.7 and 3.10.16)	3.10.5
Average roughness for hydraulic chan- nels	$\varepsilon_{av,h} = (1 - a_\varepsilon) \varepsilon_{La} + a_\varepsilon \varepsilon_{Le}$		$a_\varepsilon = 0.98 - 0.0012 n_q \sqrt{f_q}$	3.10.6	
Roughness effect on disk friction, for turbulent flow only	$f_R = \frac{C_{f,rough}}{C_{f,smooth}} = \left\{ \frac{\log \frac{12.5}{Re}}{\log \left(0.2 \frac{\varepsilon}{r_2} + \frac{12.5}{Re} \right)} \right\}^{2.15}$			Calculated with equivalent sand roughness $\varepsilon_{impeller}$ and ε_{casing}	3.10.7
Rotation of fluid with zero leakage ($Q_{sp} = 0$)	$k_o = \frac{c_u}{\omega r} = \frac{1}{1 + \left(\frac{r_w}{r_2} \right)^2 \sqrt{\left(\frac{r_w}{r_2} + 5 \frac{t_{ax}}{r_2} \right) \frac{f_{R,casing}}{f_{R,impeller}}}}$			Open side- wall gaps: $r_w = r_2$; $t_{ax} = 0$	3.10.8
Disk friction coeffi- cient	$k_{RR} = \left[\frac{\pi r_2}{2 Re s_{ax}} + \frac{0.0625}{Re^{0.2}} (1 - k_o)^{1.75} f_{R,imp} f_L f_{RS} \right] f_{therm}$				3.10.9
Effect of pump-out vanes	$f_{RS} = 0.63 + 0.6 d_{RS}/d_2$		$d_{RS} = \text{o.d. of pump-out vanes};$ if no pump-out vanes: $f_{RS} = 1.0$		3.10.10
Thermal effect on disk friction (em- pirical)	$f_{therm} = \exp \left\{ -2 \times 10^{-5} \left(\frac{v}{v_{Ref}} \right)^{1.34} \right\}$			$v_{Ref} = 10^{-6} \text{ m}^2/\text{s}$ see Chap. 13.1	3.10.11
Influence of lea- kage, valid for: $r_{sp}/r_2 > 0.3$	$f_L = \exp \left\{ -350 \varphi_{sp} \left(\left[\frac{r_2}{r_{sp}} \right]^a - 1 \right) \right\}$			* radially inwards: φ_{sp} positive; $a = 1.0$ * radially outwards: φ_{sp} negative; $a = 0.75$	3.10.12

Table 3.10 (2) Influence of roughness and Reynolds number on efficiency			Eq.
Disk friction $f_{\text{geo}} \approx 1.22$ for typical radial impellers	$\left(\frac{P_{\text{RR}}}{P_u} \right)_M = \frac{8\sqrt{2} k_{\text{RR},M} f_{\text{geo}}}{\omega_s^2 \psi_{\text{opt}}^{2.5} f_q} = \frac{31680 k_{\text{RR},M} f_{\text{geo}}}{n_q^2 \psi_{\text{opt}}^{2.5} f_q}$		3.10.13
Mechanical losses	$\frac{P_m}{P_u} = \frac{0.0045}{\eta} \left(\frac{Q_{\text{Ref}}}{Q} \right)^{0.4} \left(\frac{n_{\text{Ref}}}{n} \right)^{0.3}$	$Q_{\text{Ref}} = 1 \text{ m}^3/\text{s}$ $n_{\text{Ref}} = 1500 \text{ rpm}$	3.10.14
Hydraulic efficiency of model or baseline	$\eta_{h,M} = \frac{\eta}{\eta_{\text{vol}} \left\{ 1 - \eta \left[\left(\frac{P_{\text{RR}}}{P_u} \right)_M + \frac{P_m}{P_u} \right] \right\}}$	Calculated from measured overall efficiency and loss analysis for model and prototype	3.10.15
Friction coefficient of hydraulic channels for turbulent flow with $Re > Re_{\text{crit}}$	$c_f = \frac{0.136}{\left\{ -\log \left(0.2 \frac{\varepsilon}{r_2} + \frac{12.5}{Re} \right) \right\}^{2.15}}$	c_f is calculated with ε_{av} from Eq. (3.10.6) Tu = turbulence level; valid for $Tu < 0.1$	3.10.16
	$Re_{\text{crit}} = \frac{3 \times 10^6}{1 + 10^4 Tu^{1.7}}$		
Friction coefficient of hydraulic channels for laminar flow	$c_{f,\text{lam}} = \frac{2.65}{Re^{0.875}} - \frac{2}{8 Re + 0.016 / Re} + \frac{1.328}{\sqrt{Re}}$	$0.01 < Re < Re_{\text{crit}}$	3.10.17
Fraction of losses dependent on roughness and Reynolds number	$\frac{\zeta_{R,M}}{\psi_{\text{opt}}} = \left\{ \frac{1}{\eta_{h,M}} - 1 \right\} (a_1 - b_1 n_q \sqrt{f_q})$	$a_1 = 0.635$ $b_1 = 0.0016$	3.10.18
Correction factor for hydraulic efficiency	$f_{\eta_{h,\text{opt}}} = 1 - \frac{\zeta_{R,M}}{\psi_{\text{opt}}} \left(\frac{c_{f,a}}{c_{f,M}} - 1 \right)$	$c_{f,a}$ und $c_{f,M}$ from Eq. (3.10.16)	3.10.19
Correction factor for efficiency at: $f_{\eta_{\text{vol}}} = 1.0$ $f_{\eta_m} = 1.0$; $\rho_a = \rho_M$ $f_Q = f_H = f_{\eta_h}$	$f_{\eta} \equiv \frac{\eta_a}{\eta_M} = \frac{f_{\eta_h} \left[1 + \left\{ \left(\frac{P_{\text{RR}}}{P_u} \right)_M + \left(\frac{P_m}{P_u} \right)_M \right\} \eta_{\text{vol},M} \eta_{h,M} \right]}{1 + \left\{ \left(\frac{P_{\text{RR}}}{P_u} \right)_M \frac{k_{\text{RR},a}}{k_{\text{RR},M}} + \left(\frac{P_m}{P_u} \right)_M \right\} \frac{\eta_{\text{vol},M} \eta_{h,M}}{f_{\eta_h}}}$		3.10.20
Efficiency Q_{opt}	$\eta_a = f_{\eta} \eta_M$	f_{η} from Eq. (3.10.20)	3.10.21
Hydraulic efficiency	$\eta_{ha} = f_{\eta_h} \eta_{hM} f_{\eta_h}$ from Eq. (3.10.19) $f_Q = f_H = f_{\eta_h}$ for highly viscous fluids (Chap. 13.1)		3.10.22
The above efficiency scaling procedure is to be followed for the best efficiency point. The factor f_{η} is then applied to the entire curve: $\eta_a(Q_a) = f_{\eta} \eta_M(Q_M)$.			
f_{geo} represents the ratio of the total disk friction losses of an impeller (on <i>all</i> rotating surfaces) to those of the flat sides of front and rear shroud			

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